

ECU MODULES REPAIR EBOOK



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Multec 700 - GM
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Multec MPFI EMS –GM
Multec H VHC - GM
EEC V - ANIL AND ENIO - FORD
EEC IV - FORD
Magneti Marelli 4CFR - Ford
Magneti Marelli 4AFR - Ford
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Magneti Marelli 1 AVB - VW
Magneti Marelli 4SV
Temic 1.74
1 AVP - VW
MP 9.0 - VW
1AVI - VW
4AVP - VW
4BV - VW
7.5.20 –VW
Bosch 3.8.3
Bosch LE - JETRONIC
Bosch ME 7.4.4 Petrol
Magneti Marelli 5NP 02
Magneti Marelli 6LP1
Magneti Marelli 6LPb
Magneti Marelli 5NP 01

Bosch MP7.2

Siemens Sirius 3134

Siemens - Sirius 32

Mercedes Class A 160 and 190 - VDO MSM

Introduction

Over time and the need to improve everything, the number of electronic components in automobiles comes progressively increasing in order to improve from fuel efficiency to driver comfort.

As a result, more and more mechanical parts are transformed into electronic components.

Today the vast majority of automobiles use the electronic injection and are constantly increasing as well as the need for qualified technicians to ensure its smooth operation.

Therefore, investing in material and knowledge is no longer a matter of staying ahead of the competition, it's almost an obligation.

Next, you will learn more about injection modules electronics: the main characteristics, identify the main components, their functions, the most defects common, and more.

Multimeters

One of the main and essential tools for anyone works with electrical and automotive electronics is the multimeter, know all the functions and know how to use it correct way can help a lot when diagnosing failure or make a repair.

Instructions and Basics

There are basically two types of multimeters, the analog and digital, but we go from the emphasis to digital, because it offers greater ease when making a measurement.

Through a display we can easily see the greatness measured without the need to multiply and readings on complicated scales as in the case of analog multimeters.

We use the digital multimeter for various purposes, the most common is to perform some measures, such as example:

- Measure electrical voltage (volts - V).
- Measure electrical current (amps - A).
- Measure electrical resistance (ohms - Ω).

In addition to these examples, depending on the model, it can have options for other measures such as: frequency, temperature, semiconductors (for example a diode), transistors, capacitance, continuity (with audible signal), etc.

Regardless of the magnitude, in digital multimeters the value of the selected scale already shows the maximum value to be measured. See below for an example of the most common found for each scale:

- **Resistance scales** : 200 Ω , 2000 Ω or 2k Ω , 20k Ω , 200k Ω , 2M Ω or 2000k Ω
- **DC voltage ranges**: 200mV, 2V, 20V, 1000V or 200Mv, 2V, 20V, 1000V or 1kV.
- **Alternating voltage scales**: 200V, 750V or 200V, 750V.
- **DC scales**: 200uA, 2000uA or 2mA, 20mA, 200mA, 2A, 10 or 20A.

- **Alternating current scales** : 2A, 10A.

The selection of the scales can usually be done through a rotary key, buttons or even no key any, are the case of the digital multimeter the auto function range (auto scale), it selects the scale and the magnitude automatically.

There are also some models of multimeters with auto range which has only one voltage scale, one for current and one for resistance, it is not necessary to worry about selecting a specific scale to measure a certain voltage value.

An important detail when handling a digital multimeter is to know the correct scale for the measurement you want of. Below are some examples of quantities with their names on the scales:

- **Continuous voltage** : VCC, DCV, VDC or V with a dashed line and another continuous line over it.
- **AC voltage**: VCA, ACV, VAC or V together with the symbol - on it.
- **Direct current**: DCA, ADC or A with one line dashed and another continuous on it.
- **Alternating current**: ACA or A together with the symbol - about it.
- **Resistance**: Ohms, Ω .

Cautions when using a Multimeter

The use of a multimeter must be done with sufficient attention and never touch your fingers with any test leads during measurement to avoid an incorrect reading, or depending on what is measuring, you risk an electric coke. ever read the multimeter's manual before using.

Most digital multimeters have four terminals for connecting the test leads. Usually one is in common (COM) and the others are for each specific measurement (voltage, resistance and current). Terminals always indicate which scales they are must be used.

See the following example of how they have a multimeter with auto range function.



1 Terminal indicated with **10A MAX** - We must connect the **red** test tip for measuring current

continuous or alternating pressure greater than 200mA.

Note: Some digital multimeters do not have the option to measure alternating current. Before do the test make sure the multimeter has this option.

2 Terminal indicated with **μA / mA** - We must connect the **red** test tip for measuring

amperage (in micro or milli amps). At the maximum of 200mA.

3 Common terminal, **COM** - We must connect always the **black** test tip.

4 Terminal indicated with **V / Ohms / F / Hz** - We must connect the **red** test lead to measure voltage (continuous or alternating), frequency, diode, resistance, continuity, capacitance and hFE.

Symbology

Symbol	Unity
A	ampere (current unit)
V	volt (voltage unit)
W	watt (power unit)
Ohm	ohm (unit of resistance)
H	henry (inductance unit)
F	farad (capacitance unit)
Hz	hertz (frequency unit)

Table with symbols and quantities of electricity:

Greatness

Symbol	fraction, multiple
p	peak (1 trillionth) = 10^{-12}
n	nano (1 billionth) = 10^{-9}
μ	micro (1 millionth) = 10^{-6}
m	milli (1 thousandth) = 10^{-3}
k	kilo (1 thousand) = 10^{+3}
M	mega (1million) = 10^{+6}
G	Giga (1 billion) = 10^{+9}
T	(TB) = (1 trillion) = 10^{+12}

Introduction to electronic injection

Thanks to the electronic fuel injection system, manufacturers have managed to greatly improve performance engine efficiency, fuel efficiency and engine comfort driver.

The electronic injection system uses several sensors and actuators to provide the engine with the correct mix of air and fuel for specific operating conditions.

All of these sensors and actuators are monitored and controlled through a management module electronic.

So it's very important to know how the “brain” of the system works to have from an accurate diagnosis until you do your own repair.

Main components NTC

If it is a protection device, it is usually connected between the power input (positive and negative). In case of an overload is due to a short circuit or because of a reversal of the polarity of the battery, this component heats up and closes a short circuit between positive and negative inputs, breaking the trails of board at the module entrance, protecting the components more sensitive.



Diode

In the case of Rectifier Diodes, when a over voltage, that is, the “voltage” is exceeded, the component conducts the current in the direction of blocking the diode, producing the same effect as the previous example, the disruption of entry trails.

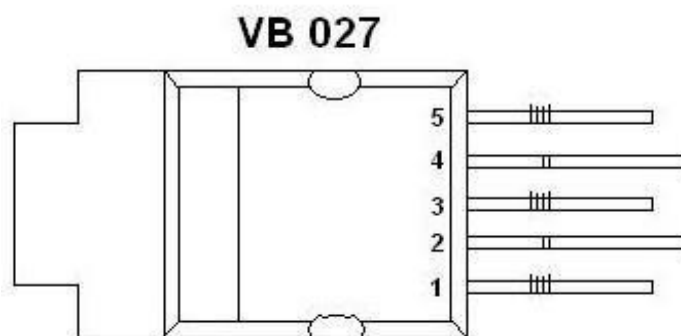
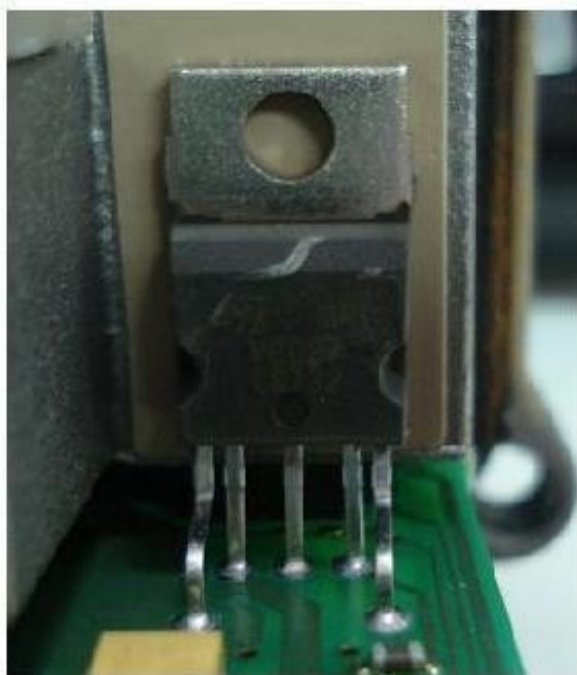
A diode has the characteristic of allowing the passage of electrical current in one direction and block in the other, to avoid a short circuit.



in injection plants the rupture voltage of the zener is 18.54 volts

Voltage regulator

The electronic circuit of the exchanges normally works with 5 volts, therefore, the regulator is one of the components responsible for the proper functioning of the modules and injection system.



Electrolytic capacitor

The capacitor is used to filter the input voltage before and after the voltage regulator. It consists internally by two dielectrics and an acidic substance, over time, this substance may leak acid causing short circuits and plate corrosion. These types of capacitors are used in filtering power supplies, low-voltage oscillating circuits



frequency, low frequency signal coupling and timer circuits.

There are two types of electronic capacitors: Those that have the metallic body (the most common) and those with the body in epoxy, similar to diodes. Some show the characteristics indicated by a letter (working voltage) and a number (value in MF): ex: 22/16.

To test electrolytic capacitors it is necessary to

firstly, to know its value in Microfarade to be able select the correct measurement scale.

Scale values in microfarade

Scale

Microfarade (μF)

X1 or X10-----From 330 to 10,000

X1K_____From 0.05 to 220

Also note that the electrolytic capacitor has polarity (+ and -). It is also found in the body of the capacitor the working voltage value.

Drive

They are the components responsible for triggering the injector nozzles, ignition coil, and solenoids through of a low power electrical signal.

To access internal functions and features, check in the Datasheet.

A coil drive normally short-circuits when the coil is defective.

The drive consists of a power transistor that can be mounted individually or in Integrated Circuit with several internal components.

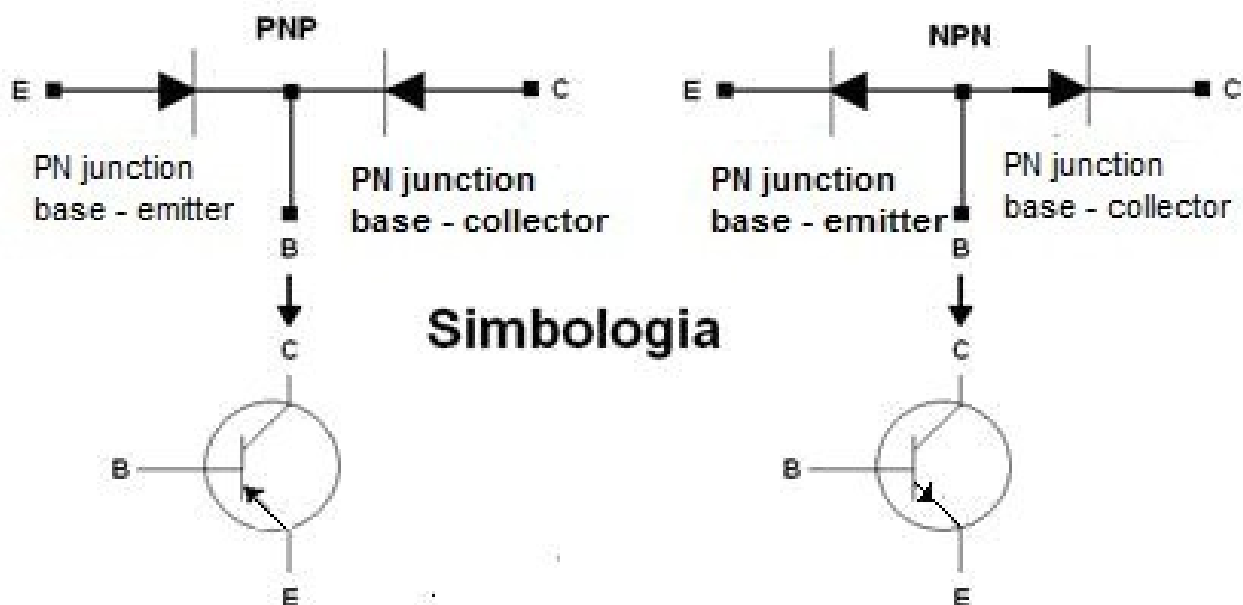


Transistor

Low power transistors are used for the switching of drives and for signal communication processed digital.

The function of the transistor is to switch electrical circuits from high power through a low power.



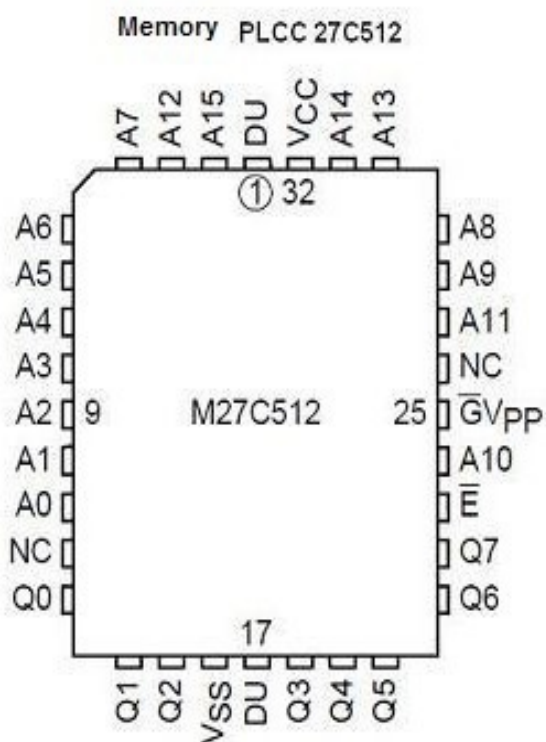


It works similar to a relay, but much more effectively.

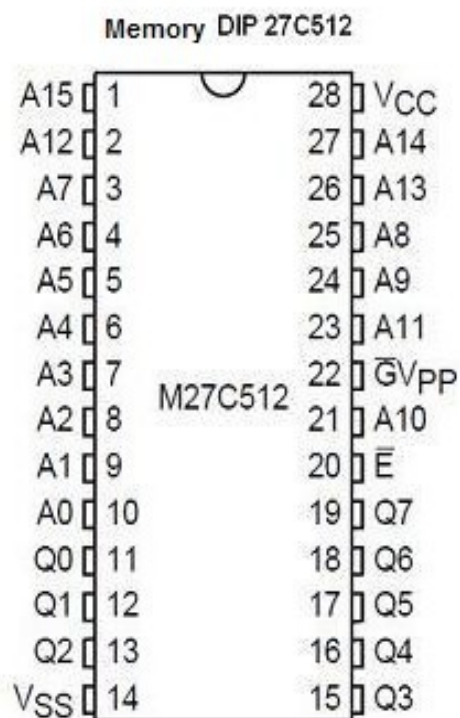
Among other functions, it is also used mainly as an output drive to drive injector nozzles, coils etc.

Eprom - ROM memory

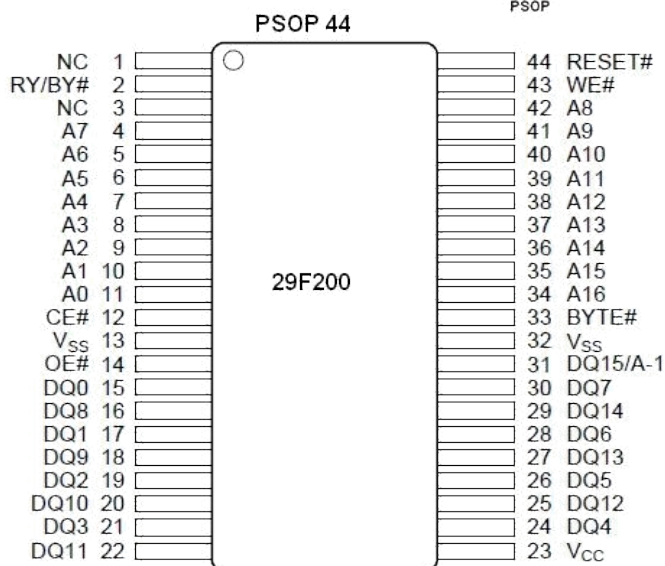
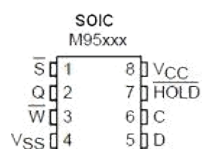
It is found in models of type DIP, PLCC, PSOP or integrated internally in the processor in Hybrid ECUs.



PLCC



DIP



They carry the parameter information to the engine operation: advance, injection time, temperature etc.

Programming can be done only by changing the chip or with the use of special equipment.

RAM memory

It is the volatile memory, where the auto parameters are injection system adaptations.

PAL decoder

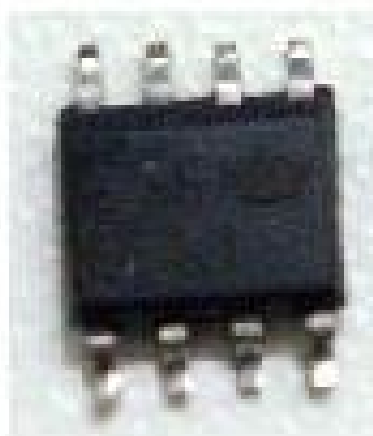
Transforms analog signals into digital signals and organizes all this information for the processor.

It is possible to find modules with a single component, or with several, with unique functions (rotation, detonation, etc.) which is the case with some Bosch power stations.

EEprom - 95160

Serial Eeprom with SOIC8 package are information such as immobilizer code is stored, engine, airbag, body module etc.

In some cases they can be reprogrammed via scanner or with the help of programmers.

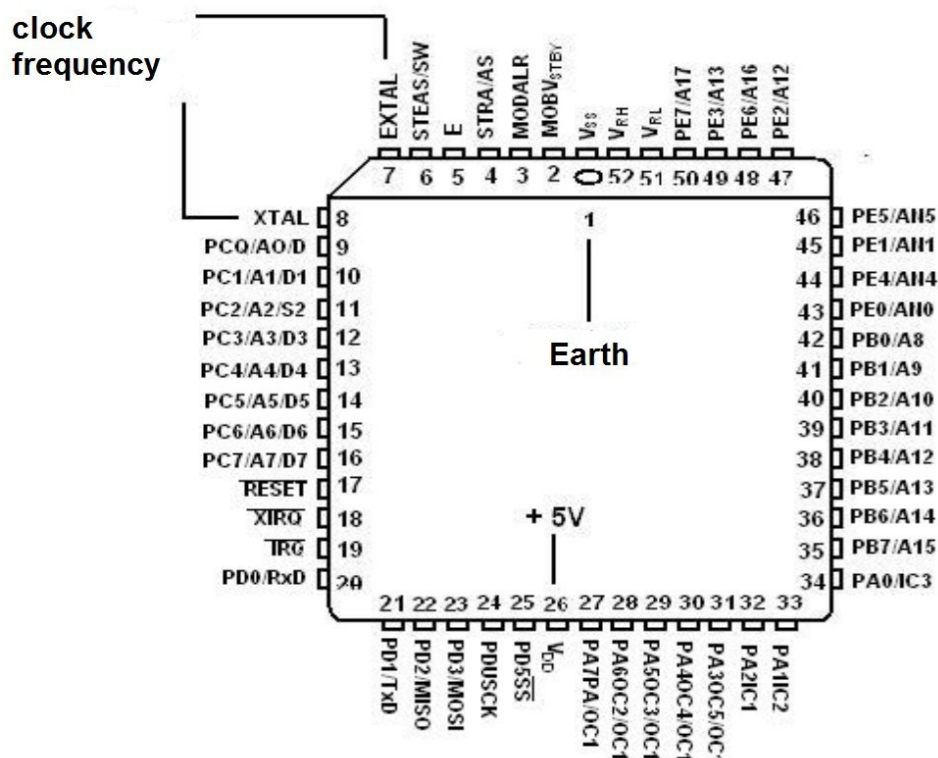


SOIC

Processor

As with a conventional computer, the processor is the injection module heart.

Contains internal software with working parameters of the system.



Crystal or Clock

In every crystal, the material it is made of has a unique characteristic of having a natural frequency. The processor uses this frequency to determine the time in which all the routines will be performed job.



Therefore, it is observed that the more modern the central the higher the clock frequency (crystal) of the processor.

Common defects (cause and effect)

Cause

Several factors can cause poor contact in welds terminals, components or broken trails.

For example:

- Time action;
- Oxidation;
- External short circuit;
- Over load, due to lack of grounding of the external components.

It is made

- Some ground or power will be missing sensors;
- Actuators will not work;
- Central does not work.

Solution

- Check the cause of the defect;
- Redo the weld or rebuild the track with wire cloaked.

Electrical Resistance

Resistors are electronic components that have the function hinders the passage of electric current through the your stuff. This difficulty is called **resistance electrical**.

Resistor Testing

Unit.....Ohm Ω

Kilo Ohm $k\Omega = 10^3$

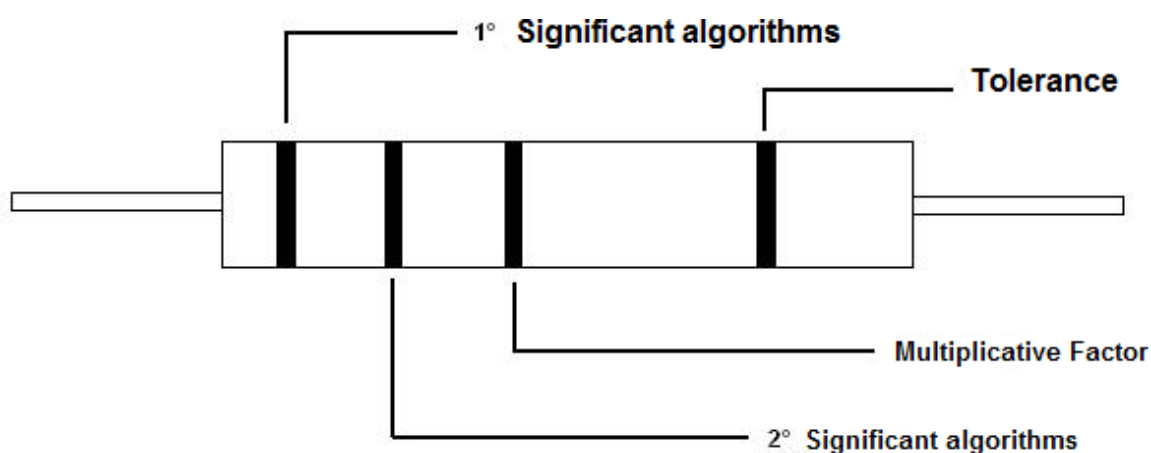
Mega OHM $M\Omega = 10^6$

There are two types of resistors, fixed and variable:

Fixed: They are resistors in which their electrical resistance does not can be changed (features two terminals.)

Variables: Are those resistors in which their resistance electrical can through an axis or stroke (rheostat, Potentiometer).

A color code is used to identify the value of a resistor, where each color and its position in the body of the resistors represent a value or a multiplicative factor.



Resistance calculation table for one resistor:

color	1°	2°	multiplicative factor	tolerance
black	0	0	x1	—
Brown	1	1	x10	1%
red	2	2	x100	2%
orange	3	3	x1.000	--
yellow	4	4	x10.000	--
green	5	5	x100.000	--
blue	6	6	x1.000.000	--
Violet	7	7	---	--
Grey	8	8	---	--
White	9	9	---	--
golden	-	-	x0,1	5%
silver	-	-	x0,01	10%
without color	-	-	---	20%

SMD resistor

SMD resistors are at least 1/3 the size of conventional resistors. Most are welded on the side of under the board by the side of the tracks, occupying much less space.

They are resistors to be mounted on the surface (SM in English *Surface Mounting*). Your reference has a encoding which can be three or four digits. The most used in electronic equipment are the resistors SMD with three.



In resistors with 3 characters, the first two numbers indicate the most important digits of the resistor, the last digit indicates the multiplication factor, in this case 100, so in the case of the resistor in the example above one has a resistance of 1000 ohms.

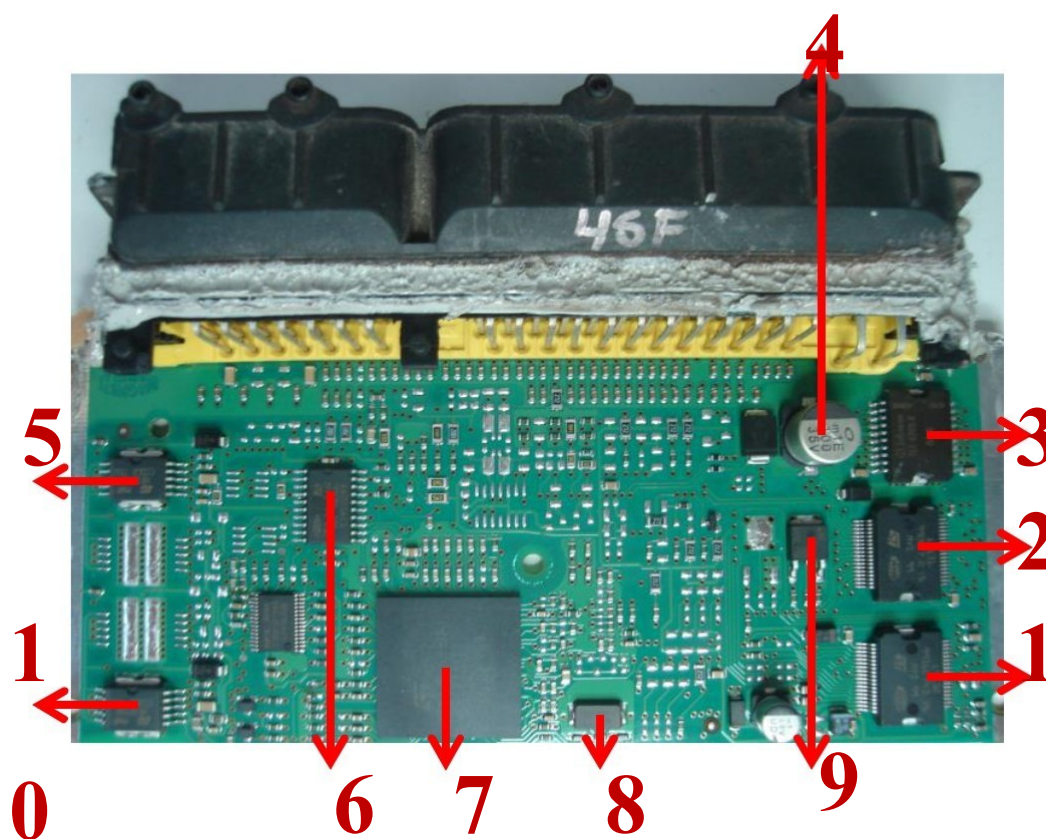
Resistor Table

	digit 1 = 1 digit 2 = 2 digit 3 = multiplier	12*100: 1200 ohms = 1K2
	digit 1 = 1 digit 2 = point digit 3 = 6	1.6 ohms
	digit 1 = point digit 2 = 2 digit 3 = 2	0.22 ohms

Electronic injection modules: Features, more defects tips and repair tips

Magneti Marelli 4SF - Fiat

Vehicles: Fiat with 1.0, 1.4 and 1.8 engines, with electronic accelerator.



Main components:

- 1** 5V regulator;
- 2** Nozzle / Fan / Main relay;
- 3** Electronic accelerator;
- 4** Capacitor;
- 5** Coil 1 and 4;
- 6** Knock sensor;
- 7** Processor - ST -10;
- 8** Crystal - 10 MHz;
- 9** Lambda probe;
- 10** coil 2 and 3.

Technical specifications

Special features

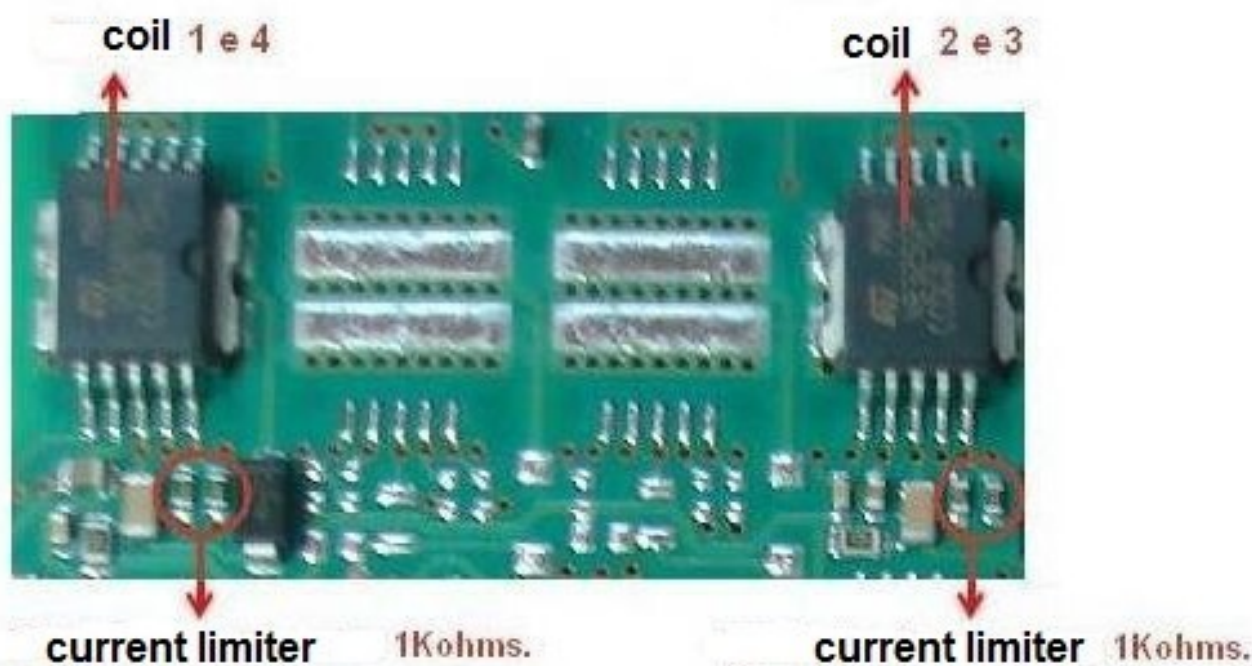
- The letter " S " after 4 means that the system has electronic accelerator;
- It is considered a hybrid plant, because it has the injection and immobilizer maps on the processor.
- Injection and immobilizer maps can only be accessed with ST10 programmer, and through Boot.

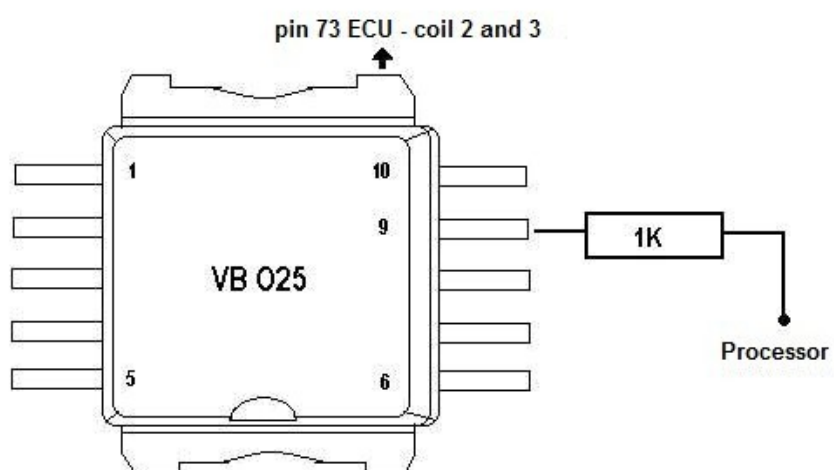
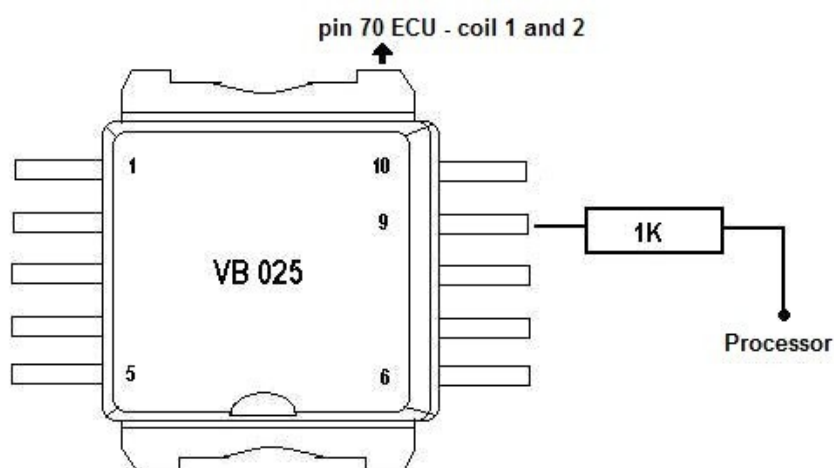
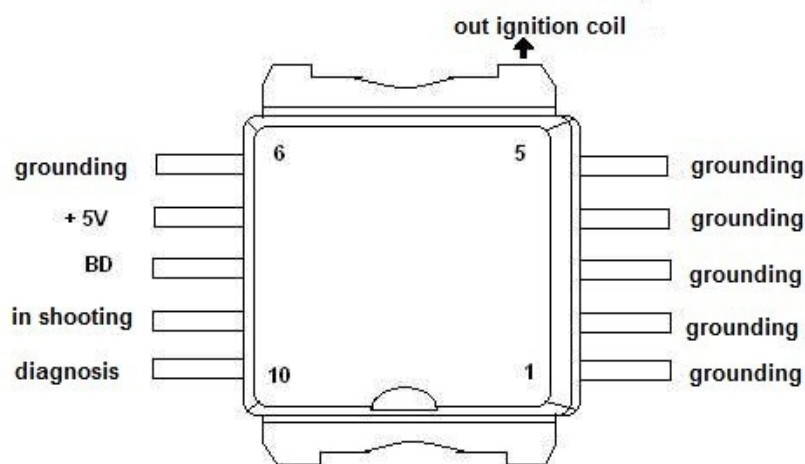
Common defects

- Failure in the DC motor drive, causing failure in the acceleration.
- Processor software failure. Many power plants may malfunction due to defect in the processor software (oscillation of idling, acceleration failure, etc.).
exchange the software for another (tele-loading).

Ignition drive - VB 025 MSP

Between point 9 of the component and the processor, there is a current limiting resistor of 1Kohms.





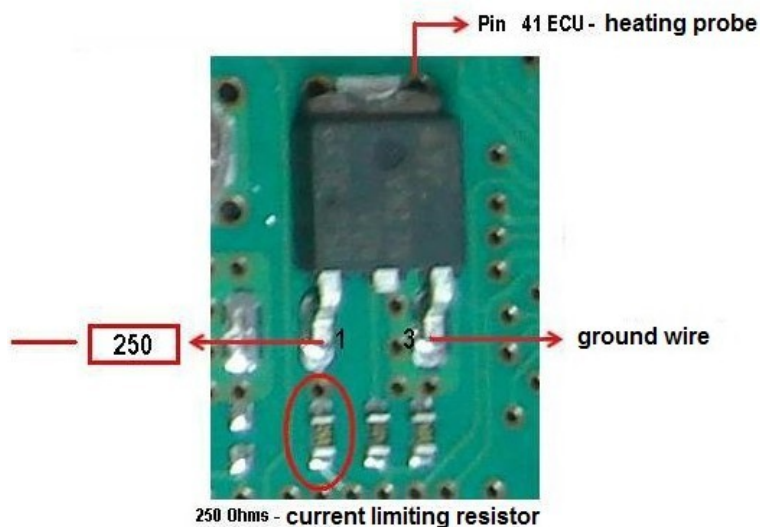
Lambda probe heating control - VDN 7N V04

This component controls the negative, by PWM, varying the

applied voltage.

Mult Drive L9131 - 36 Pins

Controls 10 components of this ECU.



Mult Drive L9131 - 36 Pins

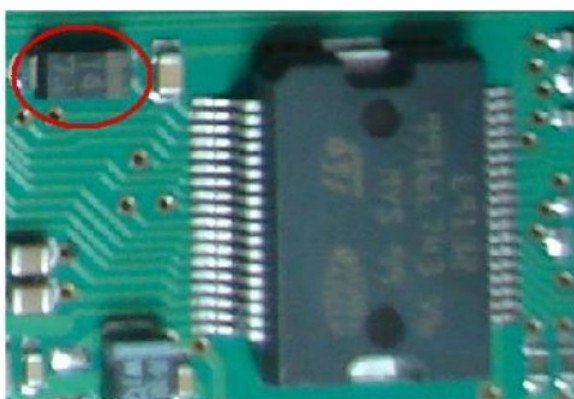
Controls 10 components of this ECU.

Drive L9132 - 36 Pins



Component Pin	Function
2	out - injection light - pin 52 Ecu
3	out - speed 2 fan - pin 32 Ecu
4	out - relay A / C - 17 ECU
12	out - speed 1 fan - pin 42 Ecu
21	out - injector 1- pin 79 Ecu
24	out - injector 2 - pin 77 Ecu
26	out - injector 3 - pin 80 Ecu
28	out - injector 4 - pin 78 Ecu
30	out - Canister - pin 43 Ecu
32	out - Cold Party System - pin 59 Ecu

Drive L9132 – 36 Pinos

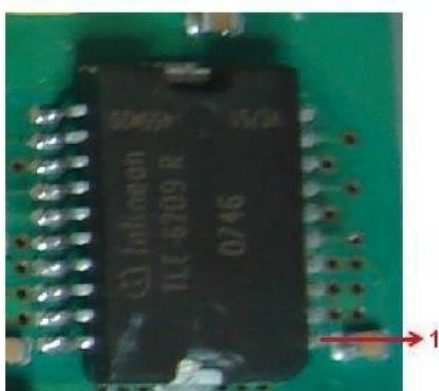


Component pin	function
2	out - main relay - Pin 5 Ecu
15	out - + 5v - TPS - Pin 10 and Accelerator Pedal - Pin 57 Ecu
16	out - + 5v - MAP - Pin 64 ECU

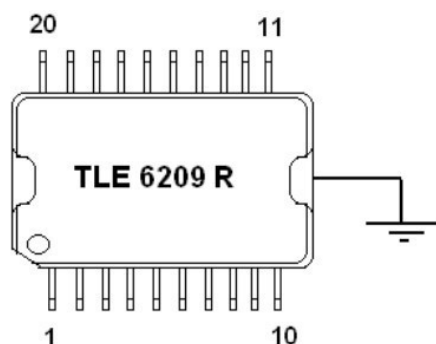
Butterfly motor control drive - TLE

6209 R

Controls the opening and closing of the throttle acceleration, by PWM.



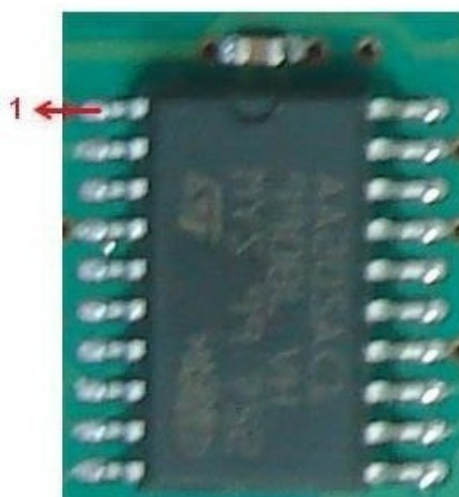
Pins 1 - 10 - 11 - 20 (Ground) are connected to the metal at the bottom of the component, where it should be soldered to the circuit board. This ensures that the component don't overheat.



Component Pin	Function
1	Grounding
2	Out 1 - Pin 53 ECU
3	Out 1 - Pin 53 ECU
4	+ 12 Volts
10	Grounding
11	Grounding
15	+ 5 Volts (Supply of logic circuit)
17	+ 12 Volts (DC motor control circuit supply)
18	Out 2 - Pin 67 ECU
19	Out 2 - Pin 67 ECU
20	Grounding

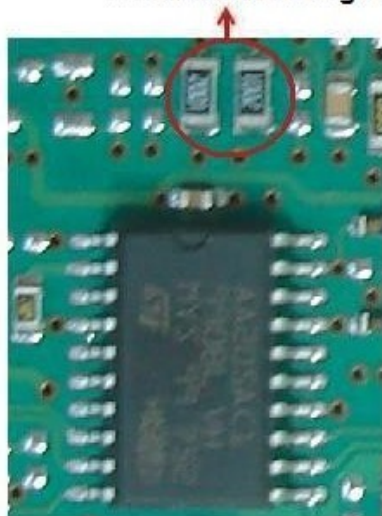
Detonation interface - AA205AC1 - H992N0430 - ST -
20 Pins

The detonation circuit has no capacitors at the input,
only 2Kohms series resistors.



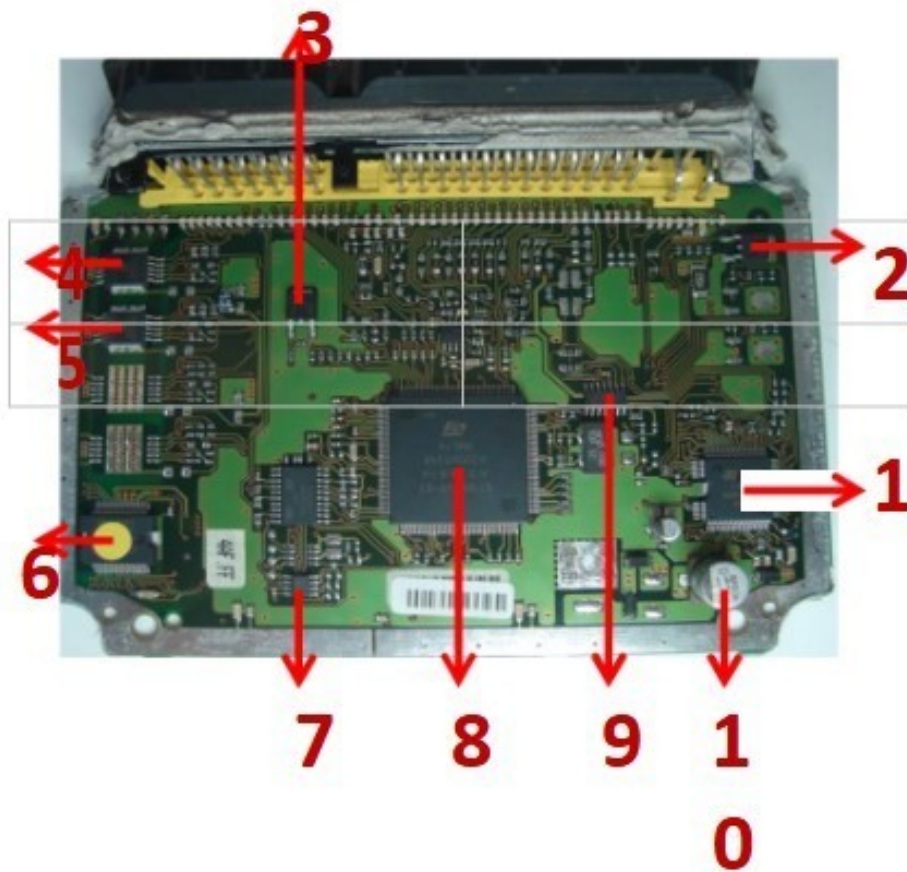
Pin	Component	Function
6	In - detonation sensor	- Pin 75 ECU
16	In - detonation sensor	- Pin 61 ECU

2K - current limiting resistor



Magneti Marelli 4AFB - Fiat

Vehicles: Palio 1.0 8V Flex.



Main components

- 1 Fan 1 and 2 / Main relay;
- 2 Lambda probe;
- 3 Cold start;
- 4 Coil 2 and 4;
- 5 Coil 1 and 4;
- 6 Nozzles / Stepper motor / Canister;
- 7 Soic 95160 / Code;
- 8 Processor - ST10;
- 9 Can network; (Note: Cars with BC).
- 10 Capacitor.

Technical specifications

Special features

- There is 4AFB with panel, box or Body Computer.

4AFB - Gasoline is not interchangeable with flex.

Common defects

- Oxidation on the plate, electro fan and cold start.

Note: AFR regulation is used, 9

for alcohol and 13 for gasoline. The cold start would only function if the AFR is below 10.5 and the temperature water and air are below 16 ° C.

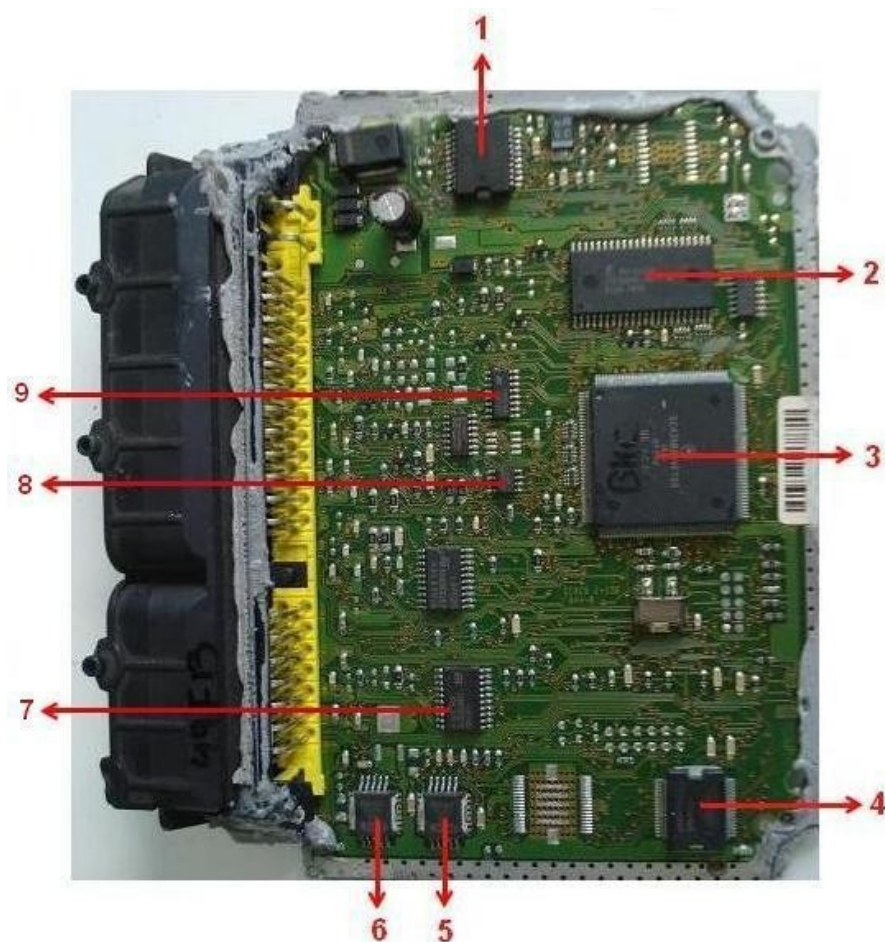
Immobilizer, use the decode.

Processor: ST10.

RPM signal: Phonic.

Coil signal: Digital.

Magneti Marelli 49FB - Fiat



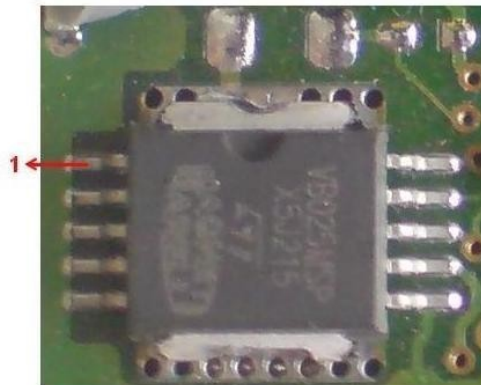
Main components

- 1** Multi functions - L9104 - 5V regulator - Relay main - Relay A / C;
- 2** Memory Rom - 29F200 AB;
- 3** Processor - Motorola - ZC 439507 MF T 20;
- 4.**MAR 9109 PD Multi Drive - 20 pins - Injectors - Canister - Stepper motor - V1 and V2 fan;
- 5** Drive ignition coil 1 and 4 - VB025;
- 6** Ignition coil drive 2 and 3 - VB025;
- 7** Detonation interface - L9119;
- 8** Soic 8 - 95040 - Immobilizer;
- 9** Blasting interface – AA205AC.

Common defects

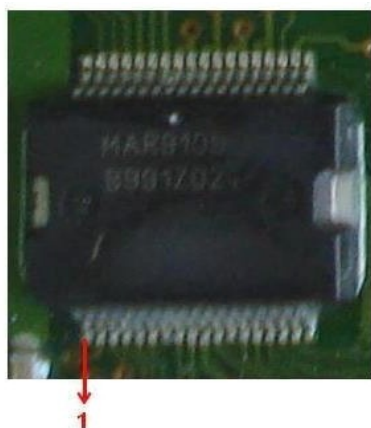
- Oxidation of the board causing fan failure or burning the drive that controls the fan;
- Burn the drive that controls the nozzle outlet.

Ignition coil drive – VB025



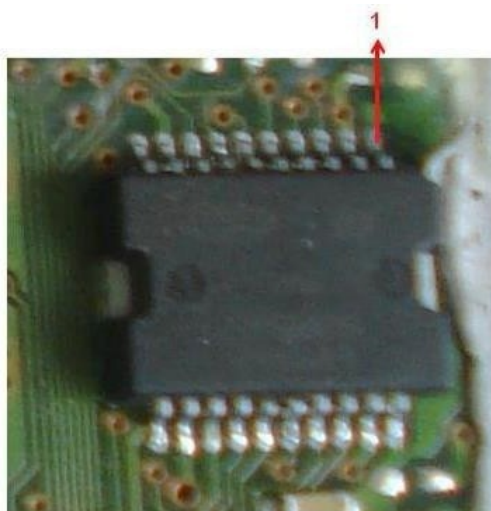
Pin	Component	Function
1/6		wire grounding
7		+ 5 Volts
8		BD
9		in firing
10		Diagnose

MAR 9109 PD multi drive - 20 pins - Injectors -
Canister - Stepper motor - Fan V1 and V2.



Pin	Component	Function
2		Out - Stepper Motor - Coil 1. Pin 65 ECU
9		out - stepper motor - bonina 1. pin 58 ecu
11		out - stepper motor - coil 2. pin 57 ecu
17		out - stepper motor - coil 2. pin 64 ecu
22/23		out - injector 1. pin 71 ecu
24/25		out - injector 3. pin 78 ecu
26/27		out - injector 4 .pine 72 ecu
28/29		out - injector 2. pin 79 ecu
30		out - canister - pin 52 ecu
31		out - V1 fan - pin 40 ecu
32		out - V2 fan - pin 14 ecu

Multi functions - L9104 - Regulator 5V- Main relay - Relay BC



Pin	Component	Function
2		Out - Main relay - Pin 15 ECU
15		Out - + 5V sensor MAP - Pin 68 ECU
16		Out - + 5V sensor TPS - Pin 60 ECU
19		Out -Relay A/C

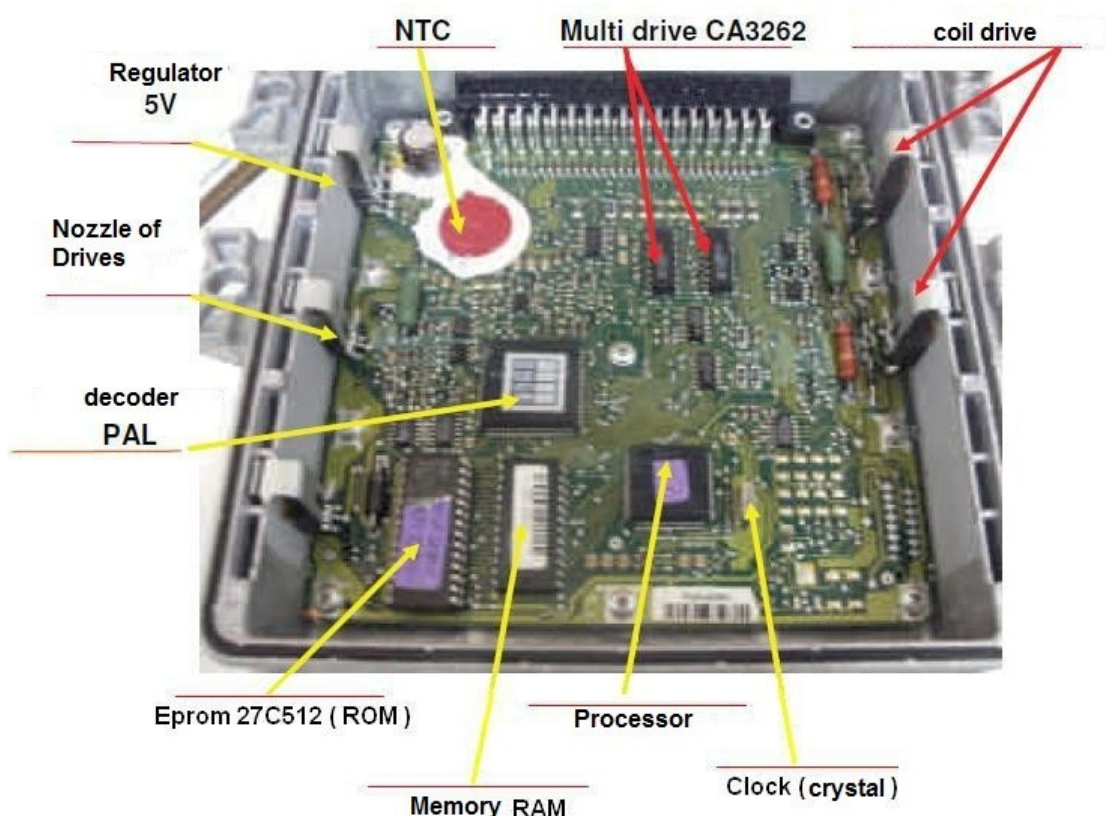
Detonating magneti – L9119

Series Resistor with pin 6



Pin	Component	Function
6		In - detonation signal - Pin 56 ECU

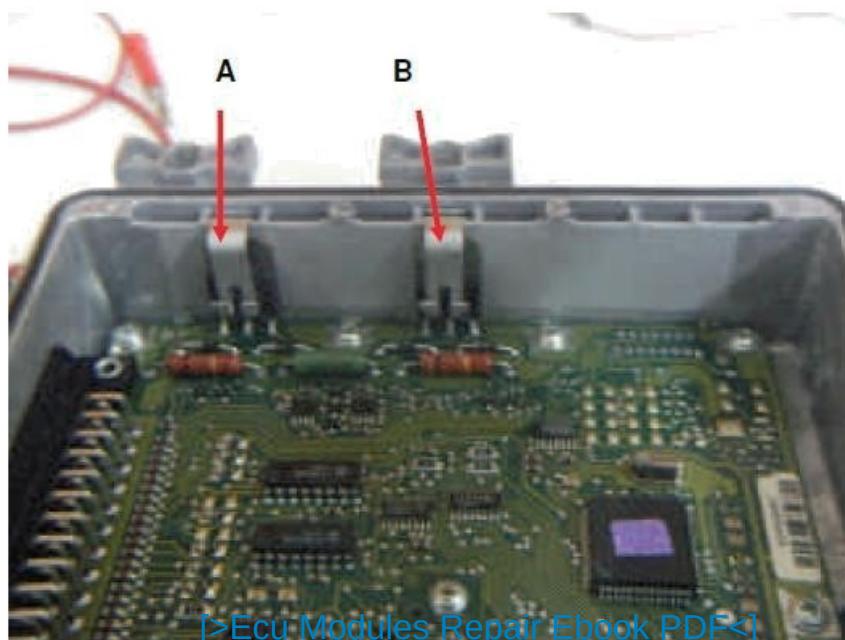
Magneti Marelli G7.11 – Fiat



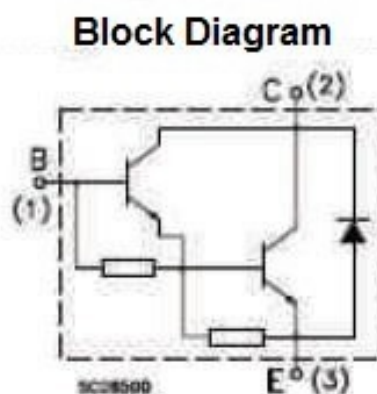
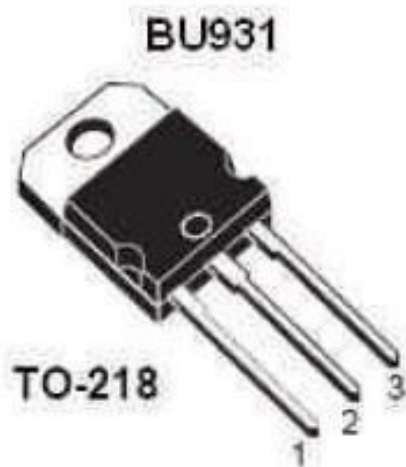
List of components

- Processor - Motorola MC68HCP11A1 - 52 pins;
- RAM memory - MHS MM3 65162S0129;
- PAL-ST B22AF627 decoder;
- BU 931 and BU 941 - Ignition Coils;
- L4947R - Voltage regulator 5 volts;
- CA3262 AE - Quadruple mult drive.

Ignition coil drive - BU 931 ou BU 941



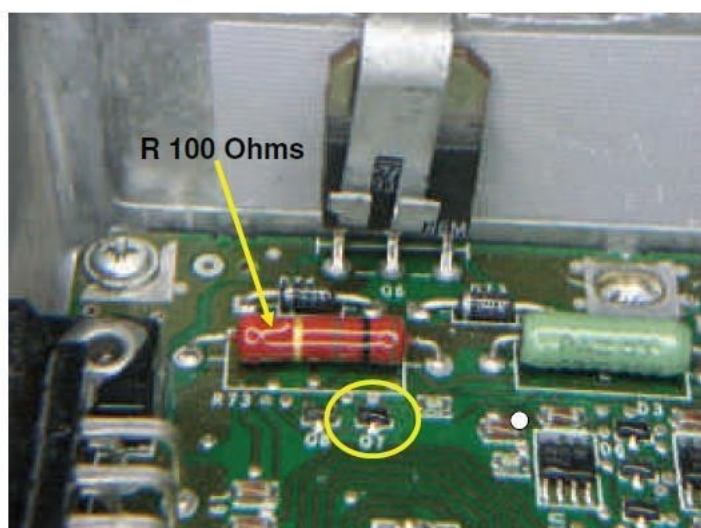
BU 931 or BU 941 - Ignition Coils



Coil A - Cylinder 1 and 4

Pin	Component	Function
Base (B)		Firing – Comes from the low power transistor Q7
Collector (C)		Pin 1 From ECU
Emitter (E)		wire grounding

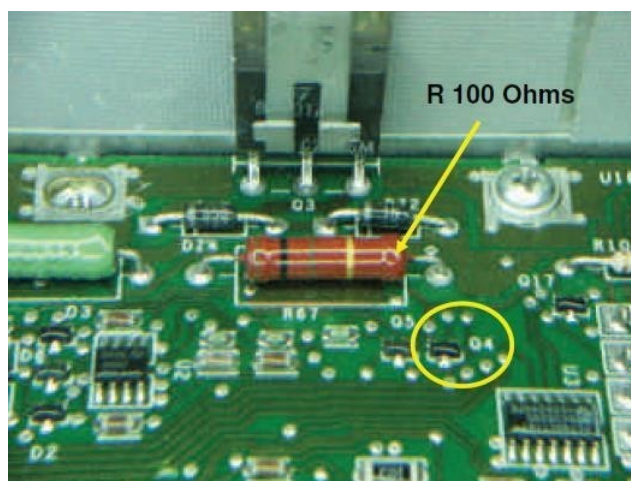
The coil A drive is driven by the PNP transistor of low power shown below, among them is the transistor current limiter of 100 Ohms.



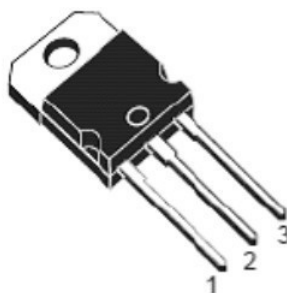
Coil B - Cylinder 2 and 3

Pin	Component	Function
	Base (B)	Firing - Comes from the low power Q4 transistor
	Collector (C)	Pin 19 from ECU
	Emitter (E)	wire grounding

The drive of coil B is driven by the PNP transistor of low power shown below, among them is the resistor current limiter of 100 Ohms.

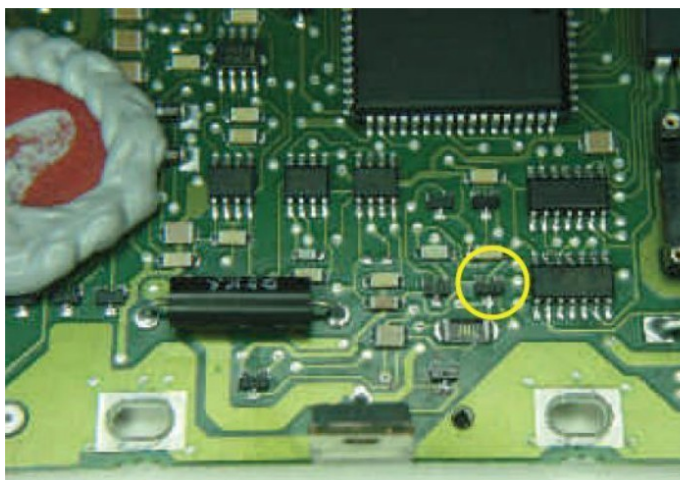


Drive Injetor – BDX 53

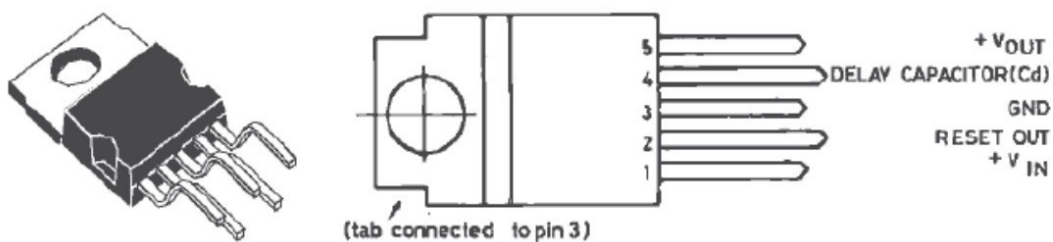


Pin	Component	Function
	1 (B)	Firing - Comes from the low power Q12 transistor
	2 (C)	Pin 18 from ECU
	3 (E)	wire grounding

The injector drive is driven by the low-voltage PNP transistor power shown below.



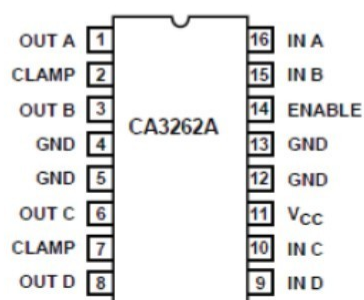
L 4947R – 5v voltage regulator



Pin Component	Function
1	In - + 12 V
3	Grounding
5	Out - + 5 V

CA 3262 AE - Quadruple mult drive

This component works similarly to NPN transistors, that is, the input (trip) is made by positive 5 volts and the output is negative.



Drive A - Diagnose - Solenoid Relay 2 and Solenoid Valve 1

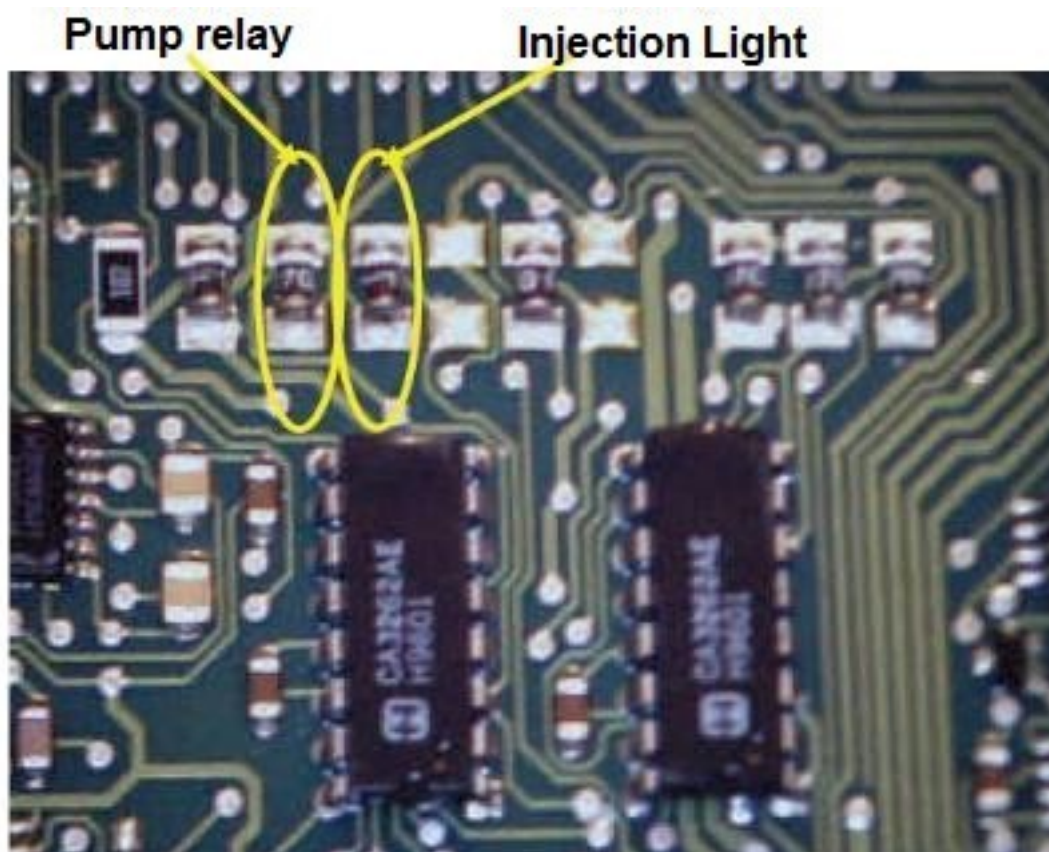
Pin	Component	Function
1 / 3		Out A - Electrovalve 1 - Pin 22 ECU
6		Out C - Diagnostic line - Pin 4 ECU
8		Out D - Electrovalve Relay 2 - pin 23 ECU

Drive B - Fuel pump relay - Injection light

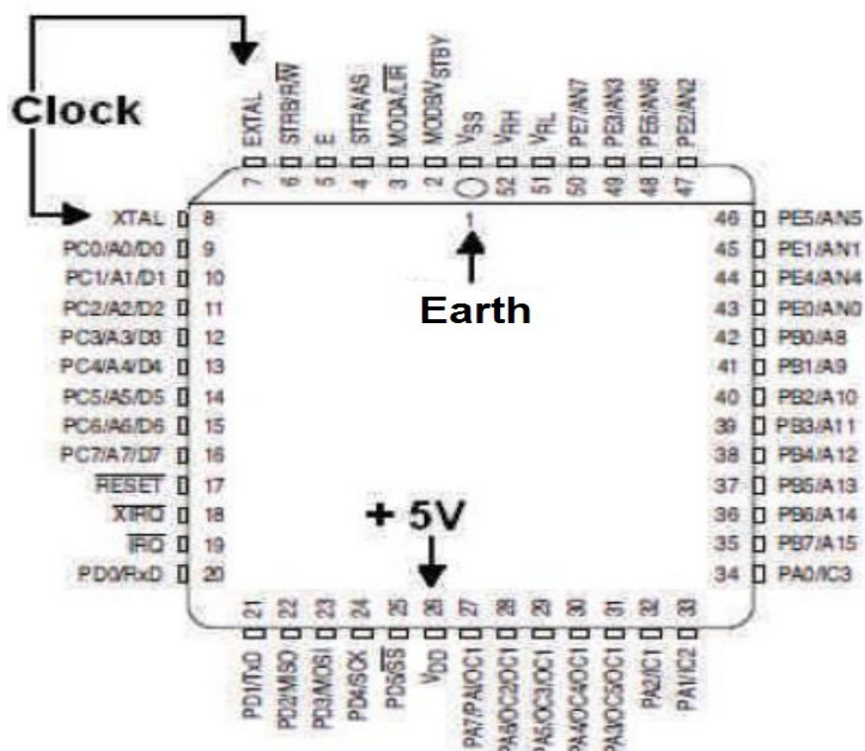
Pin	Component	Function
1		Out A - Rele the Fuel Pump - Pin 7 ECU
3		Out B - Injection Light - Pin 7 ECU
6		Out C - Pin 26 ECU

Note: The pump relay and the fault lamp

have a diode in series, as shown in the following image.

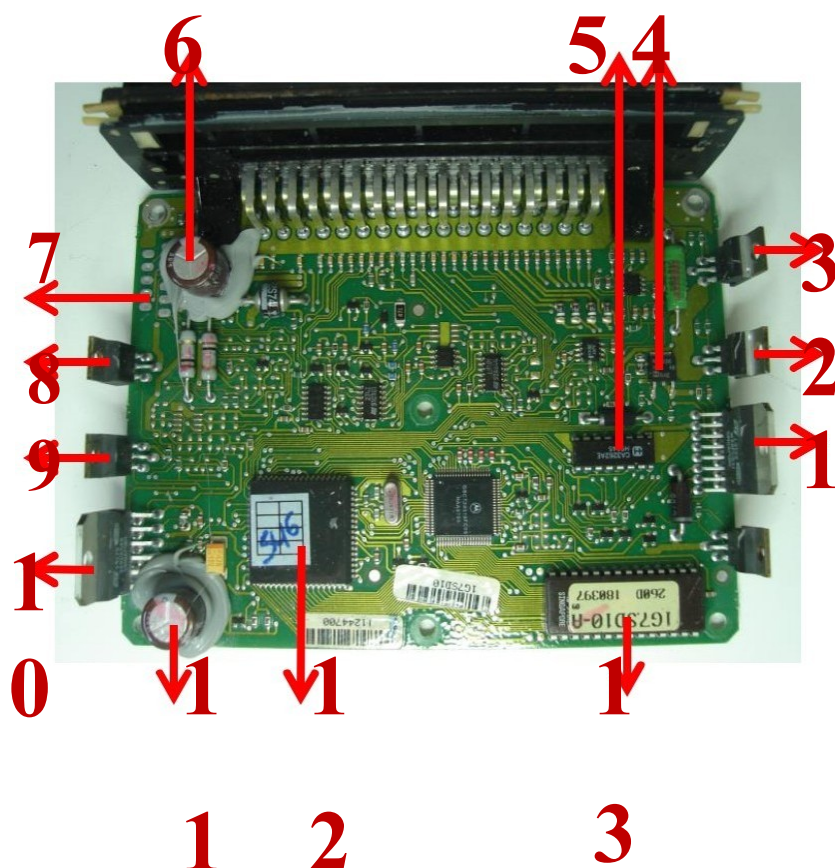


Processor - Motorola MC68HCP11A1



Magneti Marelli 1G7 - Fiat

Vehicles: Canopy, Fiorino and Uno, 1.0 and 1.5 mpi.



Main components

- 1 Stepper motor;
- 2 Coil 2;
- 3 Coil 1;
- 4 Canister;
- 5 Double relay;
- 6 Capacitor;
- 7 Nozzle drive (single point); Note: You will not have the drives 8 and 9.
- 8 Nozzle 2 and 3;
- 9 Nozzle 1 and 4;
- 10 5V regulator;
- 11 Capacitor;

12 Processor;

13 Eprom memory.

Technical specifications

Special features

- The immobilizer is on the processor.
- The immobilizer is optional.

Common defects

- The 96/97 system came out with a processor defect.

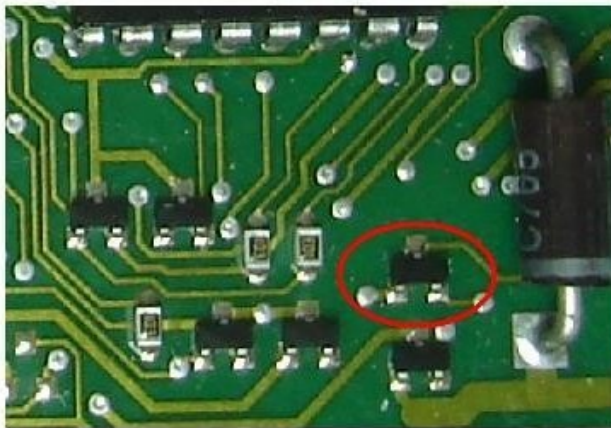
When successive failures occur, the software of the processor blocks the stepper motor, causing idle oscillation. This failure will only be repaired when the reset procedure is performed. processor, using the tool chip.

Note: Nozzle drive - Commercial - IRF 34.

Drive relay A

Activate side A of the double relay, by pin 4 of the ECU.

Caution: Between pin 4 of the ECU and the relay drive A has a diode.

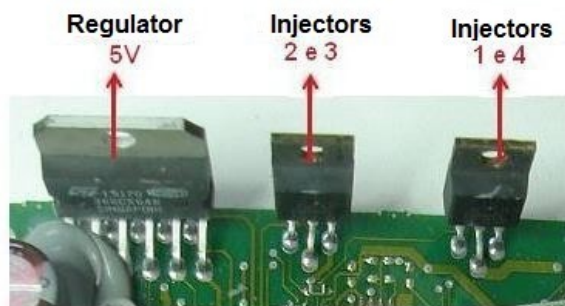


Micro transistor that drives the A relay drive

Drive Coils ignition

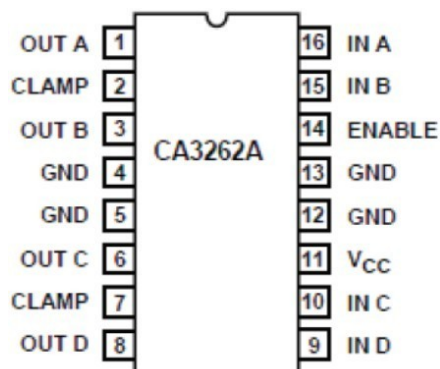


1 Kohms - current limiting



Drive CA 3262 AE - Relay A / C - Relay B - Injection light

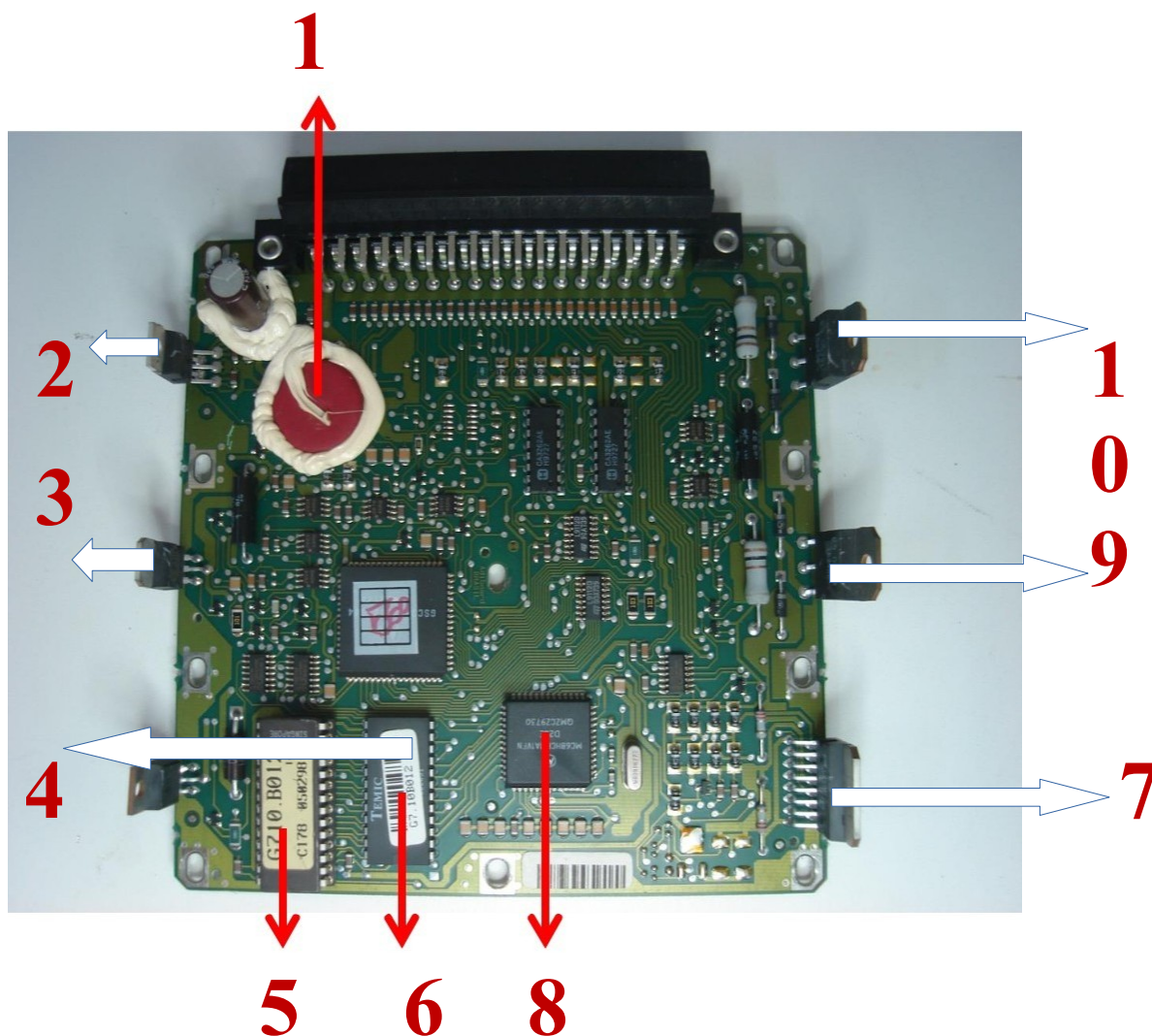
This component works similarly to **NPN** transistors, that is, the input (trip) is made by positive 5 volts and the output is negative.



Pin Component	Function
1	Not used
3	Out - Pin 24 ECU -relay of A/C
6	Out -relay B - Pin 23 ECU
8	Out - Pin 6 ECU - Injection Light

Magneti Marelli G7 - Fiat

Vehicles: Uno / Premium Families / Tempra 8V and 16V.



- 1 Main Components
- 2 Varistor;
- 3 5V regulator;
- 4 Nozzle drive;
- 5 Processor;
- 6 Eprom;
- 7 Rom memory;
- 2 Auxiliary processor - MC68HC11;

3 Coil drive 1;

4 Coil drive 2;

Technical specifications

Rotation Signal: Phonic 60-2

Transformation: Tempra 8V to 16V just change the file

One 1.0 gas. For 1.5 gas. just change the archive

OBS: 1 - When exchanging these exchanges pay attention if the car has air conditioning and sensor detonation.

2- It is not advisable to exchange gasoline for alcohol because the starting system is original cold will not work.

Eprom: 27C256 - Uno e Elba 1.5 and 1.6 from 92 to 94

27C512 - Family One and Tempra from 94 onwards

Common defects:

- Varistor firing: check for possible alternator or short circuit in the harness.

How to solve - Change the varistor.

- Disruption of the sensors' TERRA track (follow pin 31 of the nozzle).

How to solve - resold the track.

- Memory eprom or Ram accused by the scanner.

How to solve it - Change the auxiliary processor (photo 8)

- Electrolytic capacitor bursts and corrodes the tracks below his.

How to solve- Change the capacitor and rebuild the tracks.

Ignition coil drive - BU 931 or BU 941



Coil A - Cylinder 1 e 4

Pin	Component	Function
Base (B)		Firing - Comes from the low power Q7 transistor
Collector (C)		Pin 1 da ECU
Emitter (E)		wire grounding

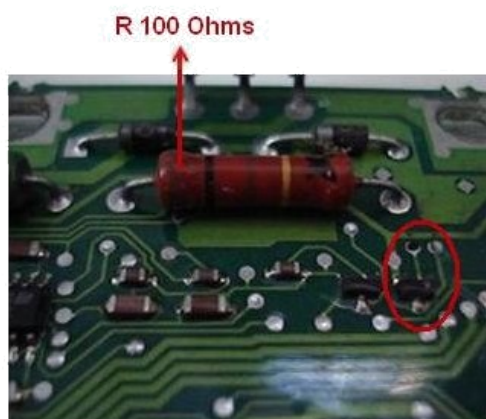
Coil B - Cylinder 2 e 3

Pin	Component	Function
Base (B)		Firing - Comes from the low power Q4 transistor
Collector (C)		Pin 19 ECU
Emitter (E)		wire grounding

The coil A drive is driven by the PNP transistor of low power shown below, among them is the resistor current limiter of **100 Ohms**.



The coil B drive is driven by the low-voltage PNP resistor power shown below, among them is the resistor current limiter of **100 Ohms**.

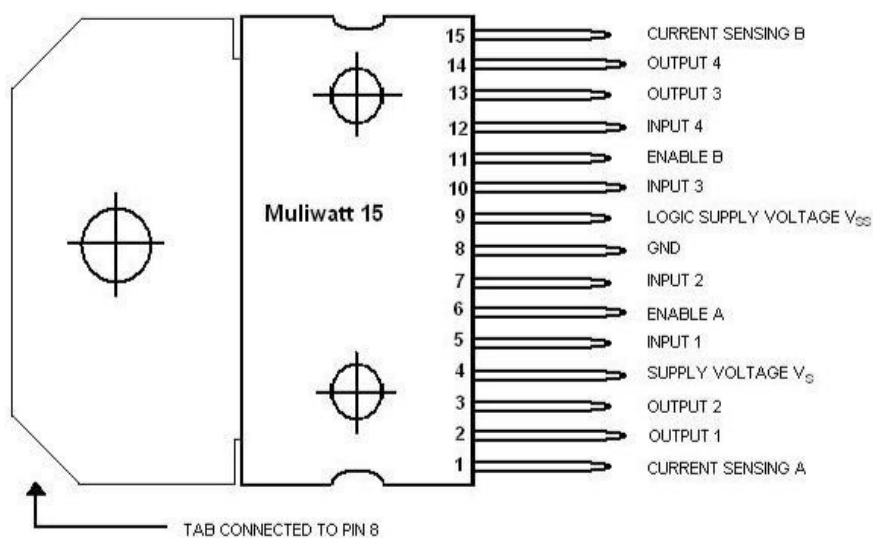


Injector drive (s)

Pin	Component	Function
1 (B)		Firing - Comes from the low power Q12 transistor
2 (C)		Pin 18 ECU
3 (E)		wire grounding

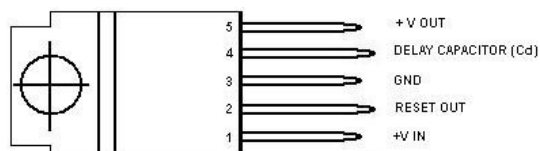
The injector drive is driven by the low-voltage PNP transistor power shown below.

L 298 N - Stepper Motor



Pin	Component	Function
1		Monitors the current of Coil A
2 - 3		Controls Coil A (pin 2 and 20 of Central)
4		+ 12V
5 - 7		Coil A Command Input A
6 - 11		Enables Coil A or B Drive
8		wire grounding
9		loading Component
10 - 12		Coil B Command Input
13 - 14		Controls Coil B (pin 21 e 3 of central)
15		Monitors the Coil Current B

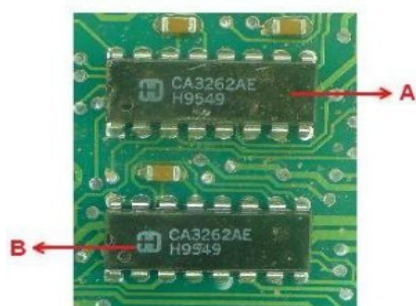
L 4947 R - 5V voltage regulator



Pin	Component	Function
1		In - + 12V
3		wire grounding
5		Out - + 5V

CA3262 AE - Quadruple Multi Drive

This component works similarly to NPN transistors, that is, the input (trip) is made by positive **5 volts** and the output is negative.



Drive A - Canister - Diagnose - Consumption Meter

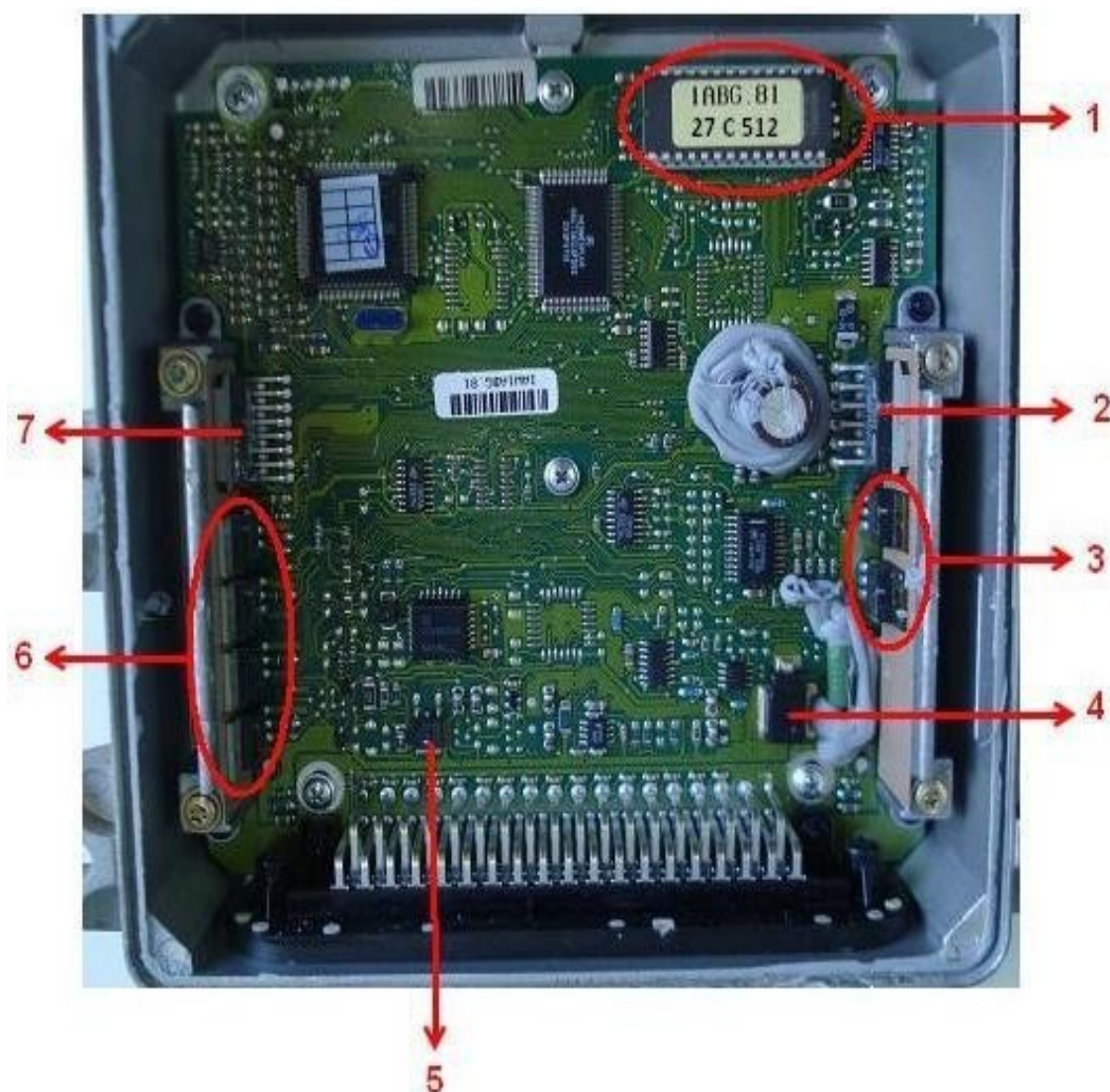
Pin	Component	Function
1		Out A - Canister or Cold Start - Pin 22 ECU
3		Out B - Canister or Cold Start - Pin 22 ECU
6		Out C - Diagnostic line - Pin 4 ECU
8		Out D - Consumption Meter - Pin 23 ECU

Drive B - Fuel Pump Relay Injection Light - Tachometer

Pin	Component	Function
1		Out A - Fuel Pump Relay Injector and Lambda probe - Pin 25 ECU
3		Out B - Injection Light - Pin da ECU
6		Out C - Pin 26 ECU
8		Out D -Tachometer- Pin 6 ECU

Magneti Marelli 1ABG - Fiat

Vehicles: Palio 1.6 16V



Main components

- 1 DIP memory;
- 2 Regulator 5 volts;
- 3 Coil 1 and 2;
- 4 Protection diode;
- 5 Canister;
- 6 Nozzles;
- 7 Idle actuator.

Technical specifications

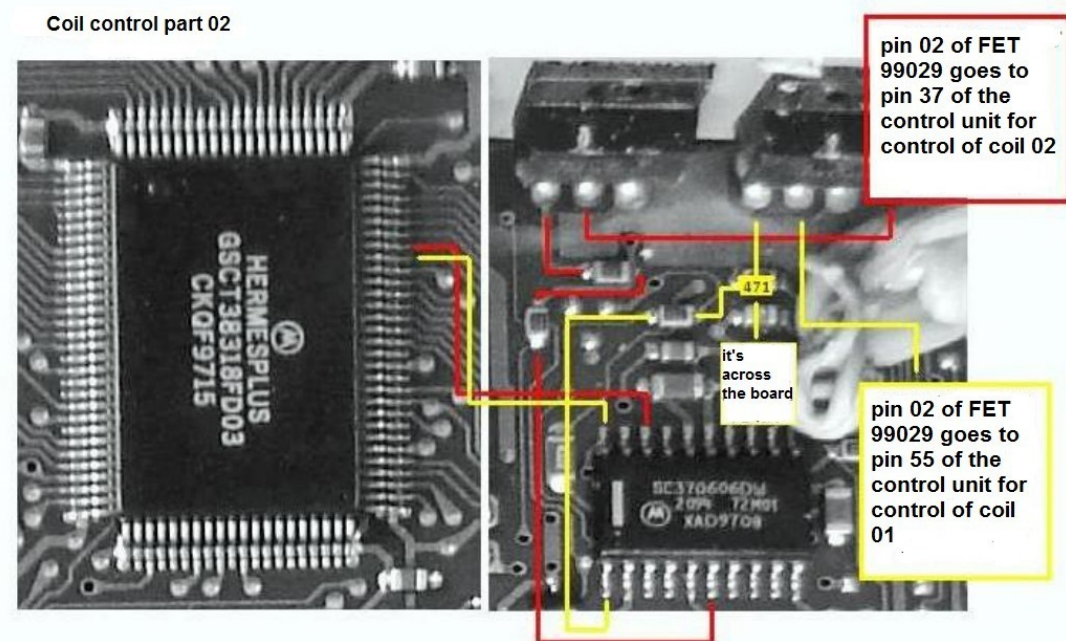
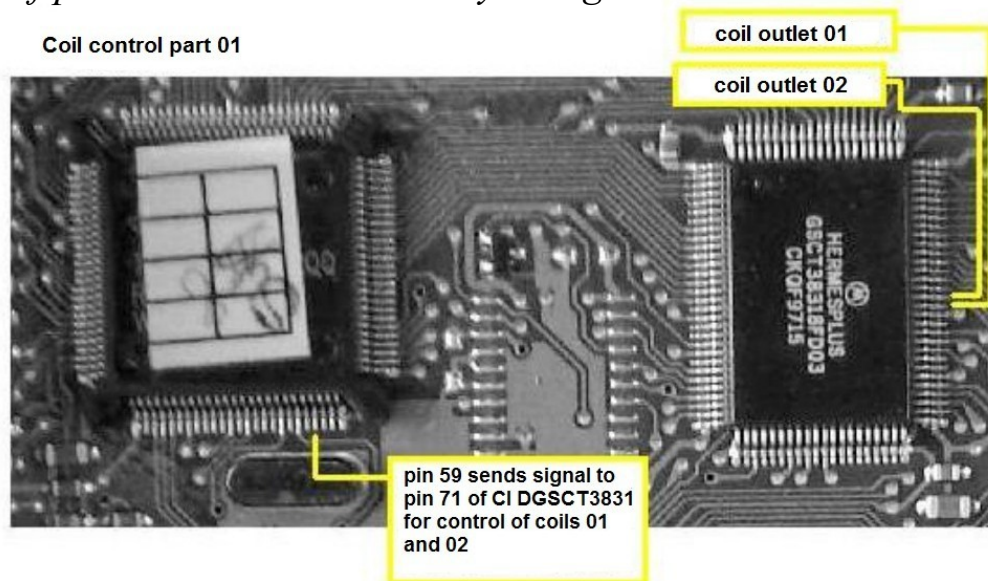
Special features

- Interchangeable with 1.6 16v brava.

Common defects

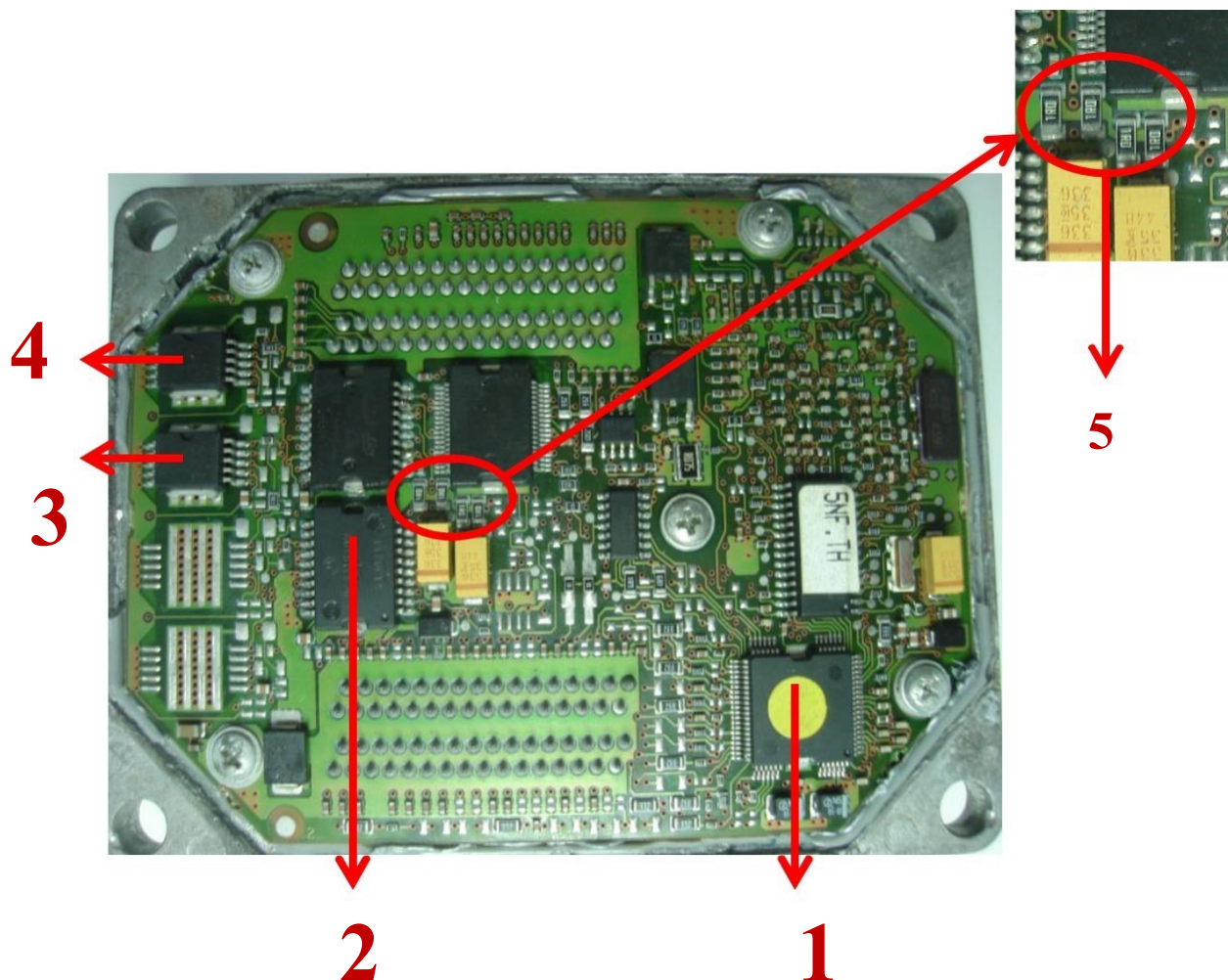
- High temperature silicone oscillation and heating temperature.

Note: This car usually mechanics leave out of point and oscillates slowly and goes out.



5NF - Fiat

Vehicles: Doblo 1.4 / 1.8 DBW.



Main components

- 1 5V regulator, electric fan;
- 2 Electronic accelerator, MC33;
- 3 Coil;
- 4 Coil;
- 5 Set of 4 protection resistors of drive n°2 (1Ω).

Technical specifications

Common defects

- Loss of acceleration - tele-loading or motor drive of step.

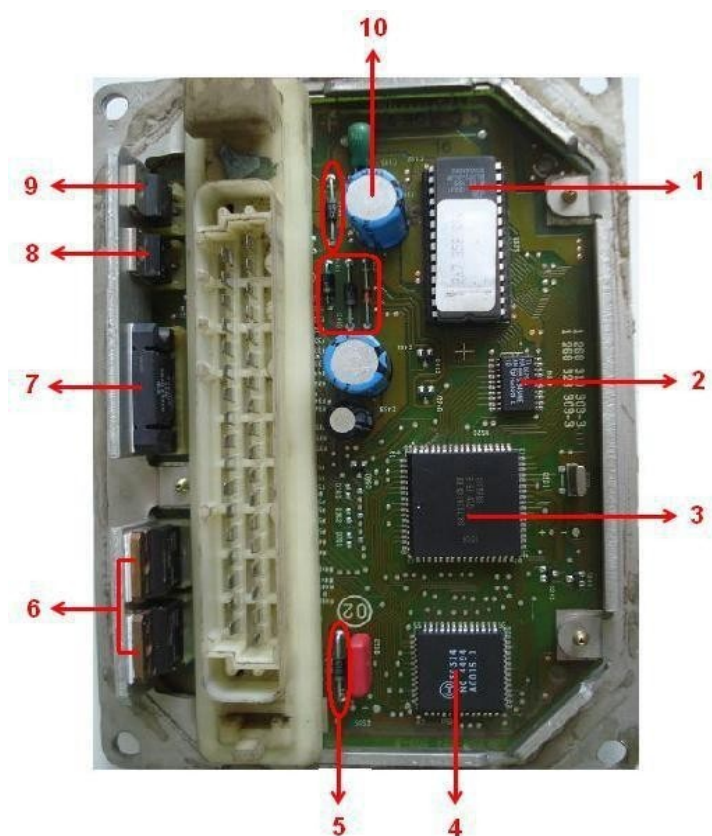
Note: This 5NF ECU before tele

loading we must remove the eprom 95160 for that perform the necessary programming and boot services.

If you do not remove the eprom 95160 and execute programming ECU crashes and repair is no longer possible.

Motronic 1.7 - Fiat

Vehicles: Type 1.6 IE 8V



Main components

- 1** EPROM;
- 2** RAM memory;
- 3** Processor;
- 4** Pulse controller: Nozzle, coil and actuator drive ML;
- 5** Protection diode;
- 6** Coil 1 and 2;
- 7** Stepper motor and pump relay;
- 8** Spout;
- 9** 5 V regulator;
- 10** Capacitor.

Technical specifications

Special features

- Pin 1 facing the processor;
- Intel 8051 processor - 68 pins;
- Eprom 27C256 with 32 kb;
- Analog coil signal;
- Phonic RPM signal 60-2;
- Has no immobilizer;
- 1 coil, rotation sensor, 1 injection nozzle.

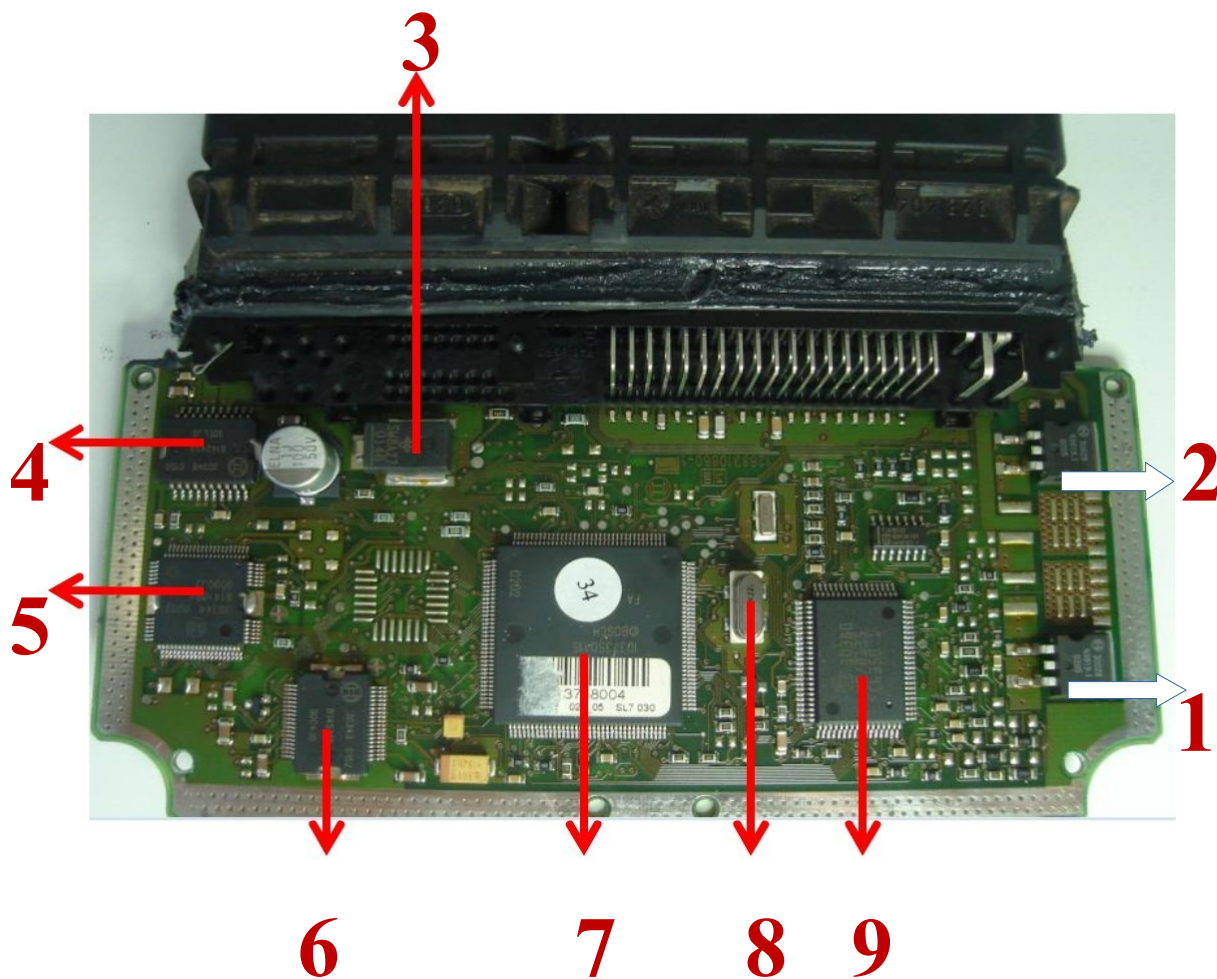
Common Defects

- Cold solder, if it is damp or oxidized;
- Nozzle and coil oscillation;
- Check that the nozzle is original and the TPS-BUTTERFLY, the two work together;
- Locked stepper motor warms up the drive motor step and ends up burning;
- Nozzle no pulses possible defect in the processor help. This processor also controls the ignition.
- ECU for Clio Mono, Peugeot mono with Bosch 3.1 are

very similar, but not interchangeable.

7.9.6 - Fiat and GM

Vehicles: Vectra 2.0 Flex and Astra Flex



Main components

- 1 Coil;
- 2 Coil;
- 3 Protection diode;
- 4 Electronic accelerator;
- 5 Nozzles / Fan / Lambda Probe;
- 6 5V regulator / pump / main relay;

- 7 Processor;
- 8 Crystal;
- 9 Electronic throttle signal converter.

Technical specifications

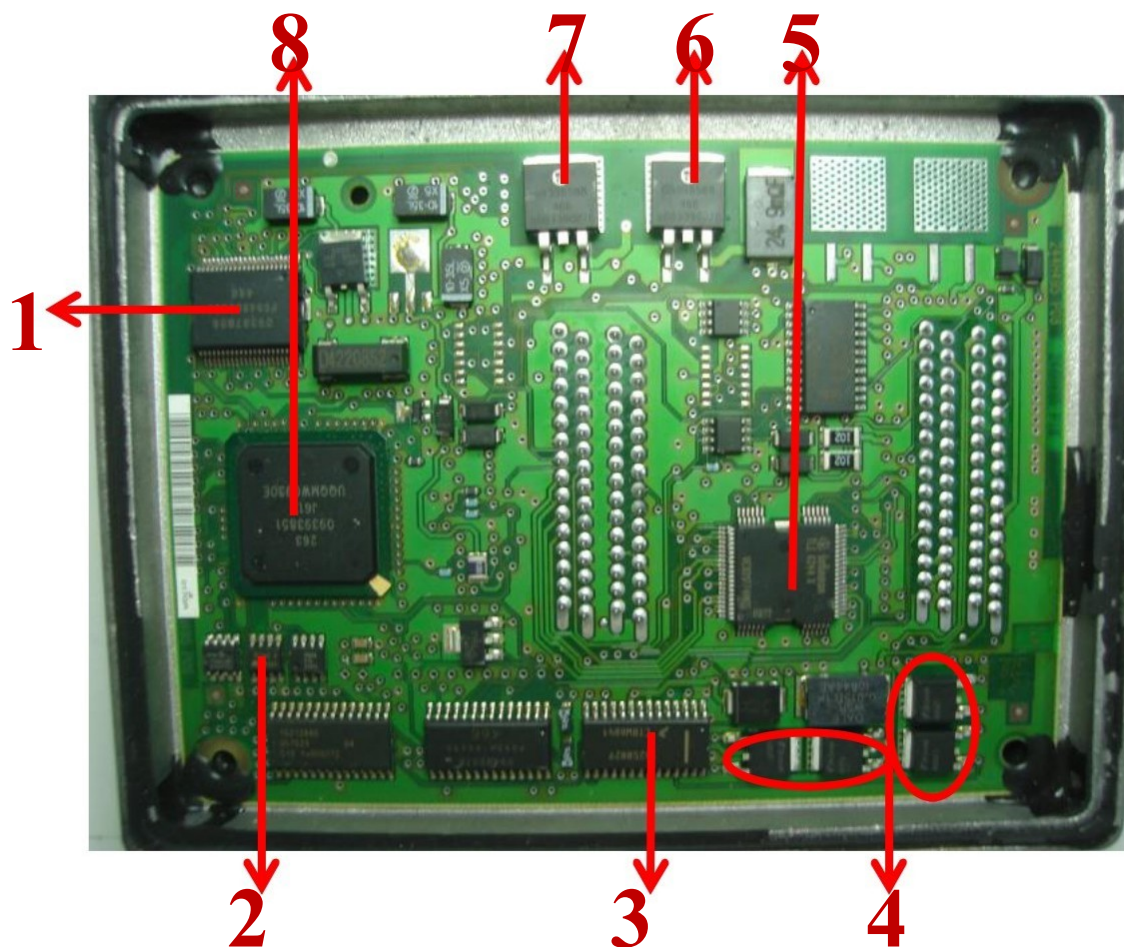
Note: To exchange the ECU from one car to the other, just change the EPROM 95080.

When reconnecting the ECU, leave the switch on for 60 seconds to adapt.

Unlike the Marelli ECU, this Bosch ECU has a Signal “converter” of the 4 tps sensors.

HSFI or MT27E - Fiat and GM

Vehicle: Stilo and Montana 1.8



Main components

1. 5V regulator;
2. Soic 95040 or 95160 code;
- 3 Electronic accelerator;
- 4 Set of transistors that assist drive 3;
- 5 Nozzle / Pump / Main Relay / Fan / Air conditioning;
- 6 Coil;
- 7 Coil;
- 8 Processor.

Technical specifications

Special features

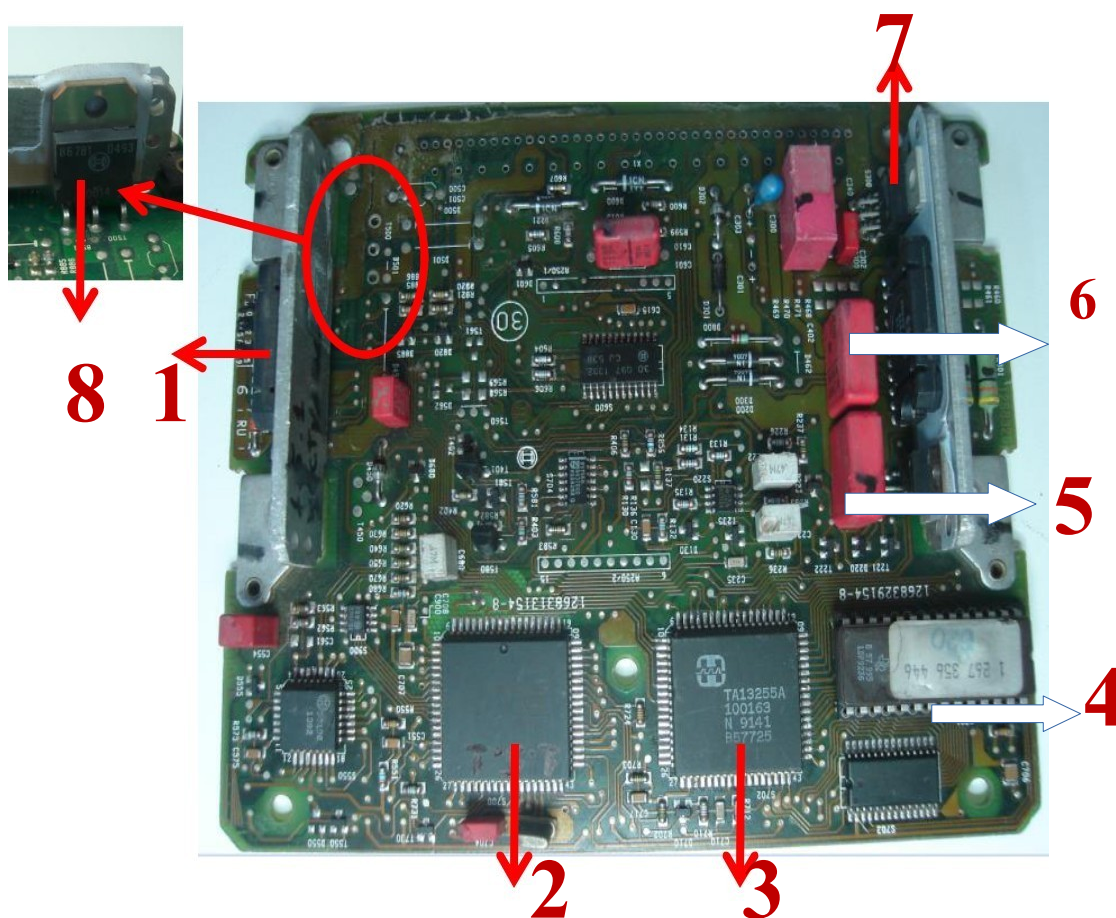
- Montana ECU different from Stilo's.

Common defects

- Defect in the butterfly body, inducing to think that the error is in the ECU.

Note: The butterfly body generally ceases to function due to failure in the SMD with the number 16250829, which is responsible for activating the body electronic butterfly.

Motronic 1.5.1 – GM



Main components

- 1 Pump / stepper motor;
- 2 Processor;
- 3 Auxiliary processor
- 4 Eprom;
- 5 Main relay;
- 6 Nozzles;
- 7 5V regulator;
- 8 Coil.

Technical specifications

Bosch Motronic M151 system, equips vehicles

Omega to gasoline 93 to 94 (Imported).

Multi Drive 30080 3992 - 21 Pins - Low Speed Corrector - Relay A / C -

Canister - Fuel pump relay - Injection light



The diode shown above is between pin 13 of the component 30080 and pin 22 of the ECU (**Injection Light**).

When you turn the ignition has pulsed signal terminals **2:03** of amplitude of 5 volts. And at 8 and 9 there is also a signal pulsed, of the same duration.

Multi Pin 21 Pin 300 3992 – Injectors



Multi Drive 21 Pins

Pin	Component	Function	Pin UCE
1		In (Pin 10) Relay A/C	
2		In (Pin 9) Correct ML	
3		In (Pin 8) Correct ML	
4		wire grounding	
5		wire grounding	
6		+ 5 Volts	
7		wire grounding	
8		Correct ML	Pin 4
9		Correct ML	Pin 4
10		Relay A/C	Pin 23
11		Canister	Pin 5
12		Fuel Pump Relay	Pin 3
13		Injection Light	Pin 22
14		wire grounding	
15			
16			
17		+ 15	
18		In (Pin 13) Injection Light	
19		In (Pin 12) Fuel Pump Relay	Low Potential Transistor T550
20		In (Pin 11) Canister	
21		wire grounding	

Left Multi Drive (Nozzles Only)

Pin	Component	Function	Note
1		In (Pin 8, 9 e 10) e (11, 12 e 13) 1, 4, 2 e 3.	negative firing
2		In (Pin 8, 9 e 10) e (11, 12 e 13) 1, 4, 2 e 3.	negative firing
3		In (Pin 8, 9 e 10) e (11, 12 e 13) 1, 4, 2 e 3.	negative firing
4		wire grounding	
5		wire grounding	
6		+ 5 volts	
7		wire grounding	
8		injectors 2 e 3	Pino 17 UCE
9		injectors 2 e 3	Pino 17 UCE
10		injectors 2 e 3	Pino 17 UCE
11		injectors 1 e 2	Pino 16 UCE
12		injectors 1 e 2	Pino 16 UCE
13		injectors 1 e 2	Pino 16 UCE
14		wire grounding	
15			
16			
17		+15	
18		In (Pin 8, 9 e 10) e (11, 12 e 13) injectors 1, 4, 2 e 3.	negative firing
19		In (Pin 8, 9 e 10) e (11, 12 e 13) injectors 1, 4, 2 e 3.	negative firing
20		In (Pin 8, 9 e 10) e (11, 12 e 13) injectors 1, 4, 2 e 3.	negative firing
21		wire grounding	

Mult drive Right

Pin	Component	Function	Note
1		In (Pin 10 e 11) ML Corrector	Pino 34 processor (Jump pin 20)
2		In (Pin 9) Relay A / C	Pin 4 B 5 8011
3		In (Pin 8) Canister	Pin 6 B 58011
4		grounding	
5		grounding	
6		+ 5 volts	
7		grounding	
8		Canister	Pin 5
9		Relay A / C	Pin 23
10		corrector ML	Pin 4
11		corrector ML	Pin 4
12		Fuel pump relay	Pin 3
13		injection light	Pin 22
14		grounding	
15		Jump pin 16	
16		Jump pin 15	
17		+15	
18		In (Pin 13) injection light	Pin 16 B 58011
19		In (Pin 12) Fuel pump relay	Pin 65 processor
20		In (Pin 10 e 11) ML corrector	Pin 34 processor Jump pin 1
21		grounding	

5 Volt Regulator - 3009 - 7 pins



Regulator 5V

Pin	Component	Function
1		+ 12 V
2		+ 12 V
3		+ 5 V
4		grounding
5		2,2 V
6		2,2 V
7		5V

Main relay drive - ON 588

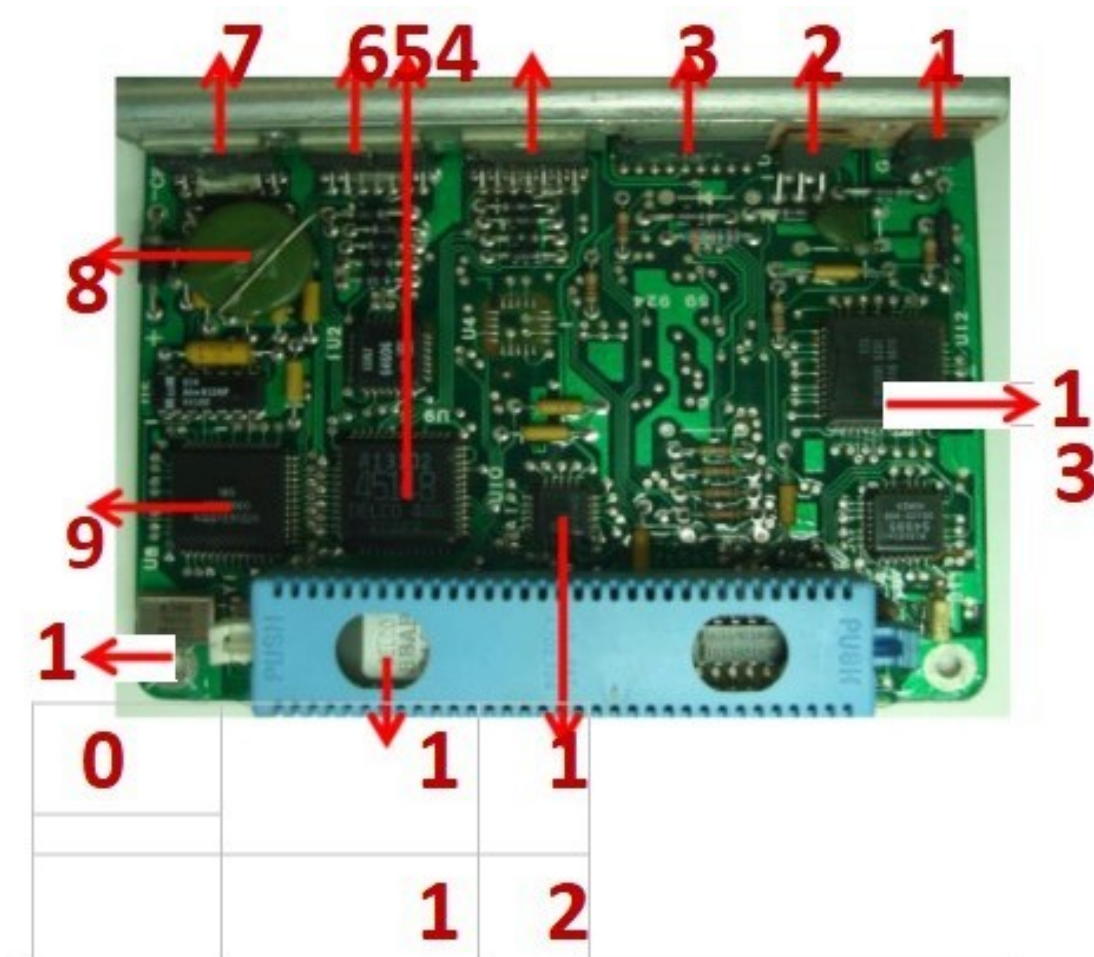


Main relay - ON 588

Pin	Component	Function
1		Firing
2		Pin 36 ECU
3		grounding

Multec 700 - GM

Vehicles: Monza, Kadett and Ipanema, from 91 to 96.



Main components

- 1 Injector drive;
- 2 16156085 - Fuel pump relay drive;
- 3 34993 - Stepper motor drive;
- 4 Delco 196 E - 16045980 - PF relay - Injection light - Flight computer - V2 fan;
- 5 Processor;
- 6 Delco 196 E - 16045980 - V1 Fan - Heating relay

collector - Ascending light - A / C relay;

7 34992 - 5V regulator;

8 NTC;

9 Processor;

10 Cristal - Clock;

11 Eprom, 27C128

Technical specifications

Special features

- Multec 700 exchanges are all the same, the difference is only in the mencil (Calibration Memory).
- Drive Beak - Commercial - **IRF 540** or **IRFZ34N**.

Common defects

- Failure in the fan relay drive;
- Bad contact at the injector secondary drive terminals and pump, causing failure when moving the ECU.

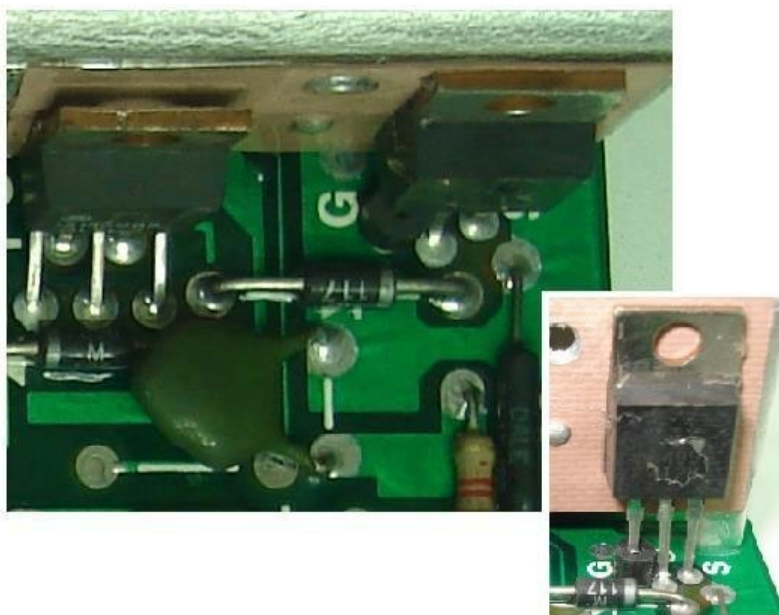
Necessary to resold the terminals.

Note:

o For the injector to pulse, it is necessary to jump the terminals D9 and D10. Doing this, passes the earthing of the D9 to D10, that is, for the injector to work is pin D10 must be grounded.

O ECUs are the same, what changes is the month.

Injector drive - 3 pins



Pin	Component	Function
1		Firing (Amplitude of 10 volts)
2		Out - injector - Pin D15 D16 ECU
3		grounding

This component is a FPN NPN transistor, with ferrite on the coupling (firing). This washer aims to eliminate electromagnetic interference.

Note: Note that the trigger has a range of 10 volts.

Secondary drive - Injector - Fuel pump



Pin	Component	Function
2		Out- Drive injector drive - pin 1 (G)
10		Out- Drive relay pump - pin 5
20		Out- Pin 9 - Delco 2

Processor



Pin	Component	Function
2		In - Clock - Crystal
9		Out - Stepper motor drive - pin 6
10		Out - Stepper motor drive - pin 9
52		In - Clock Crystal

Mencal



Kadett					
Original Chip	EPROM	Vehicle	year / model	Fuel	Mencal
BAZZ 7468	27C128	KAD/1PA 1.8 EFI	92,5 >>	A	7399
AYMM 9646	27C128	KAD/1PA 2.0 EFI	92 >>	G	5999
ARXC 9673	27C128	KAD/1PA 1.8 EFI	92 >>	G	7939
ARXF 9625	27C128	KAD/1PA 1.8 EFI	92 >>	A	7959
BKSY 2808	27C128	KAD/1PA 2.0 EFI	95 >>	G	2949
BKSJ 2765	27C128	KAD/1PA 2.0 EFI	95 >>	G	2829
7355803	27C	KADETTI 2.0 MPFI	97	G	
Monza					
Original Chip	EPROM	Vehicle	year / model	Fuel	Mencal
AYMM 5999	27C	MONZA 2.0 EFI	92 >>	G	5999
BKSJ 2829	27C128	MONZA 2.0 EFI	92 >>	G	2829
BBAB 7484	27C128	MONZA 2.0 EFI	92,5 >>	A	7419
AYMM 9652	27C128	MONZA 2.0 EFI	92 >>	G	5999
AYMM 9642	27C128	MONZA 2.0 EFI	95 >>	A	6009
APZL 9689	27C128	MONZA 1.8 EFI	95 >>	G	3659
APZL 9684	27C128	MONZA 1.8 EFI	95 >>	A	3679
BBAA 7477	27C128	MONZA 1.8 EFI	92,5 >>	A	7409

Regulator 5 volts - 34992 - 15 pins



pin componente	Function
1	+ 5 volts
2	+ 5 - 12 volts
3	+ 5 volts
4	+ 5 volts
8	+5 Map, TPS and CO Potentiometer - pin C14 ECU
11	+30 - 12 Volts
12	+ 5 volts
13	+30 - 12 Volts
14	grounding
15	+ 5 volts

Delco 169 E 1 and 2

Delco 169 E - 1

Delco 169 E - 2



Delco 169 E - 1 - PF Relay - Injection light - Computer board - V2 fan - 15 pins

In this component the input (**trigger**) is negative and its out (**out**) is also negative.

Delco 169 E - 2 - V1 fan - collector heater relay

- Up light - Relay A / C - 15 pins

Pin	Component	Function
3		Out - V1 Fan - Pin A3 ECU (Com A/C)
4		In - Firing - Collector heater relay
5		Out - Collector heater relay - pin D12 ECU
7		+ 30 - 12 volts
8		grounding
11		Out - light upward march - Pin A7 ECU
12		In - Firing - Relay A/C
13		Out - Relay A/C
15		grounding

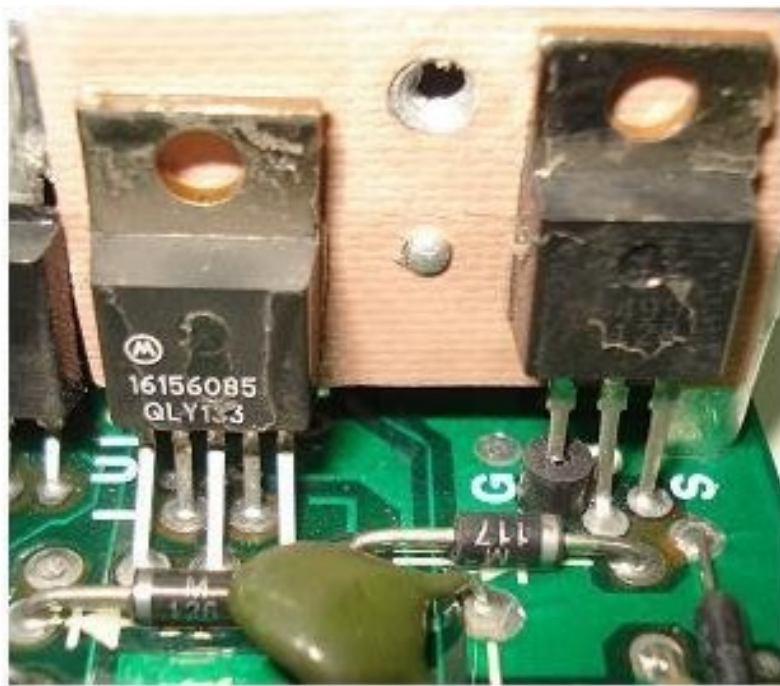
Stepper motor drive - 34993 - 9 pins



Pin	Component	Function
1		Out - Coil A - Pin C5 ECU
2		Out - Coil A - Pin C6 ECU
3		grounding
4		Out - Coil B - Pin C3 ECU
5		Out - Coil B - Pin C4 ECU
6		Firing - Coil - Amplitude of 5 volts - Pin 9 Processor
7		+30 - 12 volts
8		+ 5 volts
9		Firing - Coil Amplitude of 5 volts - Pin 10 Processor

Note: When the ignition is switched off, the ECU starts the engine step, that is, to test the drive just turn on and off the ignition. And test with the oscilloscope.

Pump relay drive and Injector drive

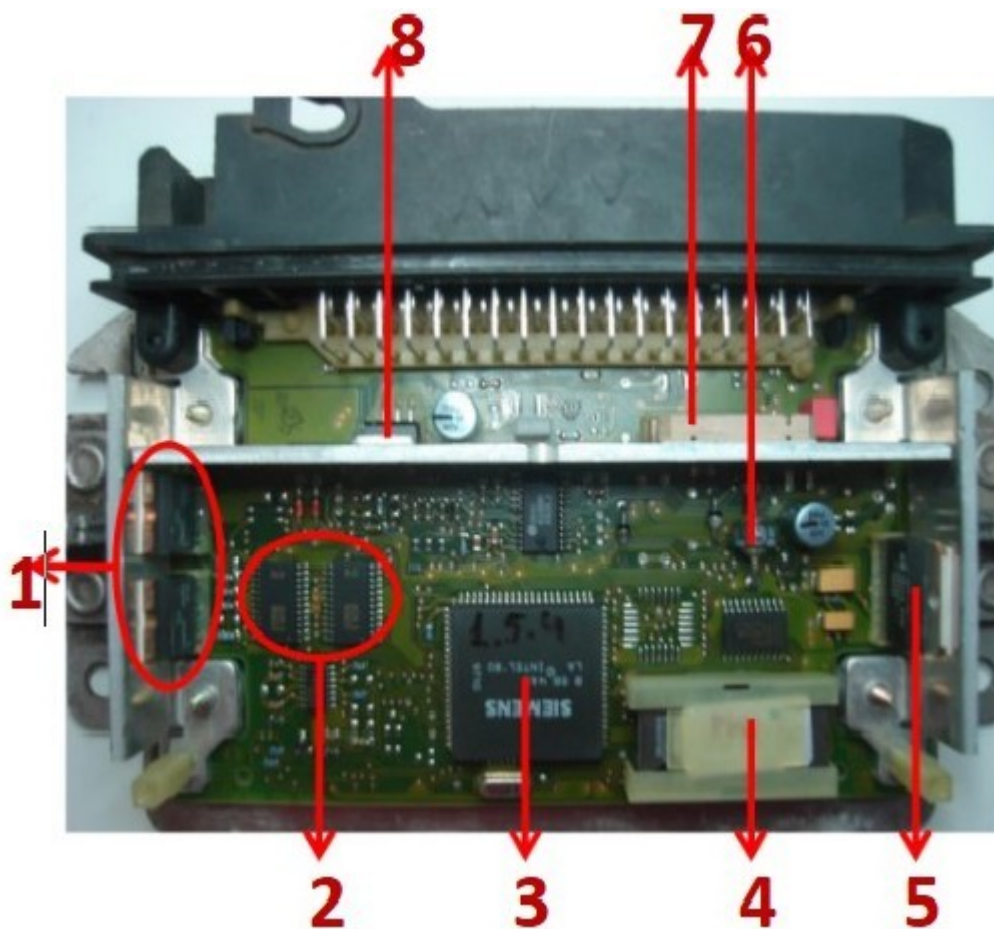


Drive 16156085 - Fuel Pump Relay - 5 pins

Input (trip) + 5 volts and output + 12 volts.

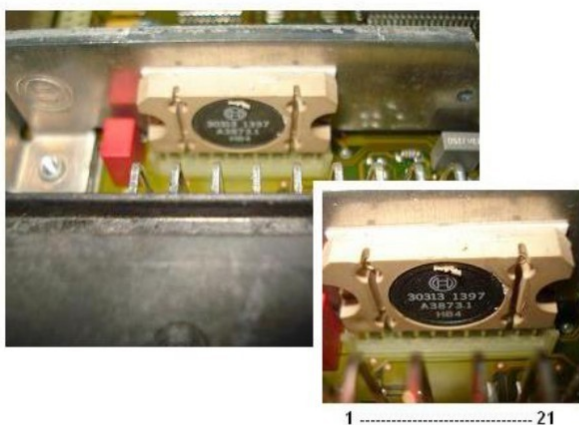
Pin	Component	Function
1		+ 15 - 12 volts
2		Out - Fuel pump relay - Pin A1 ECU
3		Out - Fuel pump relay - Pin A1 ECU
4		grounding
5		Firing (+ 5volts)

Motronic 1.5.4 – GM



Main components

- 1 Coils;
- 2 Multi fan drivers;
- 3 Processor;
- 4 Eprom;
- 5 5V regulator;
- 6 Protection diode;
- 7 Nozzles, Canister;
- 8 Stepper motor.

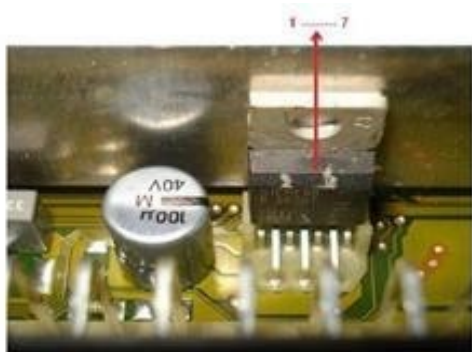


component pin	function
1	In - (Pin 10) - Canister - Pin 66 processor
2	In - (Pin 9) -spout 4- Pin 5 processor
3	In - (Pin 8) -spout 1- Pin 1 processor
4	grounding
5	grounding
6	+ 5 volts
7	grounding
8	Out -spout 1(Pin 17 UCE) processor
9	Out -spout 4 (Pin 34 UCE) processor
10	Out - Canister (Pin 5 UCE) processor
11	Out - fan speed 1 (pin 23 UCE)
12	Out - block 3 (Pino 35 ECU)
13	Out - block 1 (Pino 16 ECU)
14	grounding
15	Pin 23 processor
16	Pin 23 processor
17	+ 15 processor
18	In - (Pin 13) - Bloco 1 - Pin 2 processor
19	In - (Pin 12) - Bloco 3 - Pin 3 processor
20	In - (Pin 11) - fan speed 1 - Pin 6
21	grounding

21-pin multi drive (Injectors, Canister, Fan speed 1) - 30313;

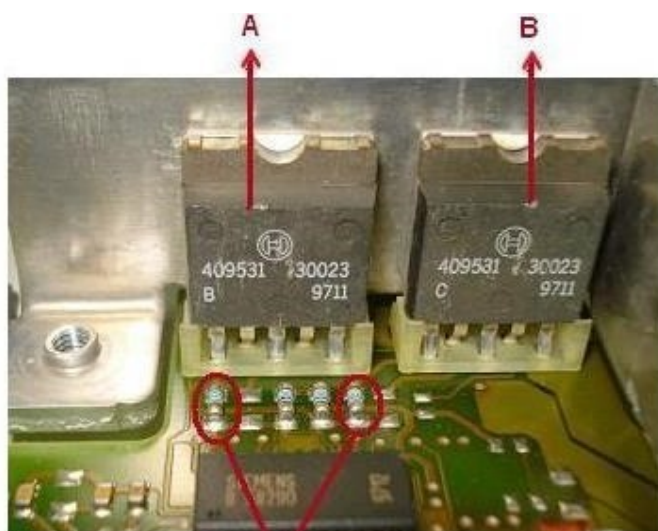
Negative shooting and negative output.

Idle Corrector Drive - 7 pins



Pin	Component	function
1		Out - Pin 26 UCE (Corrector)
2		Pin 12 of Multi Drive A and B
3		
4		grounding
5		Pulsed Signal 5v - Synchronized with the pin 7
6		+15
7		Out - Pin 4 UCE (Corrector)

Ignition coil drive - 30023 - 5-pin – Vectra



Current Limiting Resistors

Coil A - Coil 2 and 3

Pin	Component	Function
1		In - Firing - Comes from Pin 1 of (TLE 4226G)
3		Out - Pin 20 UCE
4		+ 15
5		grounding

Coil B - Coil 1 and 4

Pin	Component	Function
1		In - Finring - Comes from Pin 1 of (TLE 4226G)
3		Out - Pin 1 UCE
4		+ 15
5		grounding

5V regulator and main relay drive - 30358 or 30284 - 15 pins



Pin	Component	Function
1		
2		Out - + 5V - Connected to the Capacitor 100nF
3		Pino 12 UCE (Voltage 5V S. Butterfly , MAP , EGR of engines 16V)
4		+ 30
5		+ 15
6		Pin 46 ECU (Main relay)
7		+ 30
8		grounding
9		
10		
11		
12		+ 30
13		+ 5V
14		+ 5V
15		+ 5V

This component regulates the voltage 5 volts and also activates the main relay.

EGR control drive - BTS 5115 A (Blazer and S10)

The component shown controls the activation of the solenoid connected in series with the EGR hose,
8V Blazer and S10 engines only.

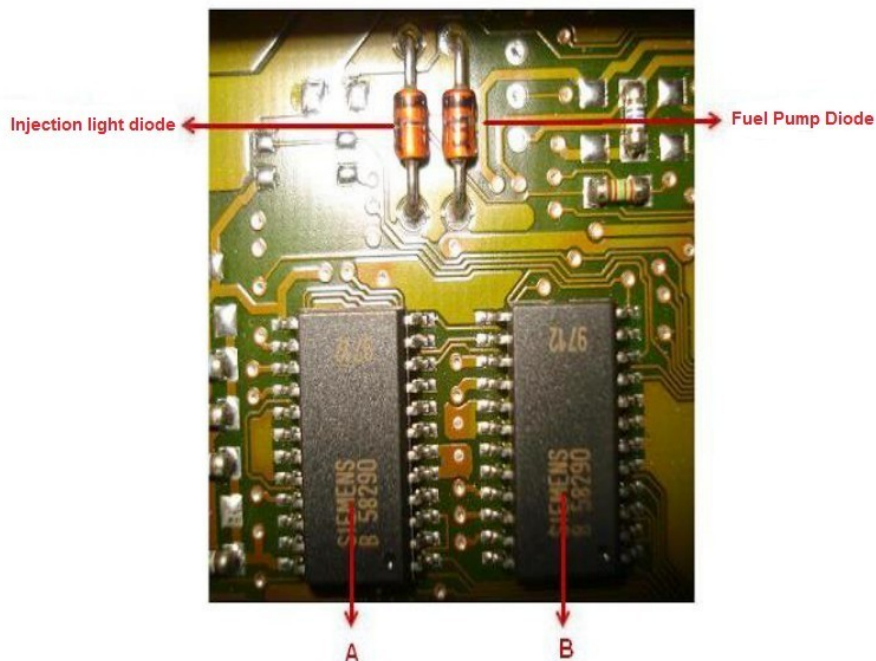
In 16V motors, use linear EGR with drive electrical, controlled by another drive (PWM).



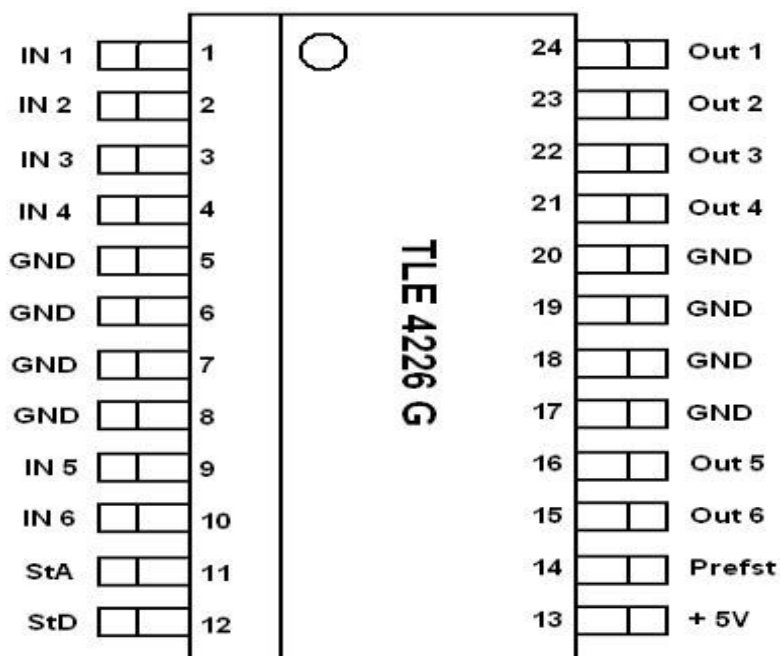
Pin	Component	Function
1		Pin 24 Multi Drive B (Firing)
2		Pin 15 ECU - Control EGR
3		grounding

24-pin mult drive - B 58108 or TLE 4226 G

In this component, the drive is made by negative and the output is positive 5V.



Note: If one of the above diodes shorted the controlled component will be connected directly. This happens due to the diodes having one of the terminals connected to the ground.



Multi Drive 24-pin A (Pump relay, injection light, ignition drive)

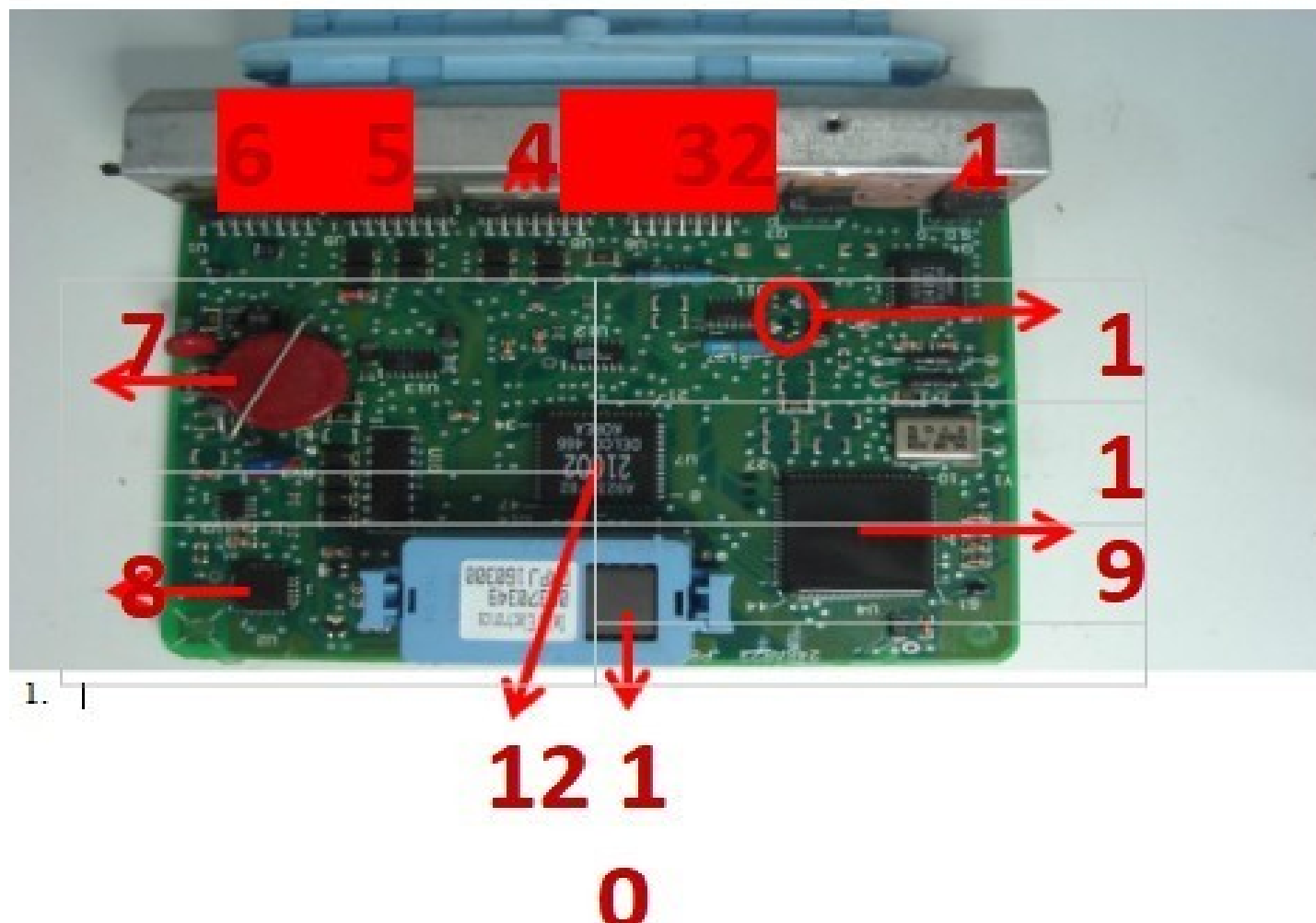
Pin	Component	Function
1		In - Drive Ignition Coil - Pin 63 Processor
2		In - Drive Ignition Coil - Pin 62 Processor
3		Pin 67 Processor
4		Pin 6 Processor
5		grounding
6		grounding
7		grounding
8		grounding
9		In - Fuel pump relay
10		In - injection light - Pin 65 Processor
11		Used by the ECU
12		Used by the ECU
13		+ 5V
14		Used by the ECU
15		Out - Pin 22 ECU - injection light
16		Out - Pin 3 ECU - Fuel pump relay
17		grounding
18		grounding
19		grounding
20		grounding
21		Out
22		Out
23		Out - Drive Ignition Coil
24		Out - Drive Ignition Coil

24-pin Multi Drive (Fan speed 2, Relay A / C, EGR control)

Pin	Component	Function
1		In - Control EGR - Pin 68 Processor
2		In - Board Computer
3		Pin 1 Processor
4		Pin 9 Processor
5		grounding
6		grounding
7		grounding
8		grounding
9		In -Relay A/C- Pin 64 Processor
10		In - Fan V2 - Pin 06 of B5796 (Behind the Plate)
11		Used by the ECU
12		Used by the ECU
13		+ 5V
14		Used by the ECU
15		Out - Pin 6 ECU - Fan speed 2
16		Out - Pin 25 ECU - Relay A/C
17		grounding
18		grounding
19		grounding
20		grounding
21		Out
22		Out
23		Out - Pin 32 ECU - Board Computer
24		Out - Pin 1 Drive Control EGR (Firing)

Multec MPFI EMS –GM

Vehicles: Corsa 1.0 MPFI / 1.6 MPFI / Omega 2.2 MPFI / S-10 MPFI



Main components

- 1** Nozzle 1 and 4;
- 2** Nozzle 2 and 3;
- 3** Stepper motor;
- 4** Fault light / turning account / U8 / Canister
- 5** Fan 1 and 2 / EGR
- 6** 5V regulator
- 7** Varistor
- 8** Lambda probe
- 9** Processor
- 10** Eprom
- 11** Coil Transistor
- 12** PLD

Technical specifications

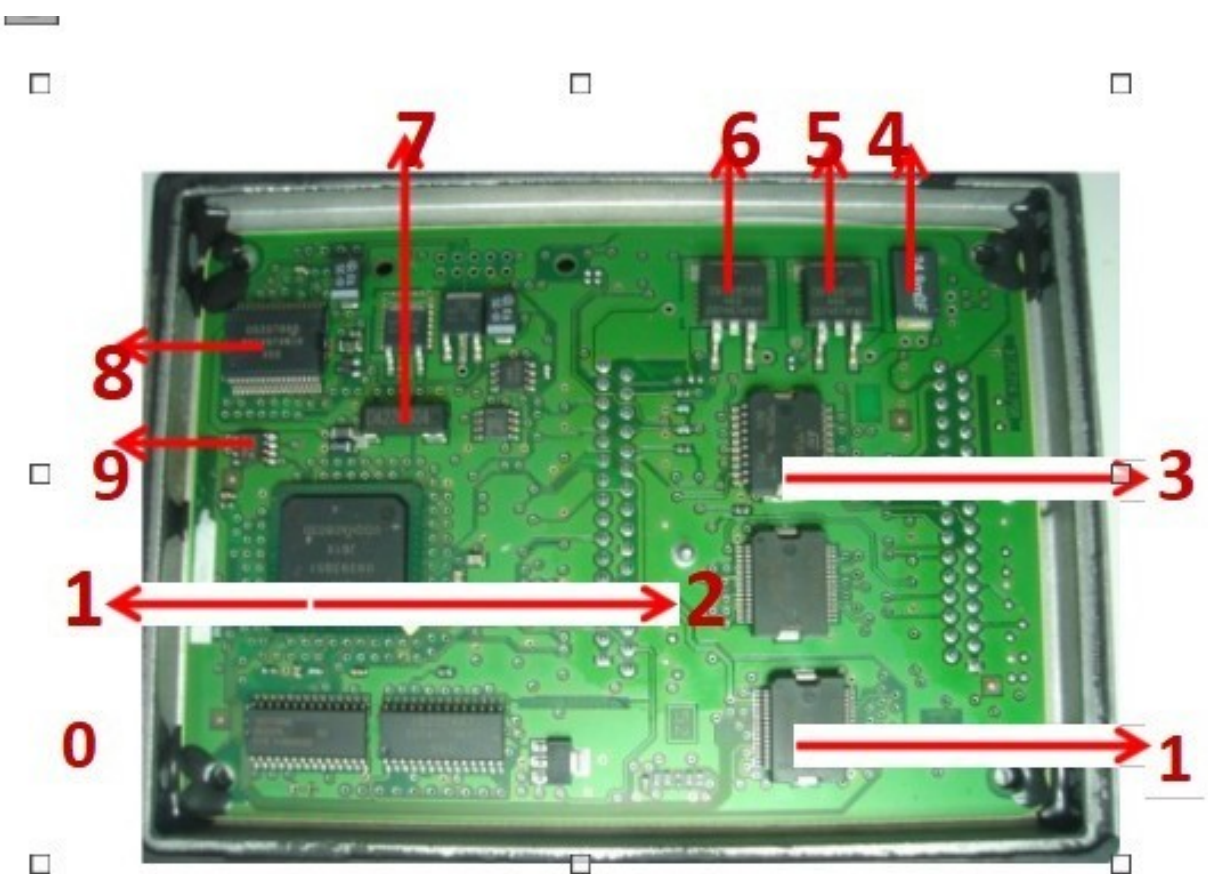
- Rotation signal: Phonic 60-2
- Transformation: Corsa 1.0 to 1.6 just change the eprom.
Corsa for Omega 2.2 needs some adjustments in the module.
- Eprom: 27C256 32Kb
- Immobilizer: When you have it is in the HC11 processor (photo 9)

Common defects

- Varistor firing: check for possible alternator or short circuit in the harness
How to solve - Change the varistor
- Direct fan:
How to solve- 1- change 2-reset processor drive
- Fan does not gun:
How to solve-1- check the track 2- check schedule eprom 3- Change drive.

Multec H VHC - GM

Vehicles: Montana 1.8, Celta 1.4 and Corsa 1.0.



Main components

- 1 Nozzles / Canister / Pump / Fan / Probe;
- 2 Cold start;
- 3 Stepper motor;
- 4 Capacitor;
- 5 Coil;
- 6 Coil;
- 7 Crystal;
- 8 5V regulator;
- 9 Soic 95040 code;
- 10 BGA processor.

Technical specifications

Special features

- Interchangeable with each other as long as the file of the injection, flex ECU and gasoline ECU are not fully interchangeable.

Common defects

- Probe does not read, SOIC corrupts the file and defects accelerated.

Note: To change the ECU from car to car, simply change the EPROM.

EEC V - ANIL AND ENIO - FORD

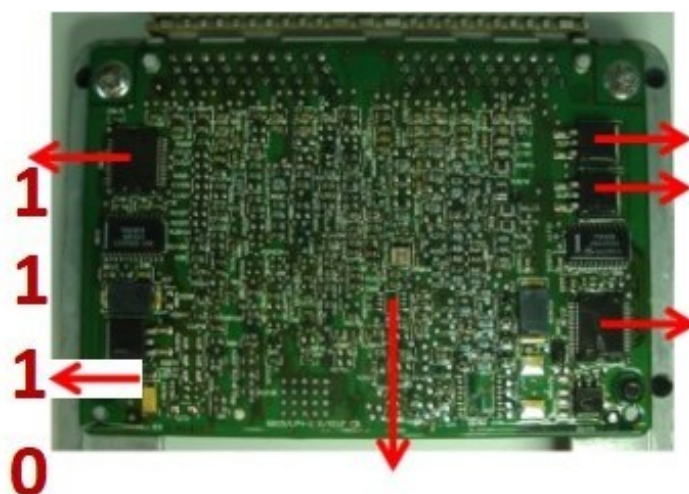
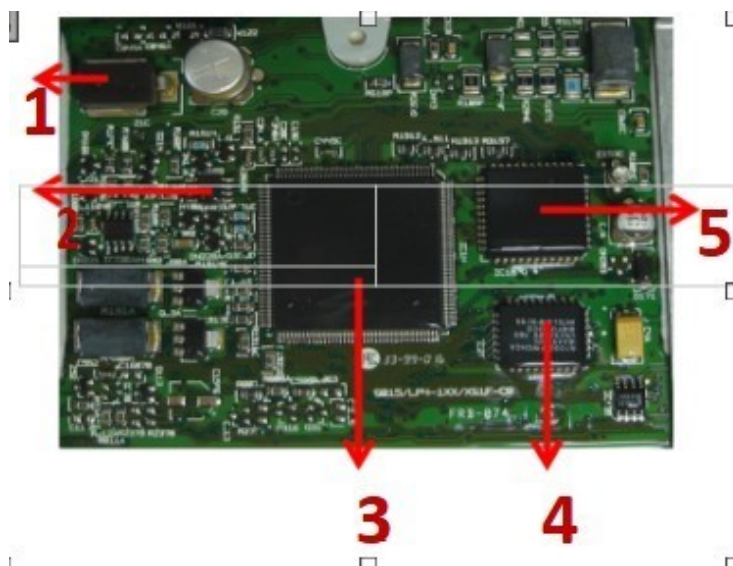
INDIGO

Vehicles: With 1.0 8V Rocan engine.

ENIO

Vehicles: With 1.6 8V Rocan engine.

FRONT



SEE IF

Main components

- 1** Protection diode;
- 2** Main rotation input;
- 3** Processor;
- 4** Eprom memory;
- 5** PLD;
- 6** Nozzle 1 and 2 / Stepper motor / Pump / Fan;
- 7** Coil (Cylinder 2 and 3);
- 8** Coil (Cylinder 1 and 4);
- 9** Soic immobilizer;
- 10** 5V regulator;
- 11** Nozzle 3 and 4.

Technical specifications

Special features

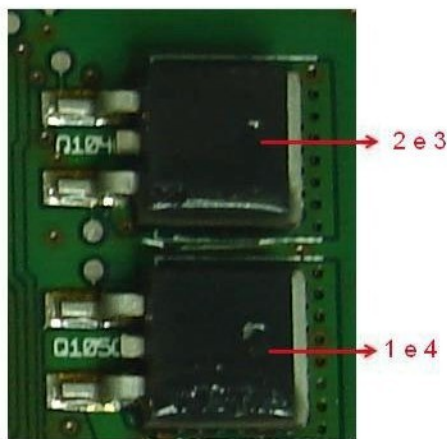
- With immobilizer.

Common defects:

- Failure to activate the fan.

Note: The Soic behind the sign, which contains the immobilizer information has the original number 90510 and its similar is 25020.

Ignition coil drive



Bobina 2 e 3

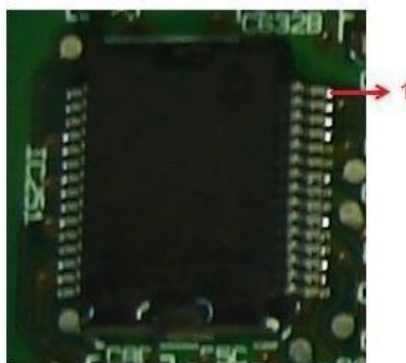
Pin	Component	Function
1		Firing
2		Out - Pin 58 ECU
3		grounding

Bobina 1 e 4

Pin	Component	Function
1		Firing
2		Out - Pino 59 ECU
3		grounding

Multi drive - 1034SE001 - 30 pins - Injectors 3 and 4 - Revolving account

Triggering by + 5 volts and output is negative.



Pin	Component	Function
1		+ 15
3		In - Firing - Injector 3
4 / 5		Out - Injetor 3 - Pin 34 ECU
6		In - Firing - Injector 4
7 / 8		Out -Injector 4- Pin 14 ECU
16		+ 5 volts
22 / 23		Out - Tachometer Panel (rev counter) - Pin 4 ECU
24		In - Tachometer
28 / 29		grounding

Multi drive - 1035SE001 - 30 pins - Canister - Injectors
1 and 2 - Fuel pump relay - Run corrector
slow - Relay A / C - V1 and V2 Cooling
 Trip done by +5 volts and negative output.



Pin	Component	Function
1		+ 15
2		Out - Canister - Pin 11 ECU
3		In - Firing - V1 Fan
4		Out - V1 Fan - Pin 13 ECU
5 / 6 / 7 / 8		Out - Corrector M.L. - Pin 21 ECU
9		Out - V2 Fan - Pin 31 ECU
10		Out - Relay A/C - Pin 54 ECU
11		Out - Fuel pump relay - Pin 53 ECU
16		+ 5 volts
17		In - Fuel pump relay
19 / 20		Out - Injector 1 - Pin 12 ECU
21		In - Firing - Injector 1
24		In - Firing - Injector 2
25 / 26		Out - Injector 2 - Pin 15 ECU
27 / 28		grounding

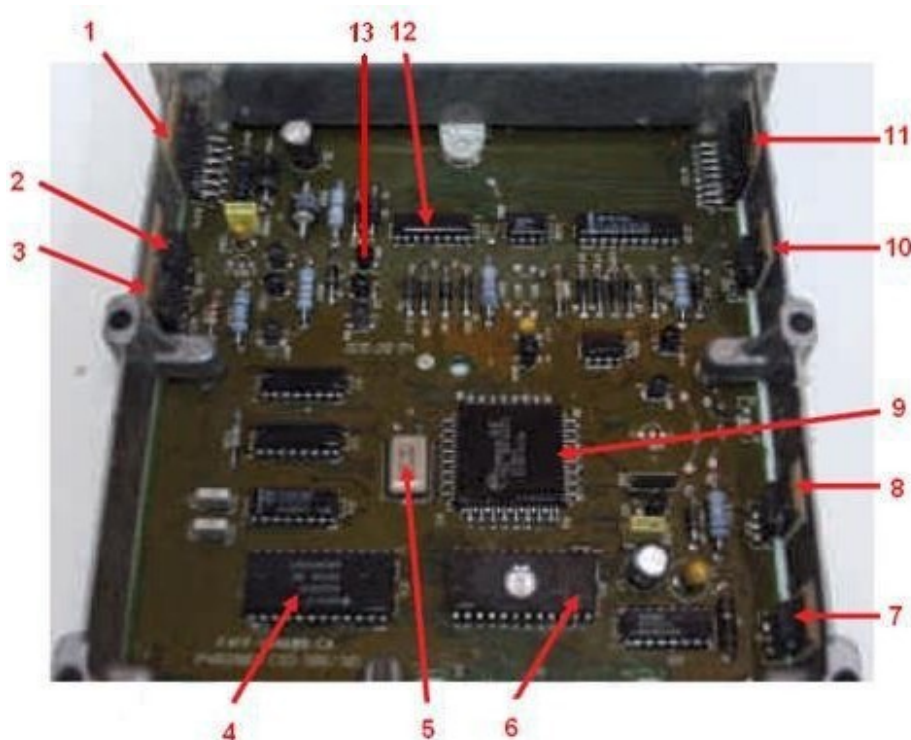
Lambda probe heating control drive



Pin	Component	Function
1		Firing
2		Out - Pin 53 ECU
3		grounding

EEC IV - FORD

Vehicles: VW and Ford from 95 to 96 monoponto, with AP and CHT (AE) engines



Main components

- 1 Injector drive;
- 2 Injection diode;
- 3 Diagnostic control drive;
- 4 RAM memory;
- 5 Crystal / Clock;
- 6 ROM memory;
- 7 Transistor - turns on / off 5 volts;
- 8 Cold start drive;
- 9 Processor;
- 10 Drive fuel pump relay;
- 11 Stepper motor drive;
- 12 Quadruple Mult Drive - Canister - A / C Relay – Similar Ca 3262.
- 13 Transistor coil.

Technical specifications

Special features

- The ECU does not control the ignition coil. This system has the TFI module.
- The quadruple multi drive can be replaced by the CA 3262, which is common in several ECUs.

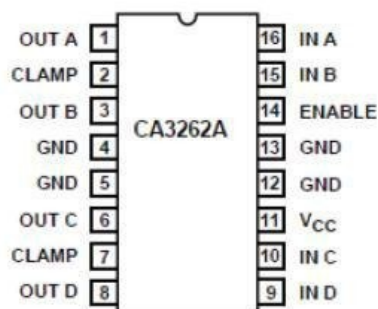
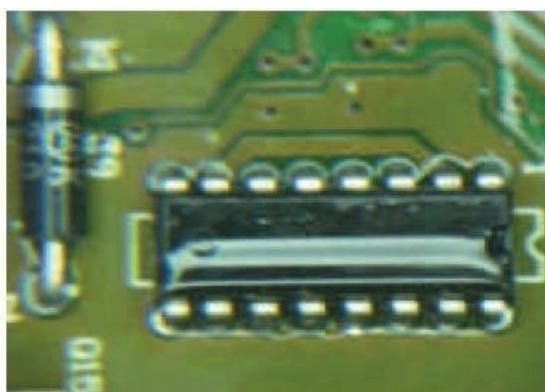
Common defects

- Injector nozzle IC;
- Idle actuator CI.

Note: This car usually shortens the whip of the nozzles and burn the CI, always check the nozzle harness and also the idle actuator. Some vehicles may present instability in operation even after replacing the defective component. You must reset the EC, passing something metallic on its pins. Thereby it is possible to erase / reset the parameters things that you can't get with a scanner.

Quadruple Mult Drive - Canister - A / C Relay - (IC 17) - Similar Ca 3262

This component works similarly to NPN transistors, that is, the input (trip) is made by positive 5 volts and the output is negative.



Pin	Component	Function
1		Out - Pin 34 ECU
3		Out - Pin 51 ECU
4		Grounding
5		Grounding
6		Out -Relay A/C - Pin 54 ECU
8		Out - Canister - Pin 35 ECU
9		In Pin 8 – Canister – Pin 25 Processor
10		In Pin 6 – Relay A/C – Pin 5 da RAM
11		+ 5 volts
12		Grounding
13		Grounding
15		In Pin 3 – Pin 24 Processor
16		In Pin 1 – Pin 11 Processor

Stepper Motor Drive - (IC 123) - 15 pins

When this component has 5 volts at the input, it has 12 volts at the output.

Pin	Component	Function
2		In – Pin 6 - Coil 1
3		In – Pin 7 – Coil 2
4		In – Pin 10 – Coil 1
5		In – Pin 11 – Coil 2
6		Out – Coil 1 – Pin 13 ECU
7		Out – Coil 2 – Pin 31 ECU
8		Grounding
9		Grounding
10		Out – Coil 1 – Pin 14 ECU
11		Out – Coil 2 – Pin 32 ECU
12		+ 15
14 / 15		+ 15

Stepper motor drive test

When the ignition of the drive is switched on, that is, we can test them only by turning on the ignition.

Pin	Component	value found when finding the ignition	value found after 5 seconds
2		pulsed signal (amplitude 5 volts)	5 volts
3		pulsed signal (amplitude 5 volts)	5 volts
4		pulsed signal (amplitude 5 volts)	0 volt
5		pulsed signal (amplitude 5 volts)	0 volt
6		pulsed signal (amplitude 12 volts)	12 volts
7		pulsed signal (amplitude 12 volts)	12 volts
8		grounding	grounding
9		grounding	grounding
10		pulsed signal (amplitude 12 volts)	0 volt
11		pulsed signal (amplitude 12 volts)	0 volt
12 / 14 / 15		+ 15	+ 15

Injector Drive - (IC 27B)

Triggering is done for more than 5 volts and the output is negative.

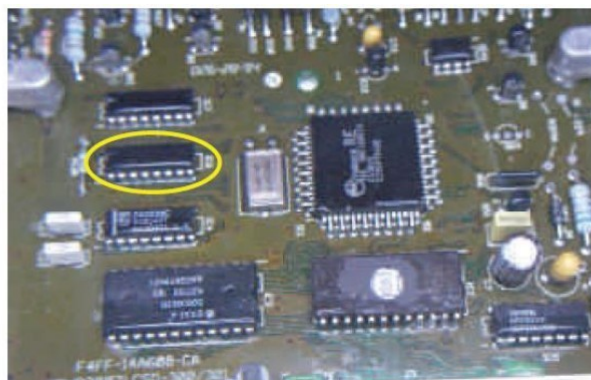
Resistor of 0,1 Ohm – Tolerance of 1%



Pin	Component	Function
2		+ 5 Volts
4		grounding wire
6		grounding wire
7		In - firing (+ 5 volts)
8		Out - injector- Pin 59 ECU
9		Out - injector- Pin 59 ECU
10		Out - injector- Pin 59 ECU

Mult drive (IC 6)

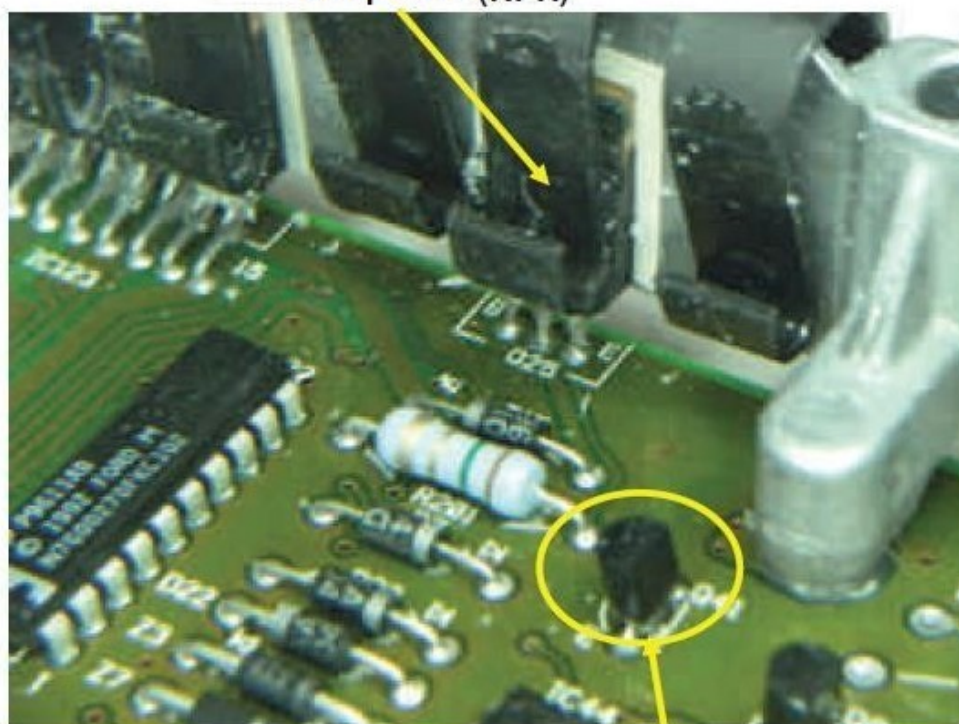
This drive has positive powder input (trigger) and output it is also positive.



Pin	Component	Function
3		+ 5 volts
4		Out - drives diagnostic drive
5		In - Pino 7 processor J
6		In - injector firing A
7		Out - drives pin 7 of the injector drive
8		grounding wire
10		+ 5 volts
13		grounding wire
16		+ 5 volts

Fuel pump drive

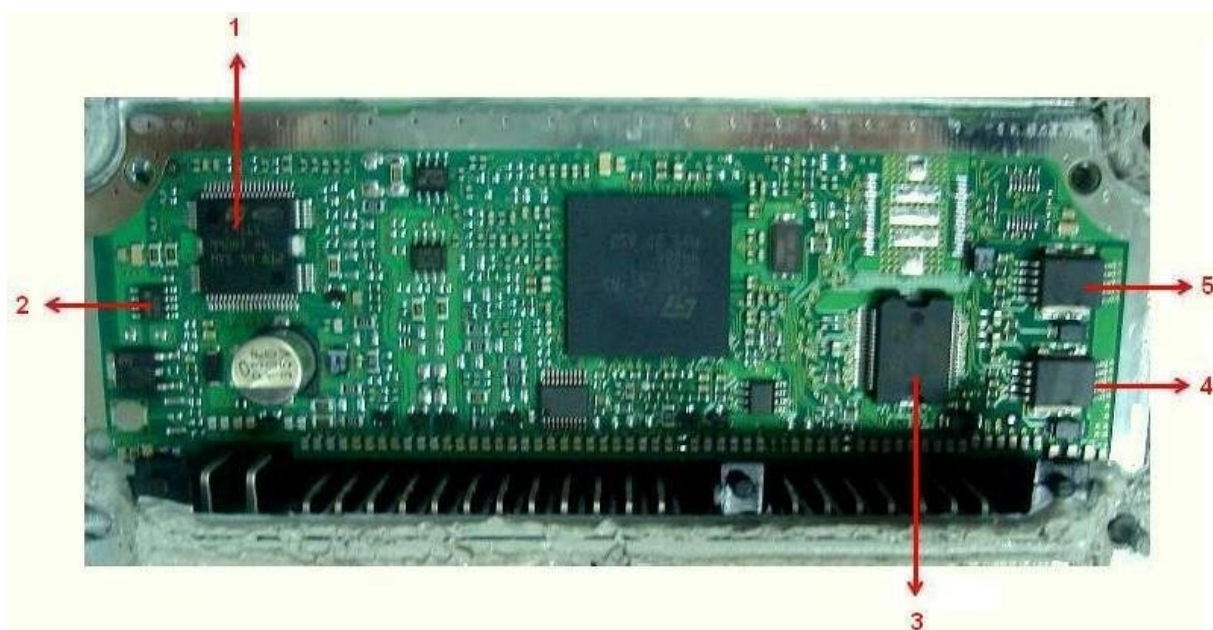
Pump Drive (NPN)



Transistor that drives the pump drive (NPN) F-41

Magneti Marelli 4CFR - Ford

Vehicles: Eco Esporte, Fiesta and Ka.



Main components

- 1 5 volt regulator, pump relay and thermostatic valve;
- 2 SOIC, code 95320;
- 3 Nozzles, Canister and Idle;
- 4 Coil 1;
- 5 Coil 2.

Technical specifications

Special features

- Without distributor;
- Immobilizer integrated into the panel;
- Pin 01 to the right;
- 4X sinusoidal RPM signal;
- Eprom on the processor;

Common defects

- Does not activate the pump relay, which is probably a defect in the bomb.

Magneti Marelli 4AFR - Ford

Vehicles: Eco Esporte, Fiesta and Ka.



Main Components

- 1 Thermostatic valve. 5 volt regulator, pump relay;
- 2 Crystal / Clock;
- 3 SOIC / code;
- 4 Beak, Canister and Slow March;
- 5 Coil 2;
- 6 Coil 1.

Technical specifications

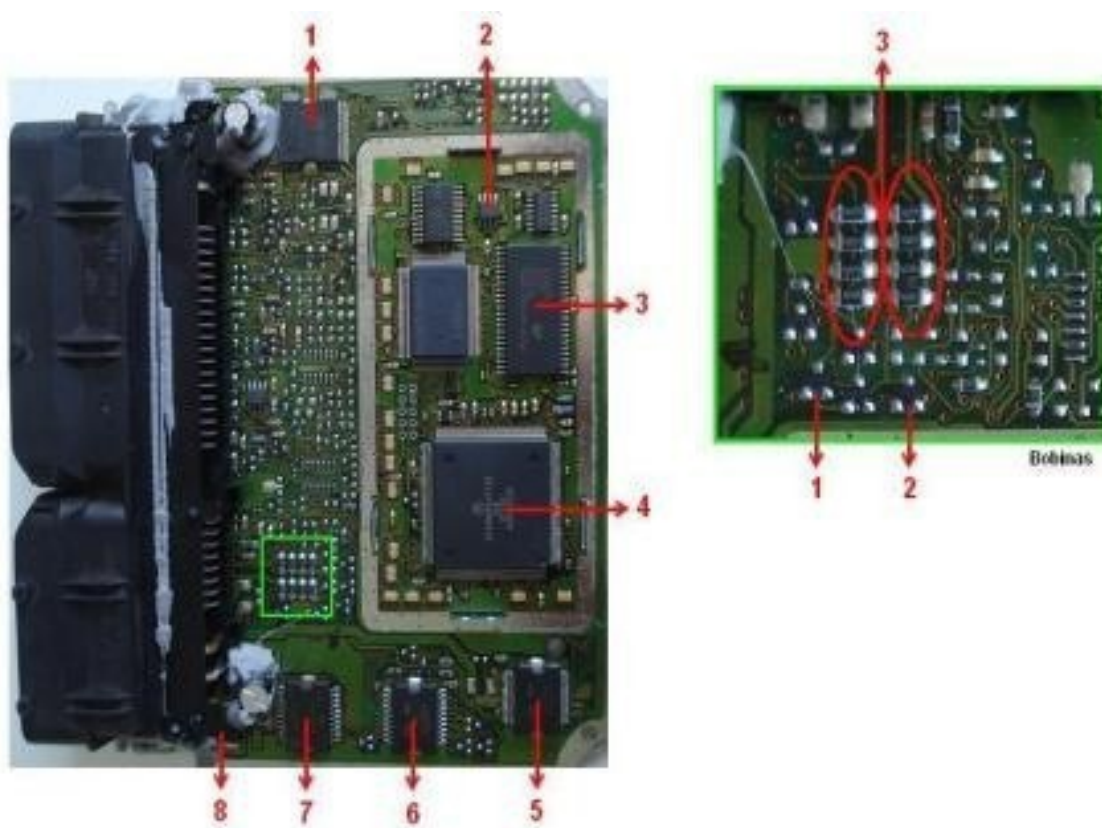
Special features

- Immobilizer integrated into the panel;
- Eprom on the processor;
- 4x sinusoidal RPM signal;
- Pin 01 to the right;
- ST processor;
- Without distributor.

Common defects

- Does not activate the pump relay, probable drive defect the pump;
- Does not trigger or is constantly activated at fan, probable drive defect or oxidation of the board.

Magneti Marelli 4LV – VW



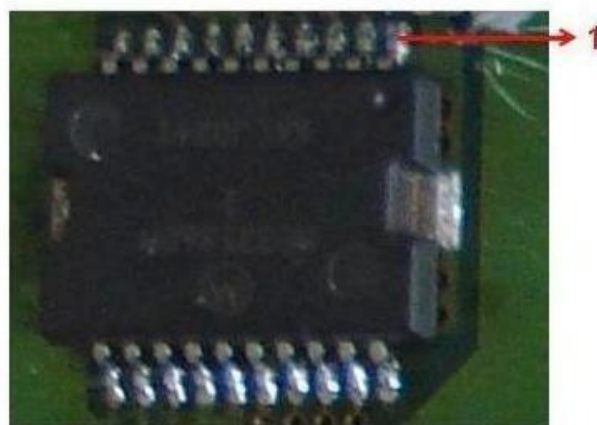
Main components

1. L9113 - Multi-function drive - 5V regulator - Relay main - A / C relay;
2. Soic 95080;
- 3 Rom memory - 29F400 BC;
- 4 Motorola processor - ZC439620VFT25;
- 5 L9110 PD - Multi drive - Canister - A / C relay - Control probe heating;
- 6 L9135 PD - Multi drive - Injectors - V2 Fan;
- 7 MC33186DH1 DC motor control drive;
- 8 Line 15 protection diode.

Coil

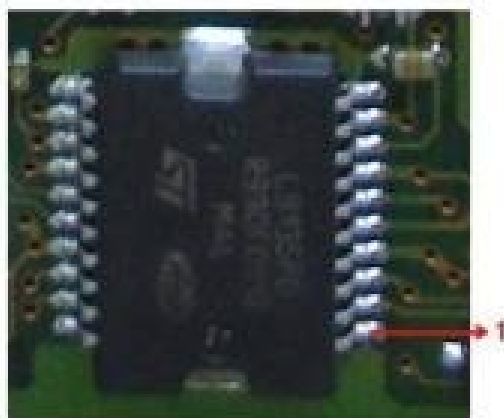
- 1 Coil signal amplifier transistor 1;
- 2 Coil signal amplifier transistor 2;
- 3 Set of coil signal resistors.

DC motor control drive – MC33186DH1



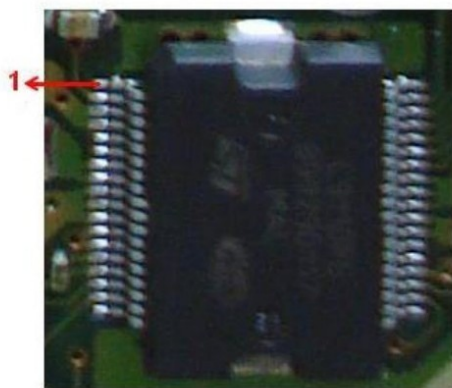
Pin	Component	Function
6/7		Out - Pin 117 ECU - Pin 3 Butterfly body
14/15		Out - Pin 118 ECU - Pin 5 Butterfly body

LP135 PD - Multi drive - Injectors - V2 Fan



Pin	Component	Function
4	Out - Injector 2 - Pin 97 ECU	
7	Out - Injector 1 - Pin 96 ECU	
14	Out - Injector 3 - Pin 88 ECU	
17	Out - Injector 4 - Pin 89 ECU	
19	Out - V2 Fan - Pin 24 ECU	

L9110 PD - Multi drive - Canister - A / C relay - Control heating probe



Pin	Component	Function
22	Out - Canister - Pin 64 ECU	
25/26	Out - relay A/C - Pin 116 ECU	
27/ 28/ 29	Out - lambda probe heating control	

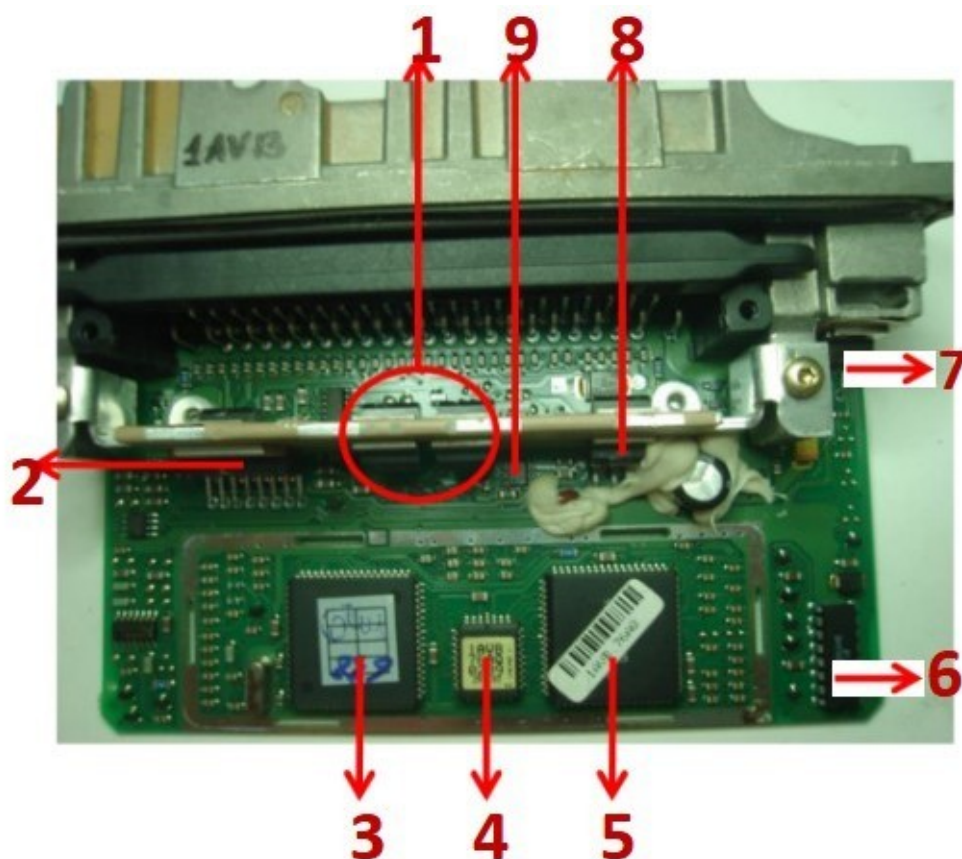
Multi-function drive - 5V Regulator - Main Relay - A / C Relay



Pin	Component	Function
8		In - Reader TPS - Pin 76 ECU
9 /10 ok		Out - +5V MAP and Phase Sensor - Pin 98 ECU
13 /14 ok		Out - + 5V TPS Pin 83 ECU
24		Out -Relay A/C - Pin 4 ECU
25 ok		Out -Light EPC - Pin 48 ECU
28 ok		Out - velocity 1 Fan - Pin 47 ECU
41		In - Rotation Sign - Pin 67 ECU
42		In - Rotation Sign - Pin 53 ECU
57 ok		Out - Main Relay - Pin 65 ECU

Magneti Marelli 1 AVB - VW

Vehicles: With AP 1.6, 1.8 and 2.0 MI engines DE 97 to 99. And for the old model 97/98.



Main components

- 1 Set of nozzle drivers;
- 2 Stepper motor;
- 3 Main processor;
- 4 Eprom;
- 5 Auxiliary processor;
- 6 Pump;
- 7 Coil;
- 8 5V regulator;
- 9 5V of the map / TPS.

Technical specifications

Special features

- Eprom pin 1 facing upwards;
- Has no immobilizer;
- Pump drive - CA3262;
- Coil drive - VB 027. (Pole does not use this drive).
- Stepper motor drive - L9122.

Common defects:

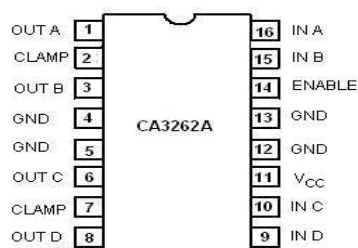
- Failure in the stepper motor drive - L9122;
- Failure in the coil drive - VB027;
- Pump drive failure - CA3262;
- Map power failure - C1 33072 (8-pin tsop)

Note: Some vehicles may show oscillation idle speed due to voltage instability (battery or alternator) and suspect a plant defect.

Drive CA 3262 AE - Relay A / C - Relay Fuel pump

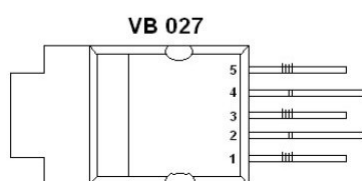
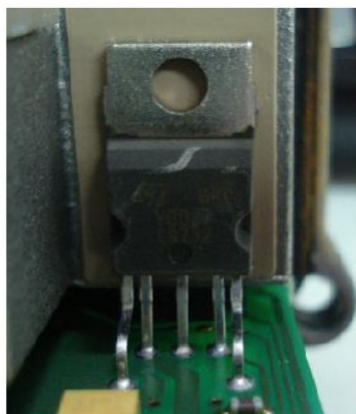
This component works similarly to

NPN transistors , that is, the input (trip) is made by **positive 5 volts** and the output is negative.



Pin	Component	Function
1		Relay A/C - Pin 20 ECU
3		Pin 28 ECU
4		Grounding
5		Grounding
6		Fuel pump relay - Pino 26 ECU
8		Pin 25 ECU
10		In Pin 6 (Relay pump)
11		+ 5 volts
12		Grounding
13		Grounding
16		In Pin 1 (Relay A/C)

Ignition coil drive - VB 027



Stepper motor drive - L 9122



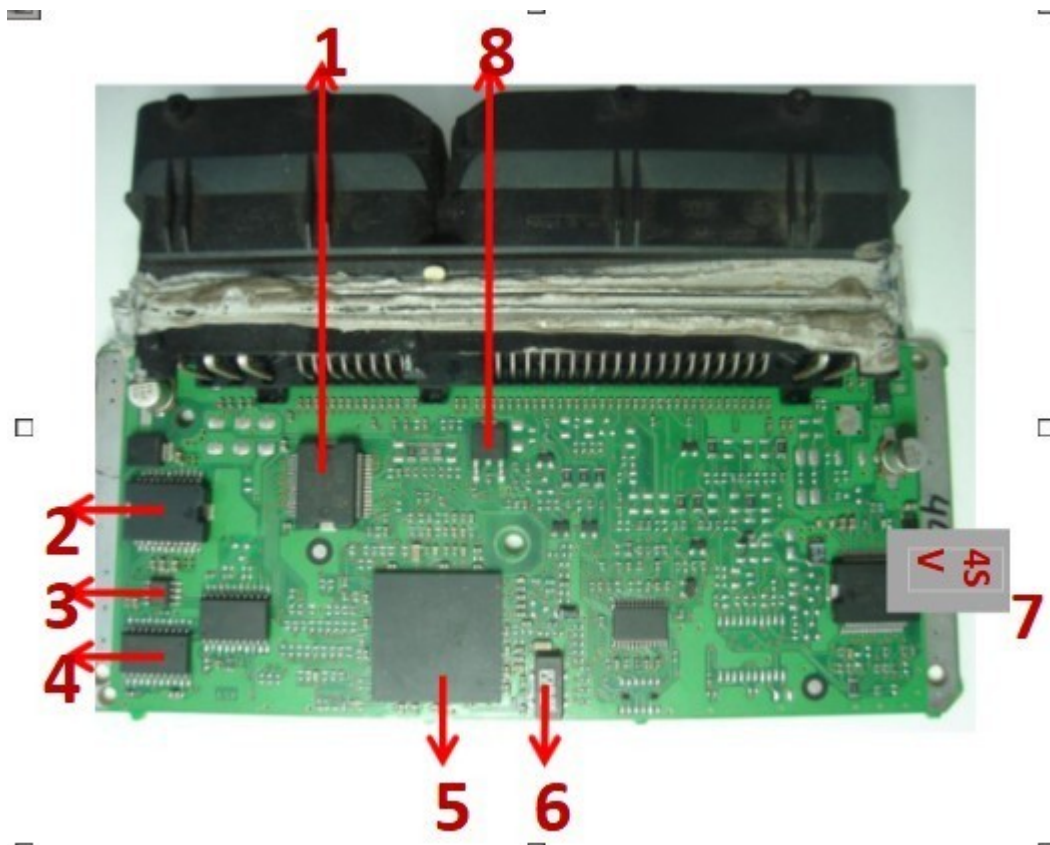
Pin	Component	Function
1		+ 5 v
2		+ 5 v
3		+ 15 v
5	In - Coil	1 - Amplitude 5V
6	Out - Coil	1 - Pin 21 ECU - Amplitude 12 V
7	Out - Coil	1 - Pin 22 ECU - Amplitude 12 V
8		Grounding
9	Out - Coil	2 - Pino 18 ECU - Amplitude 12 V
10	Out - Coil	2 - Pino 19 ECU - Amplitude 12 V
11	In - Coil	2 - Amplitude 5V

Stepper motor drive test with oscilloscope

Adjust the screen to 5V x 100ms.

Magneti Marelli 4SV

Vehicles: With EA 111 1.0 8V and 16 V engines.



Main components

- 1 Canister;
- 2 Stepper motor;
- 3 Code / Soic 95320;
- 4 Coils;
- 5 VGA processor;
- 6 Crystal;
- 7 5V regulator;
- 8 Probe.

Technical specifications

Special features

- 2nd generation of Power systems, replacing the 1 AW system 4LV.

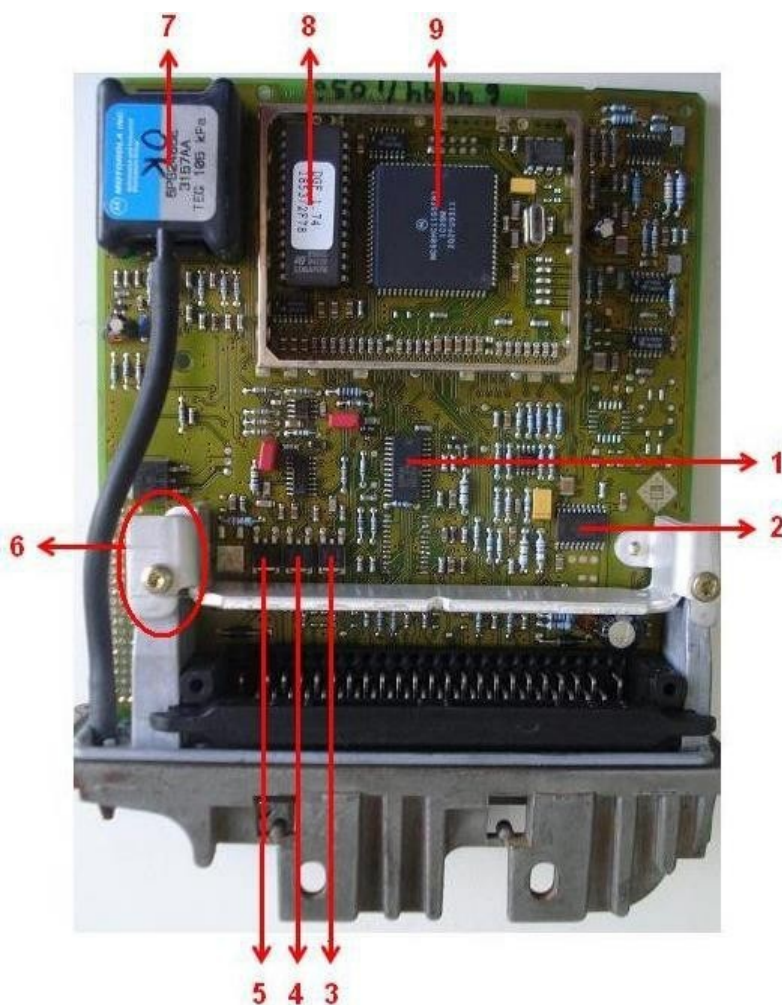
Common defects

- Loss of acceleration due to problems related to battery and alternator.

Note: Although this system replaces the 4LV system, the exchanges are not interchangeable, due to the MAP sensor be different.

Temic 1.74

Vehicles: Gol 2.0 GLX from 94 to 98.



Main components

1. TLE 4216 G - Drive relay fuel pump - Coil of ignition;
2. TLE 4262 G;
- 3 Lambda probe heating control drive - T30;
- 4 Canister drive - T30;
- 5 Idle corrector drive - T30;
6. _____
- 7 MAP
- 8 ROM memory - 27C256;
- 9 Processor - Motorola – MC68HC11G5FN1.

Technical specifications

Special features

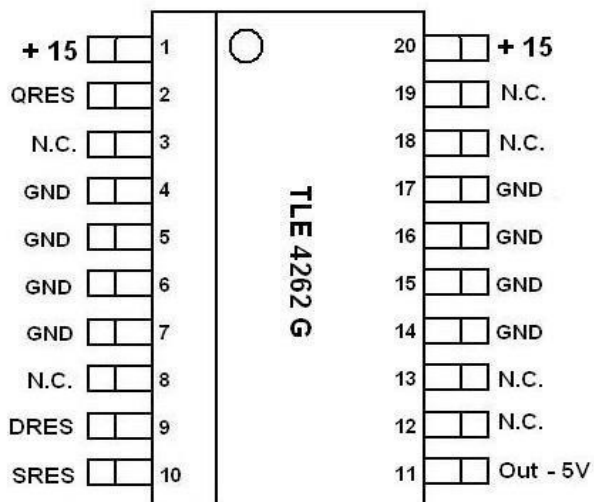
- Has the internal map

Common defects

- Fault in the idle corrector drive;
- Ignition coil drive failure;

Note: Some vehicles may show oscillation idle due to voltage instability in the ECU power, caused by defect in the relay main.

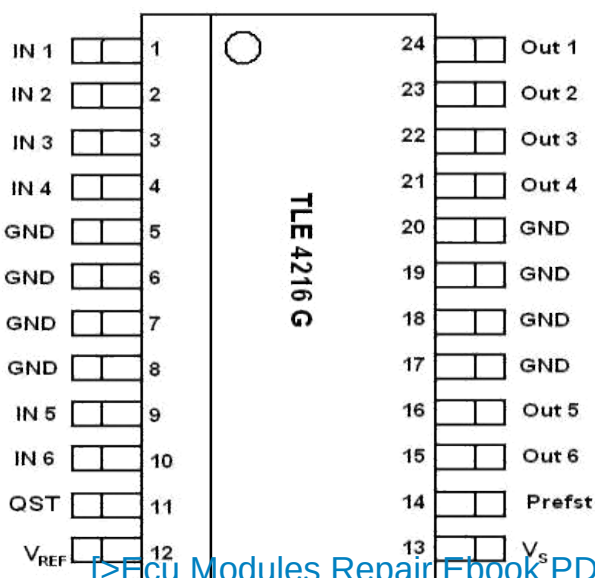
TLE 4262 G - 5 Volt Regulator



Pin	Component	Function
1		+ 15
2		Out - 5 volts
3		+ 15
4 / 5 / 6 / 7 / 14 / 15 / 16 / 17		Grounding

TLE 4216 - Fuel pump relay drive - Coil ignition

In this component the drive is done by a positive 5V and the output is negative.



TLE 4216 G –A

Pin	Component	Function
9		In - Fuel pump relay
16		Out - Fuel pump relay - Pin 31 ECU

TLE 4216 G – B

Pin	Component	Function
10		In - Ignition coil
15		Out - Ignition coil - Pin 7 ECU

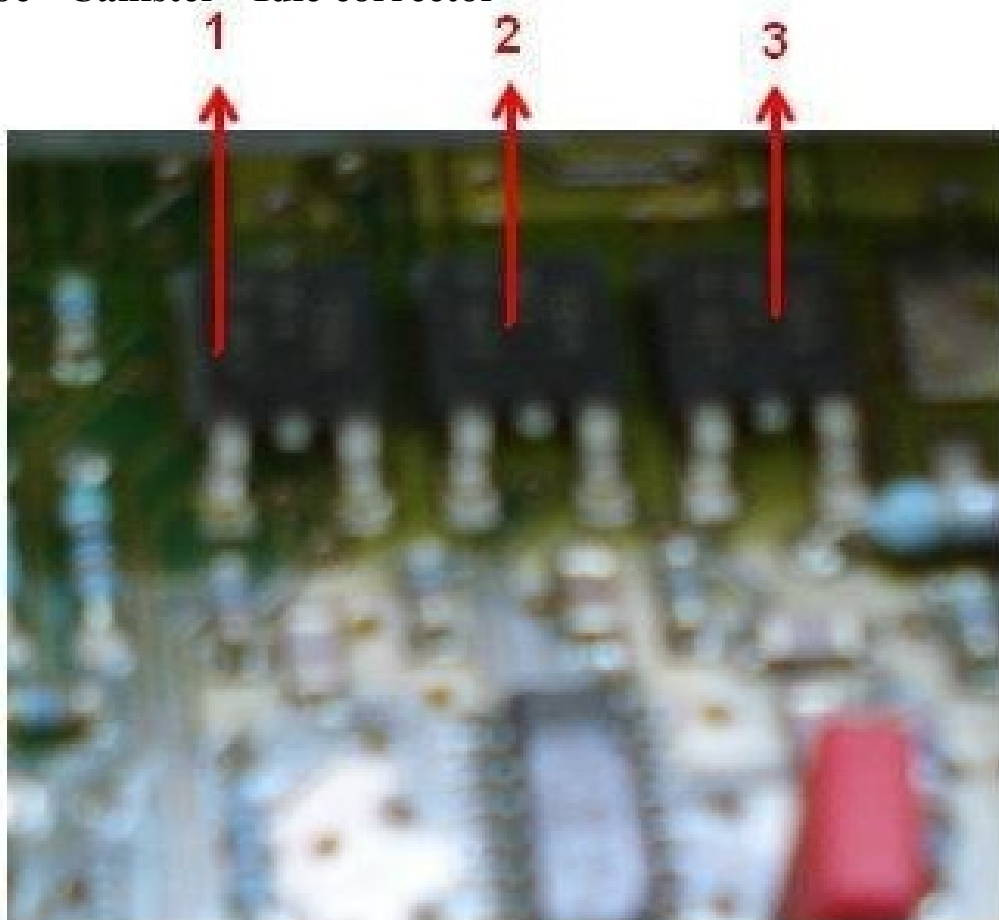
Injector Drive - BTS 121 A

This component is an NPN - Positive output trigger negative.



Pin	Component	Function
1		In - Firing
2		Out - Pin 2 ECU
3		Grounding

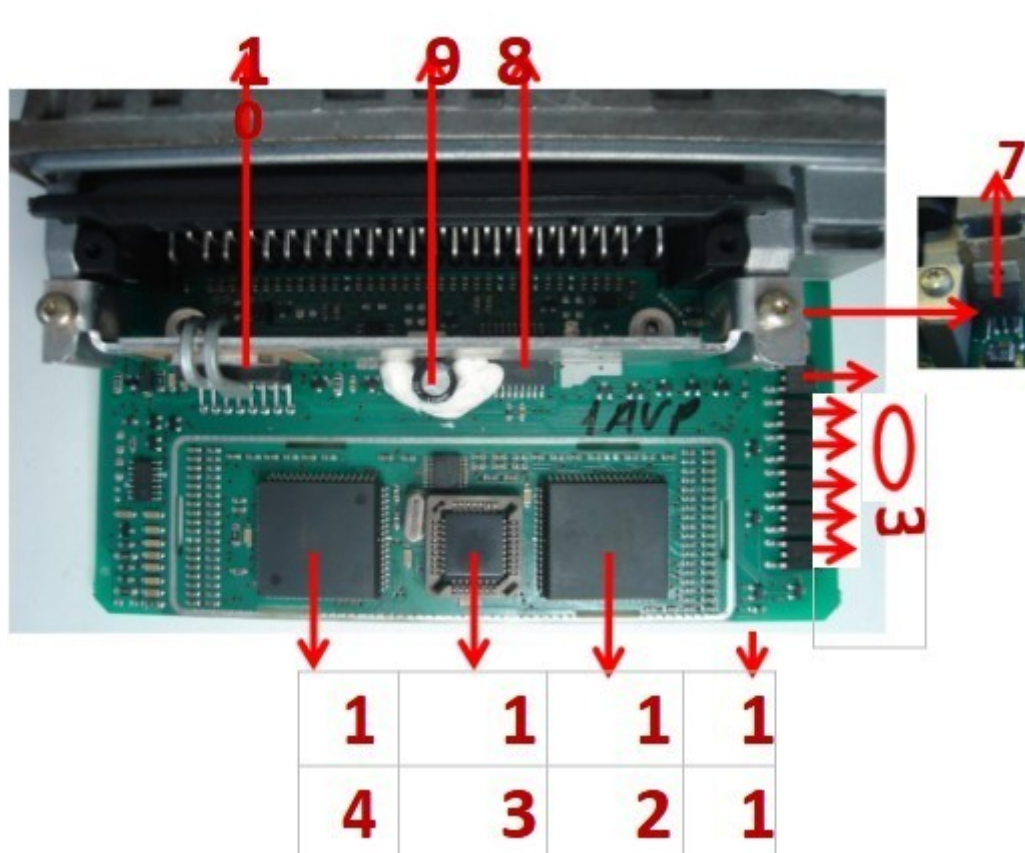
Transistors - T30 - Heating control of the probe - Canister - Idle corrector



- 1 Probe heating control;
- 2 Canister;
- 3 Idle broker.

1 AVP - VW

Vehicles: With AP 1.6, 1.8 and 2.0 MI engines from 99 onwards. And model pole old 99.



Main components

- 1 Canister;
- 2 Nozzle 4;
- 3 Nozzle 3;
- 4 Nozzle 1;
- 5 Nozzle 2;
- 6 Probe;
- 7 Coil;

- 1 5V / TPS regulator;
- 2 Capacitor;
- 3 Stepper motor;
- 4 Pump relay;
- 5 Auxiliary processor;
- 6 Eprom;
- 7 Main processor.

Technical specifications

Special features

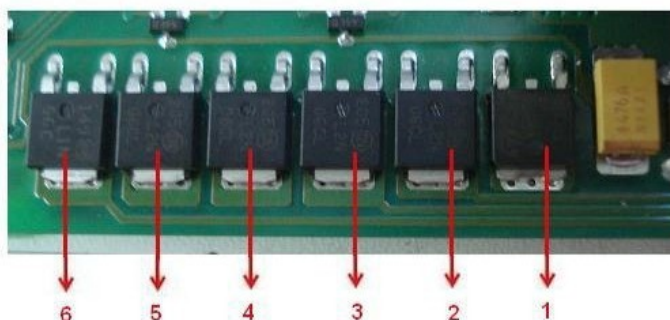
- Eprom pin 1 facing down;
- Has an immobilizer.

Common defects

- Failure in the stepper motor drive - L9122;
- Cold soldering on the multi-function drive - L9104 PD (20 pins tsop) due to heating.

Note: Some vehicles may show oscillation idle speed due to voltage instability (battery or alternator) and suspect a plant defect.

Drives

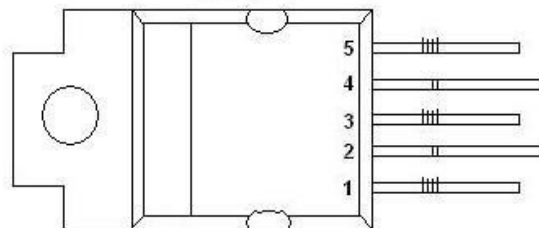


Pin	Component	Function	Pin ECU
1		lambda probe heating PWM	25
2		Injector 2	12
3		Injector 1	13
4		Injector 3	10
5		Injector 4	11
6		Canister or Cold Start	3

Ignition coil drive - VB 027



VB 027



Pin	Component	Function
1		Grounding
2		+ 5 V
3		Out - Pin 26 ECU
4		In - Firing
5		Diagnose

Note: The center of the Pole vehicle does not have the drive VB 027, due to the injection system using ignition as an external power module.

L 9104 PD



Pin	Component	Function
1		Grounding
2		Relay A / C - Pin 20 ECU
6		+ 5 volts
9		Pin 2 ECU
10		Grounding
11		+ 5 volts
13		+ 30
14		+ Hall - Pin 30 ECU
15		+5 volts - Butterfly - Pin 9 ECU
16		+5 volts MAP - Pin 8 ECU
18		IN - Main relay drive - Pulsed signal +
19		Out - Main Relay - pin 26 ECU - Pulsed Signal
20		Grounding

Stepper motor drive - L 9122



Pin	Component	Function
1		+ 5V
2		+ 5V
3		+ 15
5		In - coil 1 - Amplitude 5V
6		Out - coil 1 - pin 21 ECU - Amplitude 12 V
7		Out - coil 1 - pin 22 ECU - Amplitude 12 V
8		grounding wire
9		Out - coil 2 - pin 18 ECU - Amplitude 12 V
10		Out - coil 2 - pin 19 ECU - Amplitude 12 V
11		In - coil 2 - Amplitude 5V

Stepper motor drive test with oscilloscope

Adjust screen to 5V x 100ms

Auxiliary Processor - GSCT - 38318 PG 06

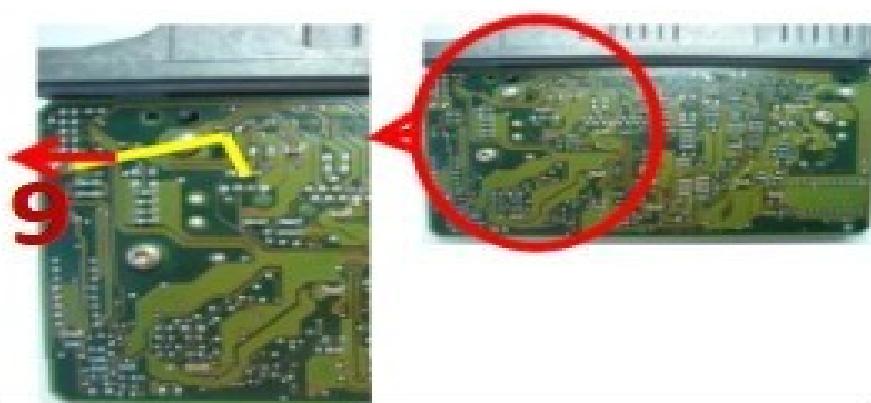
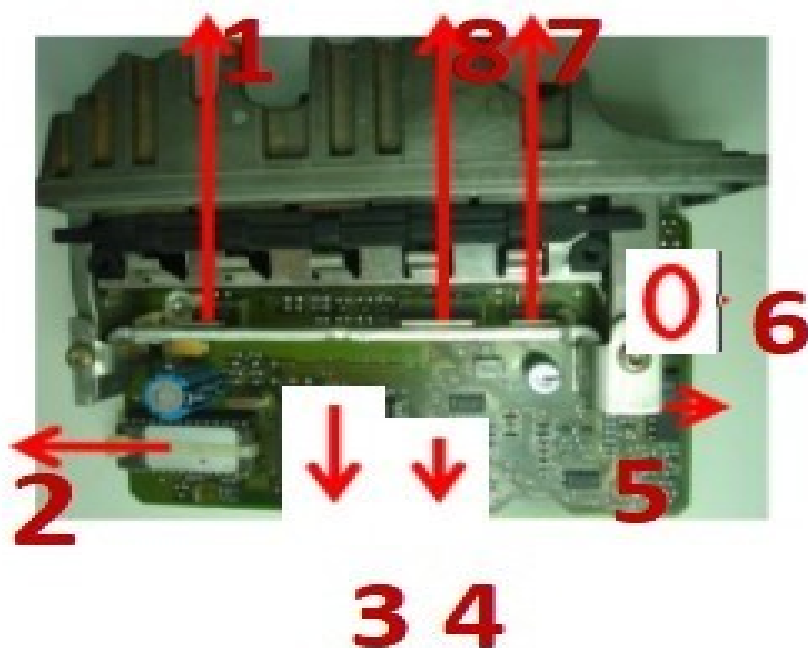


Pin	Component	Function
48		Out - rev counter (Panel)

MP 9.0 - VW

Vehicles: Gol 1.0 8V MI and Kombi 1.6 8V MI

FRONT



SEE IF

Main components

- 1** 5V regulator;
- 2** Eprom DIP;
- 3** Processor;
- 4** Knock sensor;
- 5** Coil;
- 6** Diodes;
- 7** Stepper motor;
- 8** Nozzle mult drive;
- 9** Jumper for defective coil repair.

Technical specifications

Special features

- There is MP9 without immobilizer, but most are with immobilizer. The Kombi ECU is interchangeable with that of the Goal and vice versa, as long as the programming of the EPROM, also check the knock sensor drive.

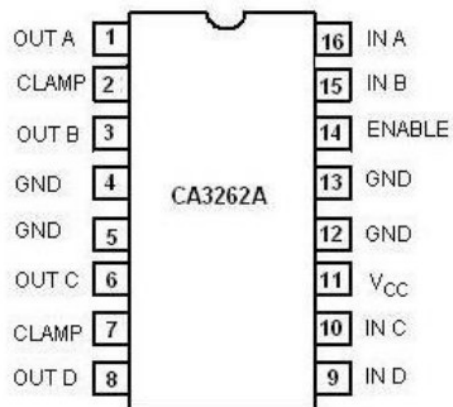
Common defects

- Coil CI and bad grounding.

Note: Burning in the coil IC, due to lack of grounding the ECU with the battery. We should do a between the ECU housing, the battery negative and the coil housing. Check that the coil heats the ignition module, if it gets hot we should replace it or repair it too.

B57942 or 3262 - Ignition coil, fuel and Canister.

This component works similarly to NPN transistors, that is, the input (trip) is made by positive 5 volts and the output is negative.



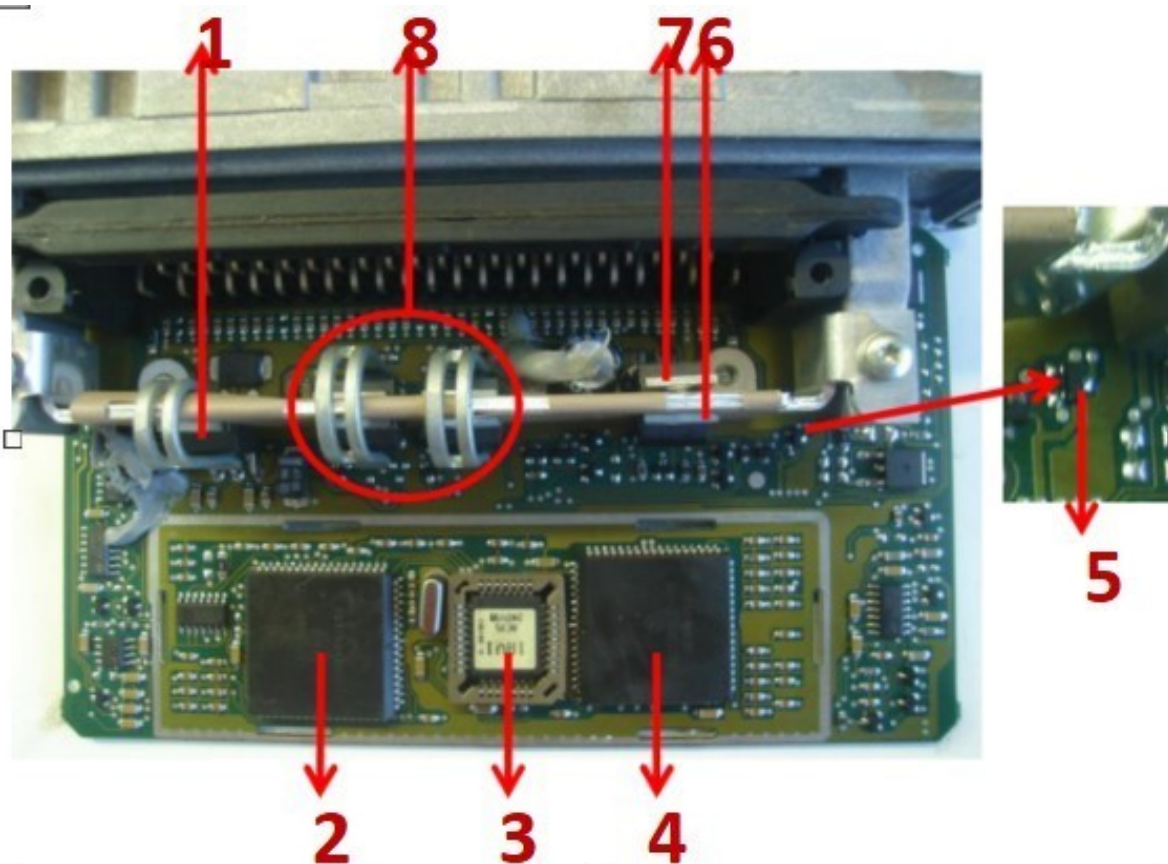
Pin	Componente	Function
1		Out - pump relay
3		Out - ignition coil
6		Out - Canister
8		Out - Realy A/C

Mult drive injectors

Pin	Component	Function
1		Input - Device - Output 3
2		Input - Device - Output 5
3		Out - Nozzle 4
4		+ 12 V
5		Out - Nozzle 3
6		Grounding
7		+ 5V
8		Grounding
9		+ 5V
10		+ 5V
11		Nozzle outlet 2
12		Diagnosis or reset
13		Nozzle outlet 1
14		Input - Device - Output 11
15		Input - Device - Output 13

1AVI – VW

Vehicles: Gol 1.0 16V



Main components

- 1 5V regulator;
- 2 Main processor;
- 3 Eeprom;
- 4 Auxiliary processor;
- 5 Coil transistor;
- 6 Canister;
- 7 Stepper motor;
- 8 Set of nozzle drivers.

• AVS is without immobilizer and AVI with immobilizer, a lock on the connector is the differential between one and the other. Eliminating hangs the AVS ECU works in place of the AVI. Attention to jumper on the ECU immobilizer.

Common defects

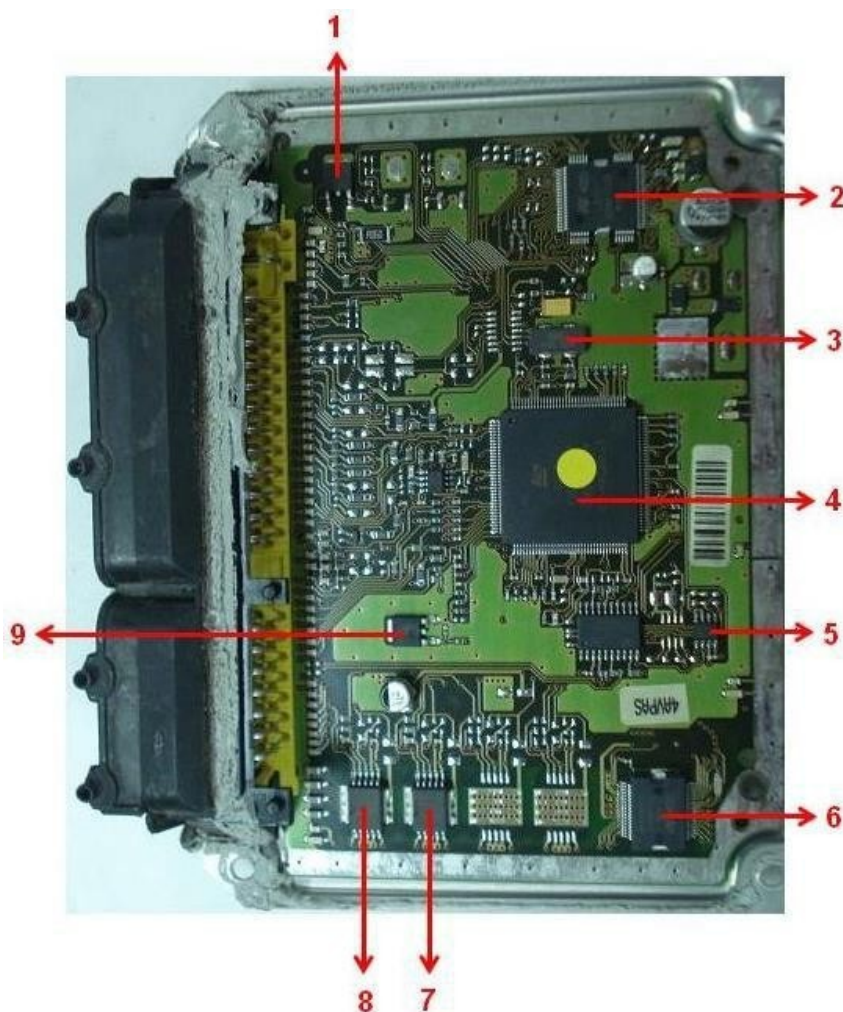
- Ci sensor map heats up;
- Ci stepper motor, without fully functioning;
- Bad contact in the eprom socket;
- Ci pump and Ci coil.

Note: The eprom 27C010, or 27C1001 can be replaced by 29F010. The main processor sends the ignition pulse signal directly to the coil in the front, we have two responsible SMD resistors for “filtering” the signal, and at the bottom we have two more 47Ω resistors that also do this job.

Follow the track from pin 24 to the main processor.

4AVP - VW

Vehicles: Gol, Saveiro 1.6 flex



- 1 Probe;
- 2 Cold start, pump relay;
- 3 Crystal;
- 4 Processor;
- 5 SOIC
- 6 Canister nozzles, stepper motor;
- 7 Coil 1;
- 8 Coil 2;
- 9 Cold broken relay.

Technical specifications

Special features

- Gol 1.6 8V Flex

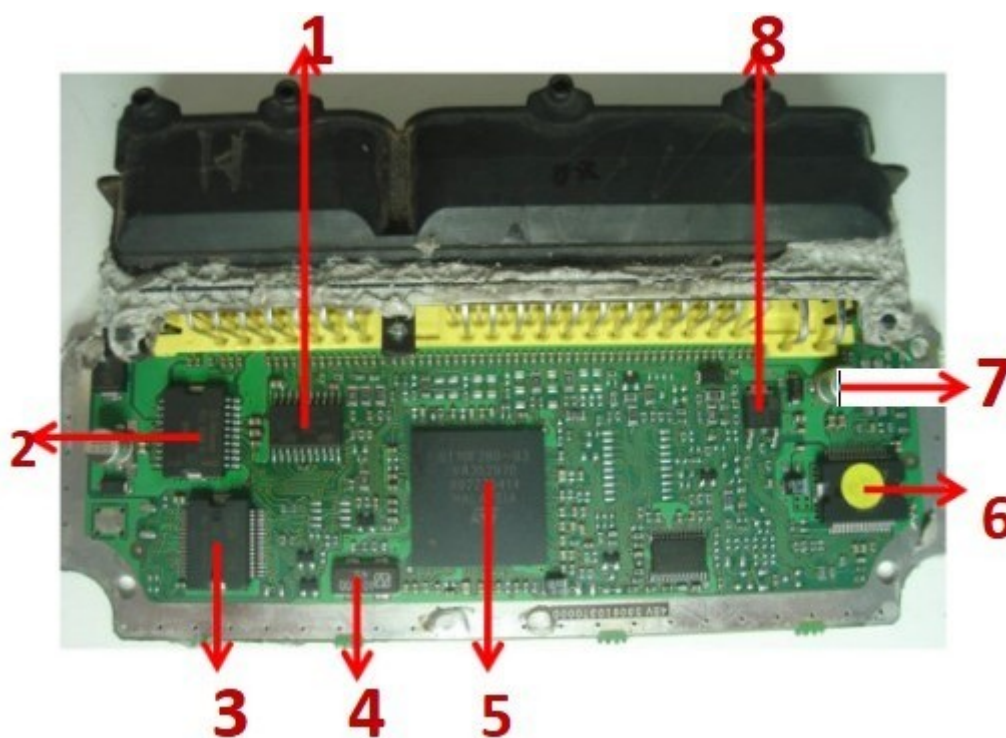
Common defects

- Ignition coil

Note: This ECU easily corrupts the ST processor, requiring only reprogramming again. This processor is a fundamental part of functioning of the ECU.

4BV - VW

Vehicles: Fox and Gol 1.0 8V



Main components

- 1 Coils;
- 2 Electronic accelerator;
- 3 Nozzle / Fan / Canister / A / C;
- 4 Crystal;
- 5 BGA processor;
- 6 5V regulator;
- 7 Capacitor;
- 8 Probe.

Technical specifications

Special features

- Gol 1.0 and Fox 1.6 8V

Common defects

- Ignition coil and throttle body *Note: This ECU easily corrupts the ST processor, requiring only reprogramming again.*

7.5.20 –VW

Vehicles: Goal flex.



Main components

- 1 Canister probe nozzles, cold start, fan;
- 2 Butterfly body;
- 3 5V regulator.

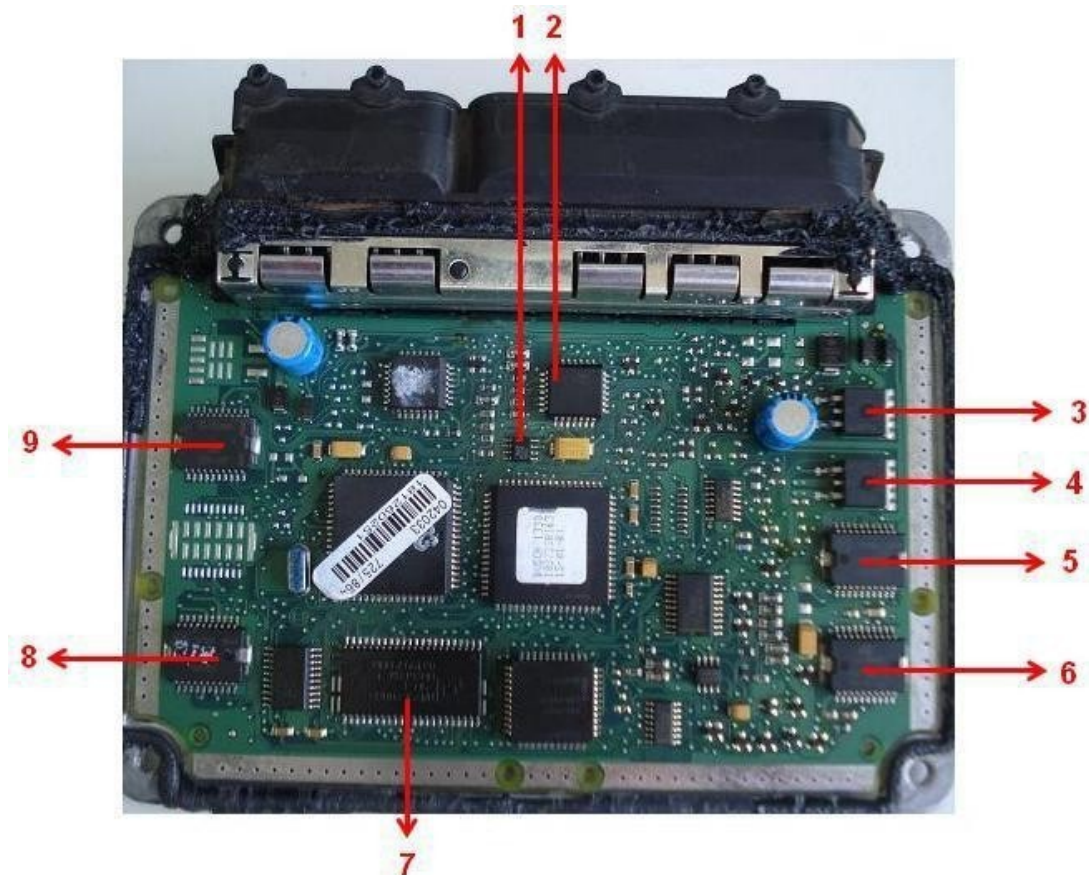
Technical specifications

Common defects

- Cold start / nozzle drive.

Bosch 3.8.3

Vehicles: Golf 2.0, Bora 2.0 and Gol turbo.



Main components

- 1 Soic 95040, Code;
- 2 Regulator 5 volts;
- 3 Probe 2;
- 4 Probe 1;
- 5 Canister;
- 6 Ignition complex;
- 7 Eprom;
- 8 Nozzles;
- 9 Idle corrector

Technical specifications

Special features

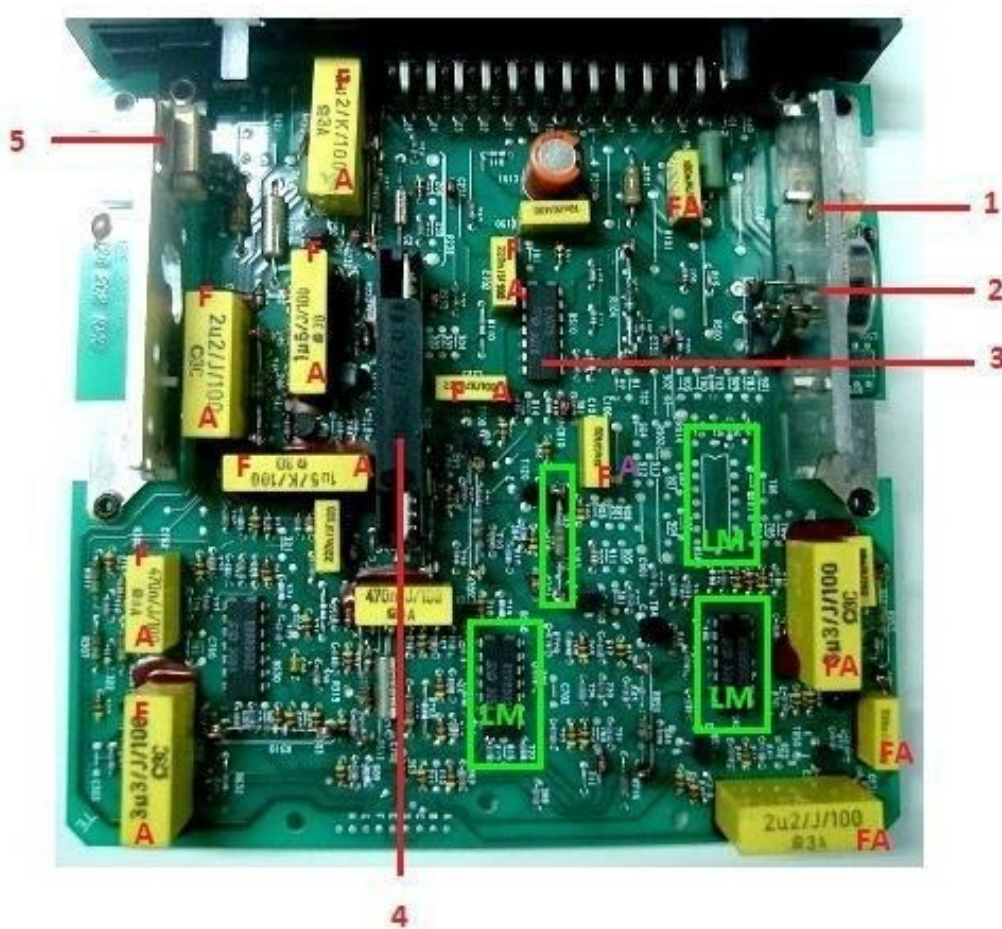
- 3.8.2 Passat 1.8 20V.
- 3.8.3 Gol 1.0 Turbo.
- 5.9.2 Bora 2.0.

Common defects

- When you close a short, the trails break easily, ignition system in the complex ECU is fragile.
- Lambda probe closes short and damages the ECU.

Bosch LE - JETRONIC

Vehicles: Gol GTI 1994 Santana Executive Kadet GS



Main Components

- 1** Ignition;
- 2** Ignition;
- 3** P2IGN;
- 4** Visual parameters module;
- 5** Injector nozzles.

Technical specifications

Special features

- No communication with scanner, ignition system for Santana and Versailles independent from the injection plant;
- Immobilizer does not exist in this system;
- Non-electronic analog processor;
- Analog non-electronic eprom;
- Injection light yes;
- RPM hall sign, distributor;
- Good trails;
- Cold welding, yes mainly in F and A;

Location: At the feet of the ride.

Application details: 007, VW line and 006, GM line.

Common defects: Cold solder across the plate, coil Ci burnt, open beak.

Note

1♦ blink and a sudden cut, or a sudden failure. That is from electromagnetic interference, probably spark plug, cable, coil, or any other current leak.

DO NOT rule out a problem with the EZK ignition module or even short on the whip. Another detail is the thread that goes to the coil, it has a grounding mesh, for

“FILTER” the signal to the coil but it is NOT going on next to the coil signal, this generates a short.

In the air flow harness, if the vehicle is shorted, it can not get and work with a super excess and flow is without function.

The knock sensor socket is easily replaced with that of the butterfly sensor, or even the broken or very open detonation.

If the Water temperature sensor is applied wrong, the vehicle may have ignition pulsed at the nozzles, however, do not use excess fuel.

Weak coil, or high resistance in the cables interferes with starting the vehicle.

Electrical conductivity in the cooling water or even alternator overload interferes with the vehicle after of hot.

Technical Notes

Nozzle drive: Darlington transistor, NPN (code commercial VM = TIP122 or TIR 127).

Attention, for the insulating plastic with the heat sink plate heat, and also with the fixing clamp. Test the CI with the multimeter in DIODOS, the measurement should be between 0.4V at 0.8V. If the nozzle transistor is blown, the resistance will be low or short-circuit between Collector and Emitter.

✓A TIP31 can also be used;

✓Remember that the CI CJ07S controls the nozzles.

If the nozzles lock open, REMOVE the plate and test the following components, LM2902A, LM2901A and LM2902B, can be replaced by an LM2902N (number commercial).

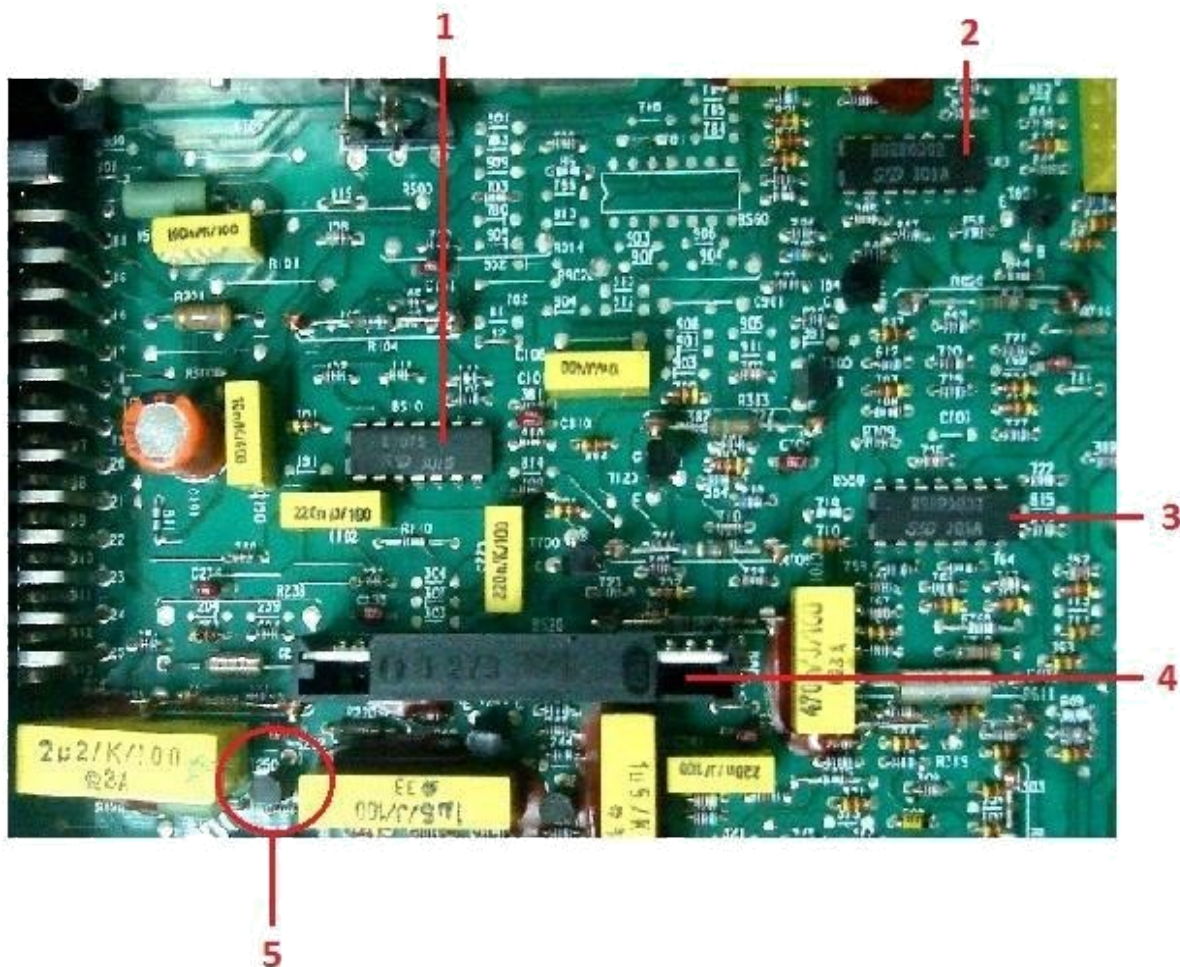
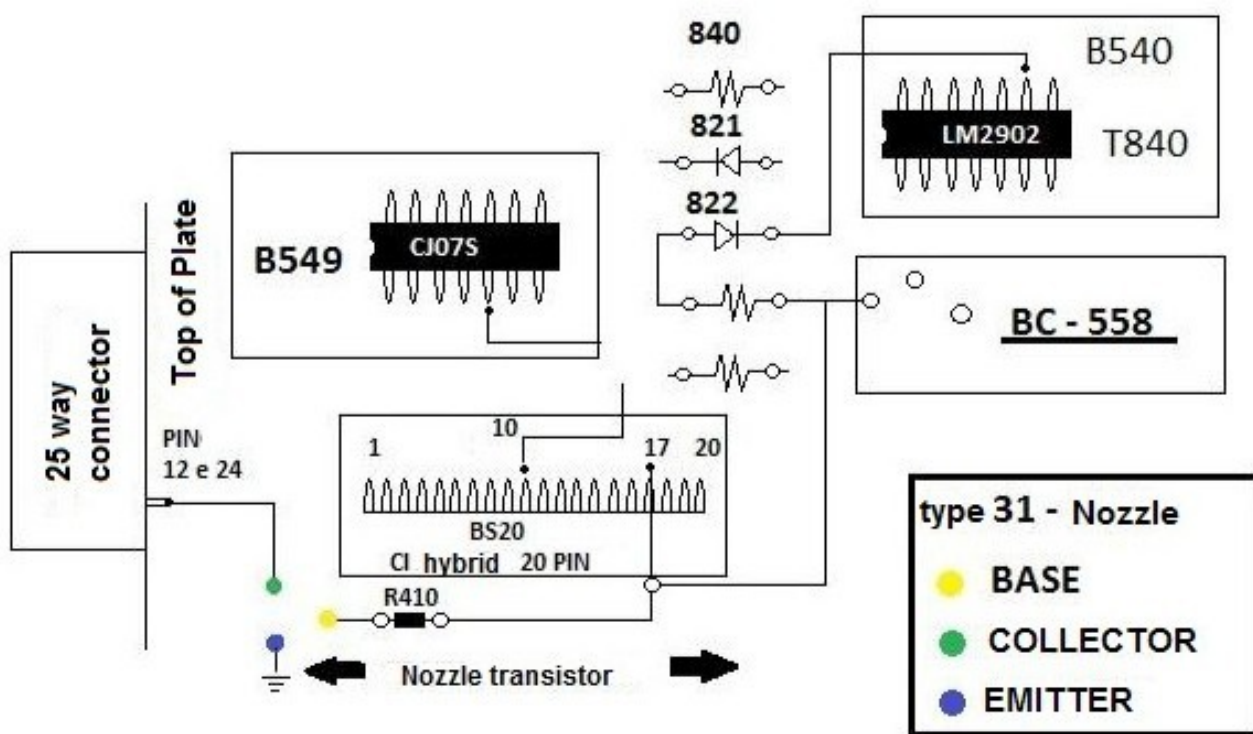
Ignition drive: Due to overheating (usually the shorted coil causes this), the solder are in bad contact. Another problem is the FET-drive ignition is burnt out due to this overheating or short in the whip. The ignition drive on the Marelli- G7 you can even replace this drive. Following the pin sequence A, B and C of the plate with pin 1, 2 and 3 of the new FET.

Airflow: The car has too much fuel, or it picks up and die. When you turn the flow off and on again, the vehicle it works. Usually just resoldering points F and A marked in the photo. Also check the electronic whip of the car.

Ignition: Below FET, there is the BC547 resistor, (NPN) that interconnects with the ignition system, EZK. The FET ignition is isolated and commercially we can find such as BIP109, 125B.

P1: Point where it connects with the ignition system. It may happen to burn the resistor or the diode.

P2: Injection unit without function. When you burn the component CJ07 the unit goes out completely, nor nozzle and no coil.



- 1** Controls nozzle pulse, and influences the ignition pulse also, if you have a problem you can leave the nozzles open;
- 2** Responsible for the beak pulse, in case of problem beak is open too;
- 3** Causes excess fuel;
- 4** Responsible for sending signals to the air flow, causing excess fuel;
- 5** Component T250 has a problem and no longer reads their flow.

Bosch ME 7.4.4 Petrol

Vehicles: Peugeot Citroën 1.6 16V and 1.4 16V family.



Main Components

- 1 5volts regulator;
- 2 TPS converter;
- 3 Coil 1;
- 4 Coil 2;
- 5 Nozzles, electro and Canister.

Technical specifications

Special features

- Soic 95160 and psop 29f400;
- Immobilizer integrated with BSI / CSI, decodes only in gasoline vehicles. (stops the air conditioning).

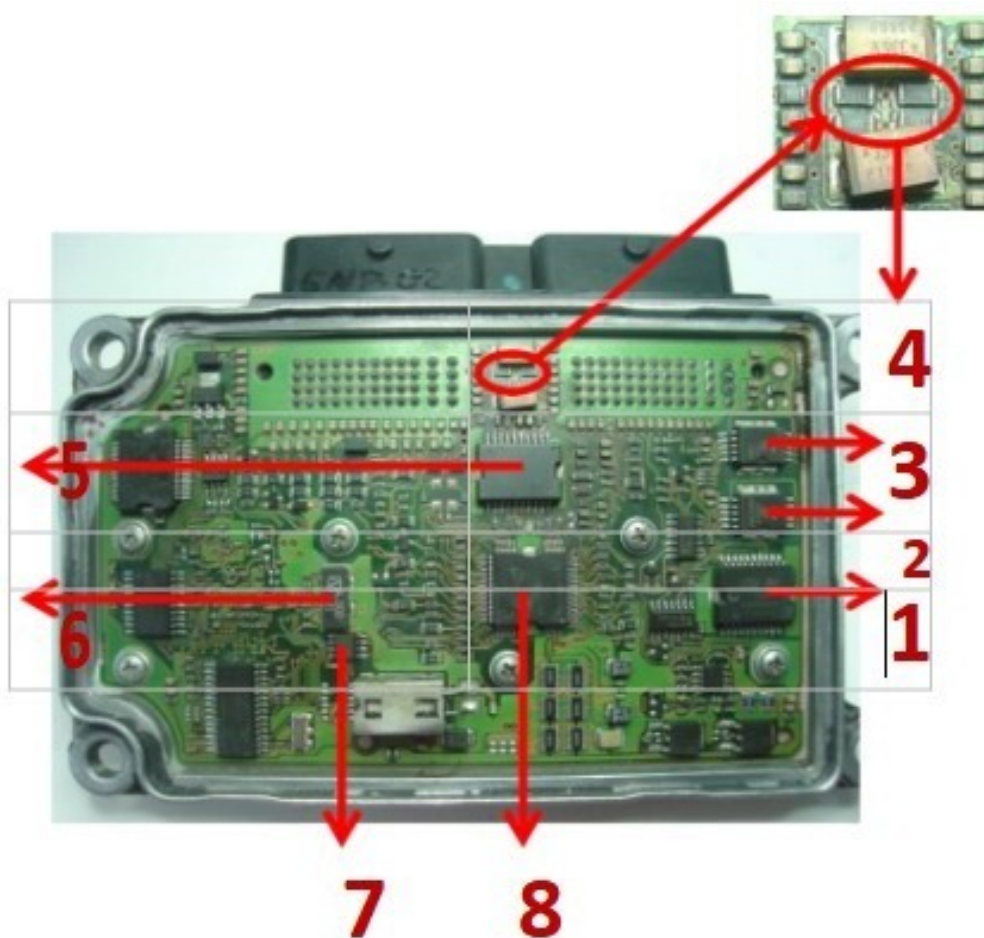
To code the entire system, the
Peugeot Citroën;

Common Defects

Check the immobilizer system carefully.

Magneti Marelli 5NP 02

Vehicles: Peugeot Citroën Family



Main components

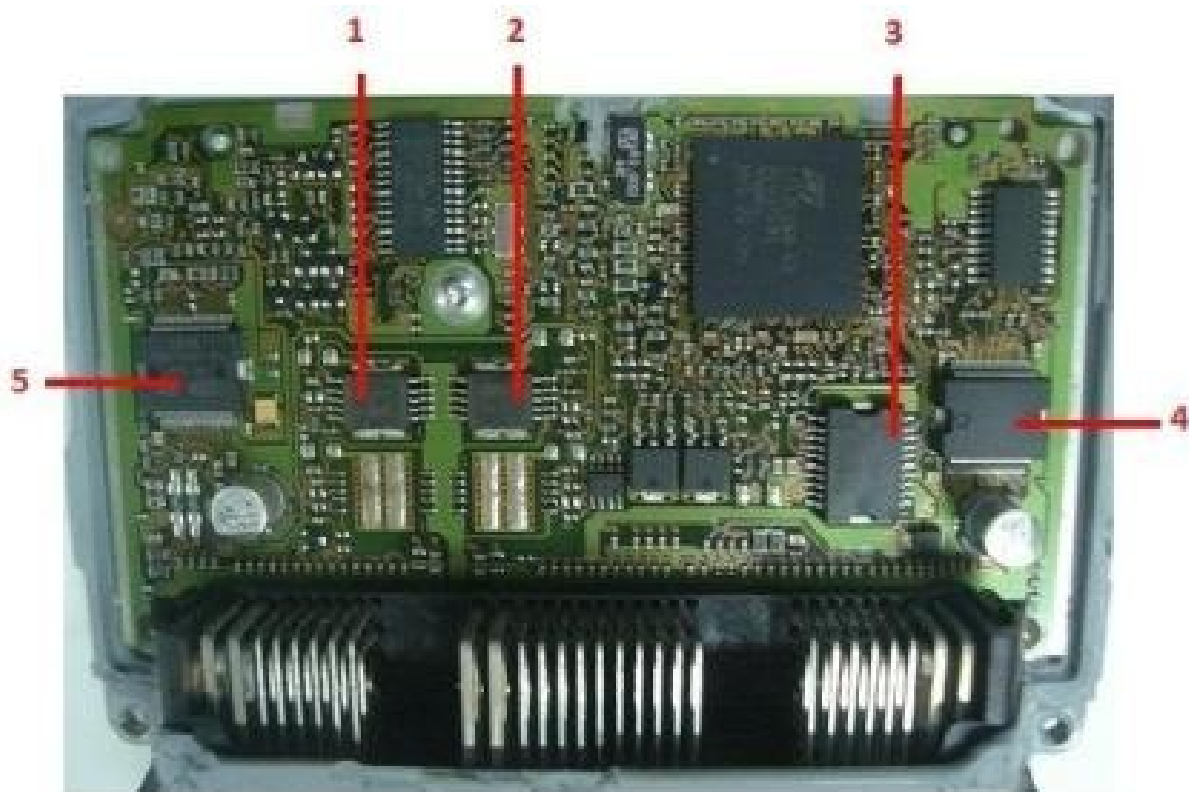
- 1 Nozzles;
- 2 Coil 3 and 4;
- 3 Coil 1 and 2;
- 4 Set of protective resistors for drive No. 5 (1Ω);
- 5 Electronic accelerator;
- 6 Crystal;
- 7 Soic 95160 / Code;
- 8 5V regulator.

Common Defects

- Eco mode
- Lose acceleration

Magneti Marelli 6LP1

Vehicles: Peugeot Citroën Family (Petrol System)



Main Components

- 1 Coil 1;
- 2 Coil 2;
- 3 Boboleta body;
- 4 Nozzles, Canister and Electro Relay;
- 5 5volts regulator.

Technical specifications

Special features

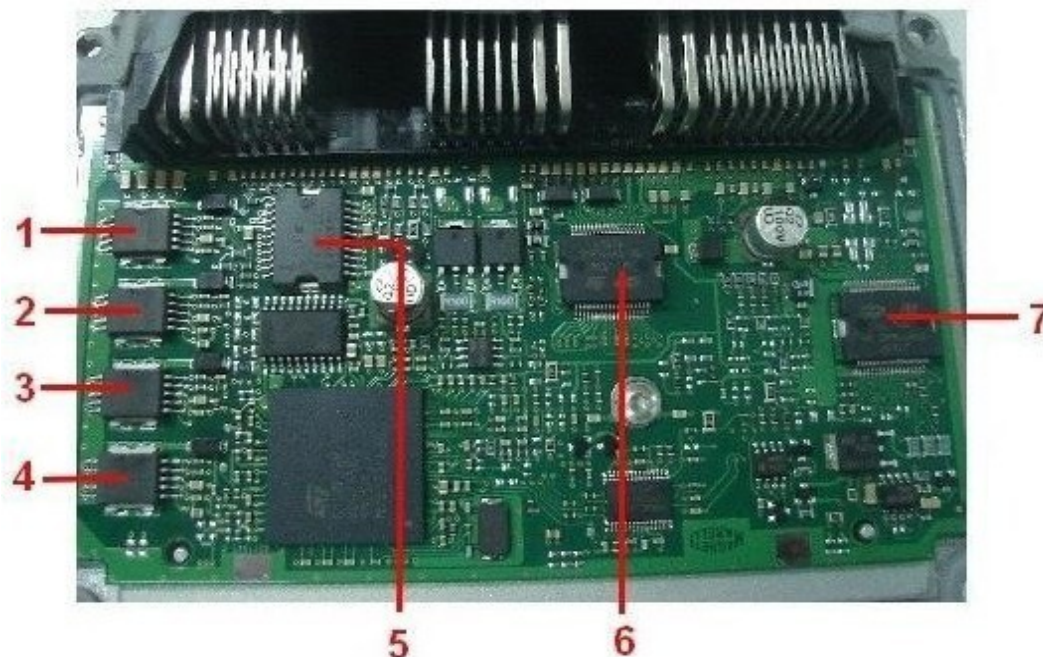
- ST processor, with integrated immobilizer system;
- ECU RESET in bank 1 and 2;

Common Defects

- Check the immobilizer system carefully;
- Be careful when exchanging from 6LP1 to 6LPb, usually something is malfunctioning.

Magneti Marelli 6LPb

Vehicles: Peugeot Citroën Family (Gasoline Systems)



Main Components

- 1 Coil;
- 2 Coil;
- 3 Coil;
- 4 Coil;
- 5 Butterfly body;
- 6 Nozzles, Relay and Canister;
- 7 5volts regulator.

Technical specifications

Special features

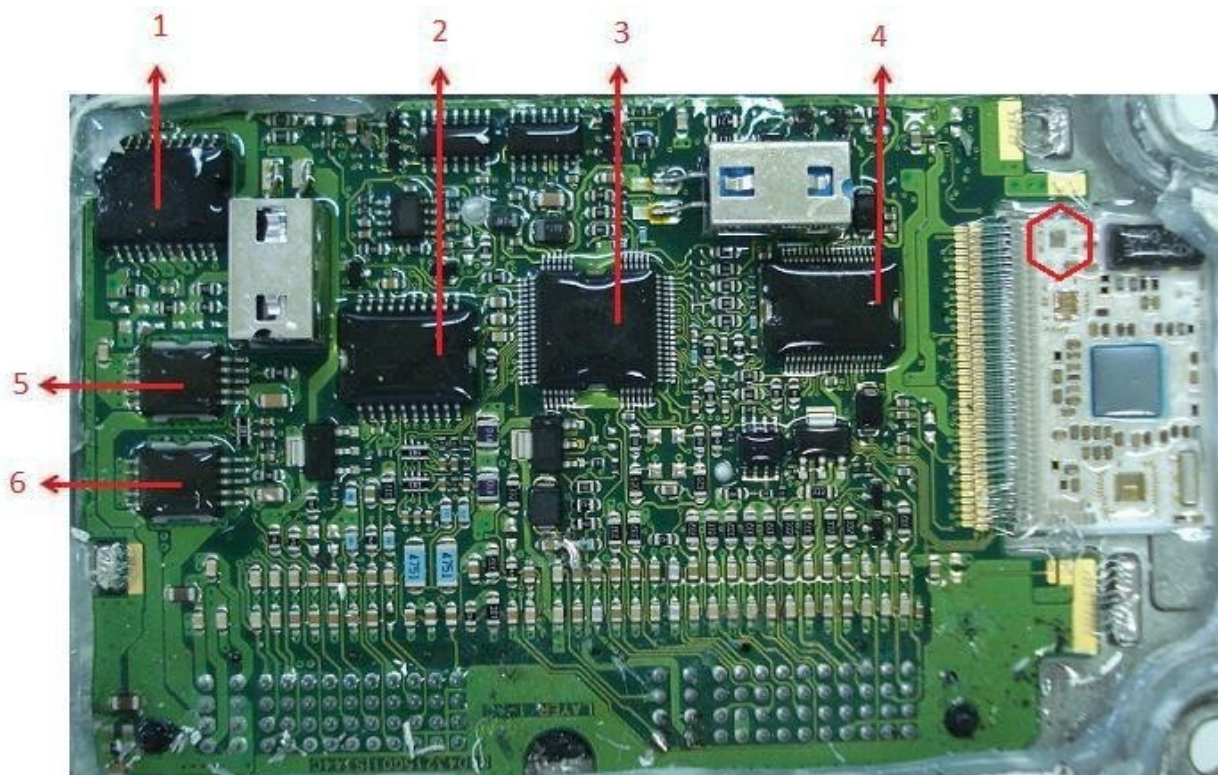
- ST processor, with integrated immobilizer system;
- ecu RESET in bank 1,2 and 3;

Common Defects

- Carefully check the immobilizer system on the immobilizer repair manual;
- Be careful to exchange from 6LP1 to 6LPb, usually something is malfunctioning.

Magneti Marelli 5NP 01

Vehicles: Clio 1.0 16V and Peugeot 206 1000 16V.



Main Components

- 1 Butterfly body;
- 2 Electro Nozzles;
- 3 5volts regulator;
- 4 Fan;
- 5 Coil 1;
- 6 Coil 2;
7. SOIC 95160.

Technical specifications

Special features

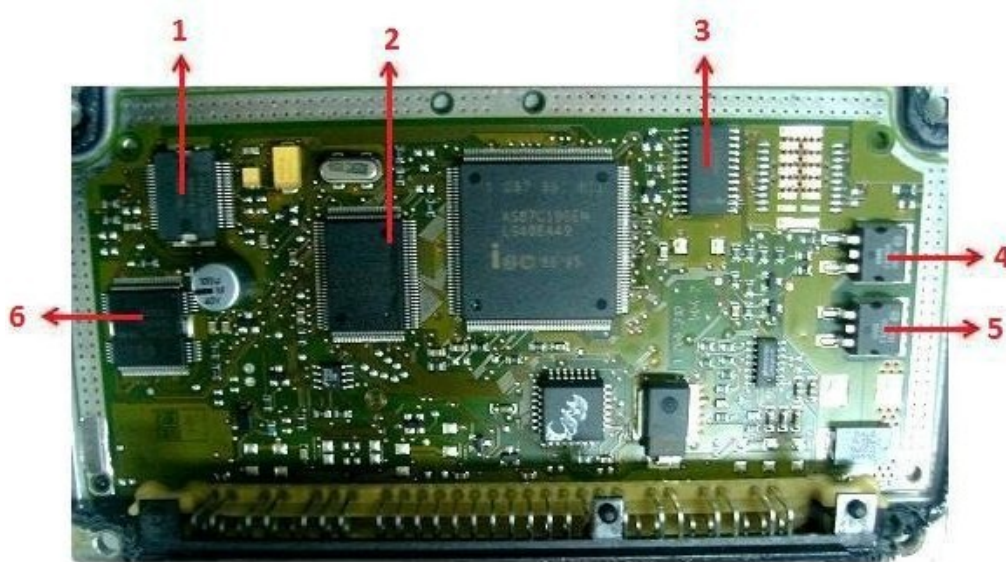
- 1 Semi hybrid;
- 2 Integrated with UCH, decode through procedures;
- 3 Pin 01 side of the connector;
- 4 Phased phonic RPM signal;

Common Defects

- 5NR1 is different from 5NR2 and is not interchangeable with 5NP because of the air conditioning system.

Bosch MP7.2

Vehicles: Peugeot Citroen Gasolina Family;



Main Components

- 1 5volts regulator;
- 2 TPS converter;
- 3 Idling;
- 4 Coil 1;
- 5 Coil 2;
- 6 Fan, Nozzle, Air conditioning and Canister.

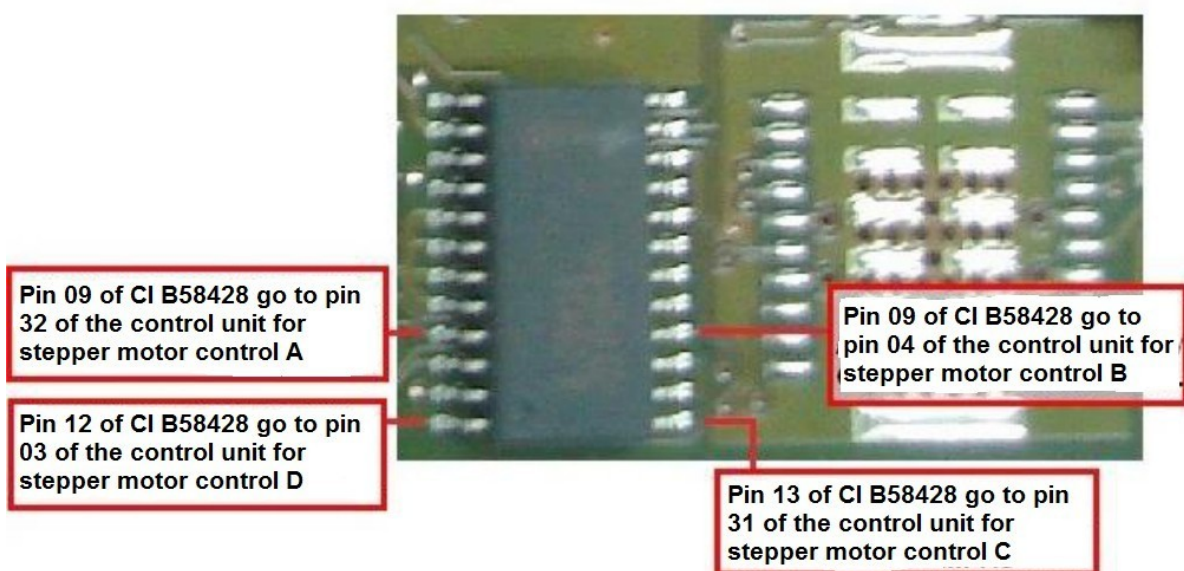
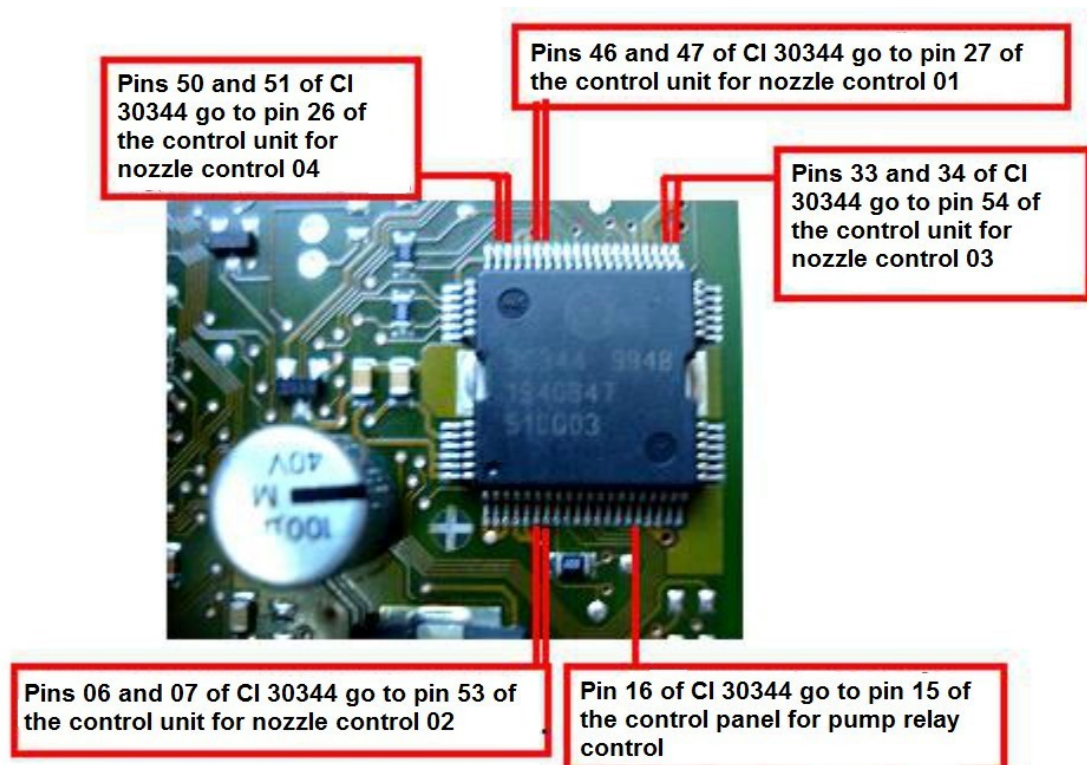
Technical specifications

Special features

- With soic 24C02
- Immobilizer integrated with BODY, reading password at BSI, decodes only on gasoline vehicles. To encode Toto the system is necessary the Peugeot scanner Citroen.

Common defects

- Carefully check the immobilizer system on the immobilizer repair manual.



Siemens Sirius 3134

Vehicles: Scenic, Clio and Sandero



Main Components

- 1 Coil 2;
- 2 Coil1;
- 3 Butterfly body;
- 4 5volts regulator;
- 5 Nozzles.

Main Components

1. Processor.

Technical specifications

Special features

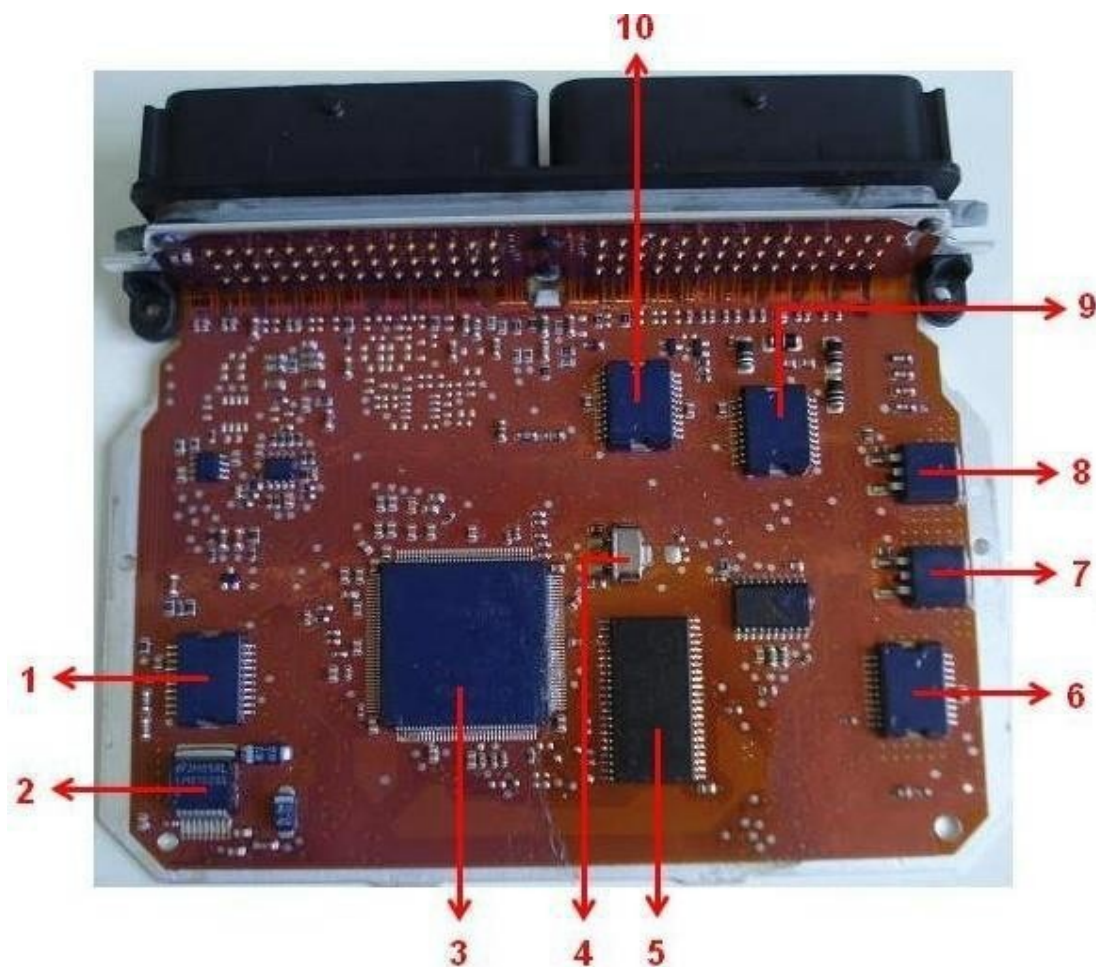
- Applied in Flex systems, some vehicles have been applied Sirius 3132 and these are gasoline only;
- Processor integrated with UHC.

Common Defects

- Plate oxidation
- Failure of the immobilizer program (a reset must be PSOP F400)

Siemens - Sirius 32.

Renault

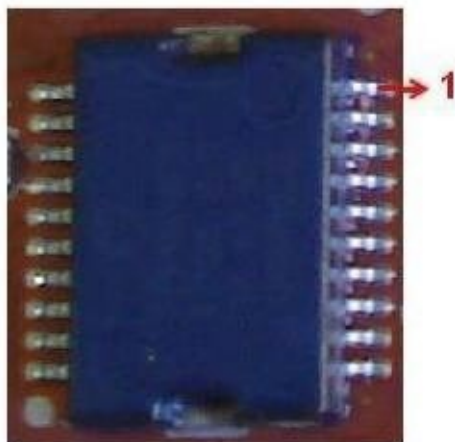


Main components

- 1 Injector drive - Motorola - TY94084 FB - 20 pins;
- 2 5V regulator;
- 3 Processor;
- 4 Crystal - Clock - 8MHz;
- 5 Rom memory - 29F200 BB;
- 6 Multidrive - Canister - Motorola - TY94082 FB - 20 pins;
- 7 Ignition drive - Cylinder 1/4;
- 8 Ignition drive - Cylinder 2/3;
- 9 Multidrive - Main relay - Fuel pump relay - V1
- Cooling - ST - ATM 398 - 20 pins;
- 10 Stepper motor drive - ST - U705 SD1C03 - 20 pins.

Injector Drive - Motorola - TY94084 FB - 20 pins

On the bench, the injection sequence is 2 to 2.



Pin	Component	Function
1 - 10 - 11 - 20		grounding
2		Out -Injector 2 - Pin 90 ECU
4		In - Injector 2
8		In - Injector 4
9		Out - Injector 4 - Pin 89 ECU
12		Out -Injector 3 - Pin 60 ECU
13		In -Injector 3
14 -15		+ 5v
17		In -Injector 1
19		Out - Injector 1 - Pin 59 ECU

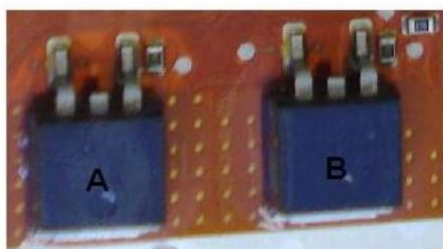
Pin	Component	Function
2		
12		Out - Canister - Pin 4 ECU
19		Out - Lambda probe heating control

5V Regulator



Pin	Component	Function
3		+ 30
4		+5V - TPS - Pin 75 ECU
5		Grounding
6		+5V
7		+5V
8		+5V
9		+5V

Ignition Drives



Drive A

Pin	component	Function
1		Firing
2		Out - Pin 1 ECU -Cylinder 2 / 3
3		Grounding

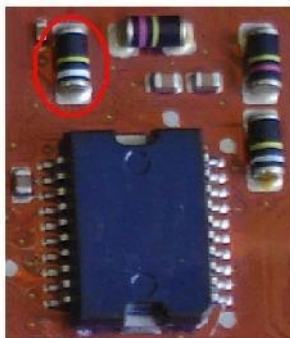
Drive B

Pin	component	Function
1		Firing
2		Out - Pin 32 ECU - Cylinder1/ 4
3		Grounding

Multidrive - Main relay - Fuel pump relay - V1 Cooling - ST - ATM 398 - 20 pins

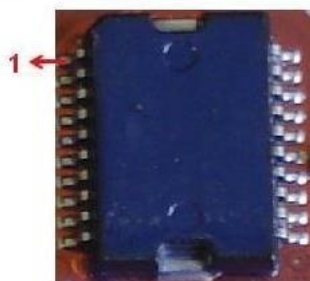
Between pin 5 of the component and pin 68 of ECU. They are protection diode.

Fuel Pump Diode



Pin	Component	Function
4	Out - V1	Cooling
5	Out - Fuel Pump Relay	- Pin 68 ECU
6	Out - V1	Cooling
14	Out - Giro Account	
15	Out - Main relay	- Pin 39 ECU

Stepper motor drive - ST - U705 SD1C03 - 20 pins

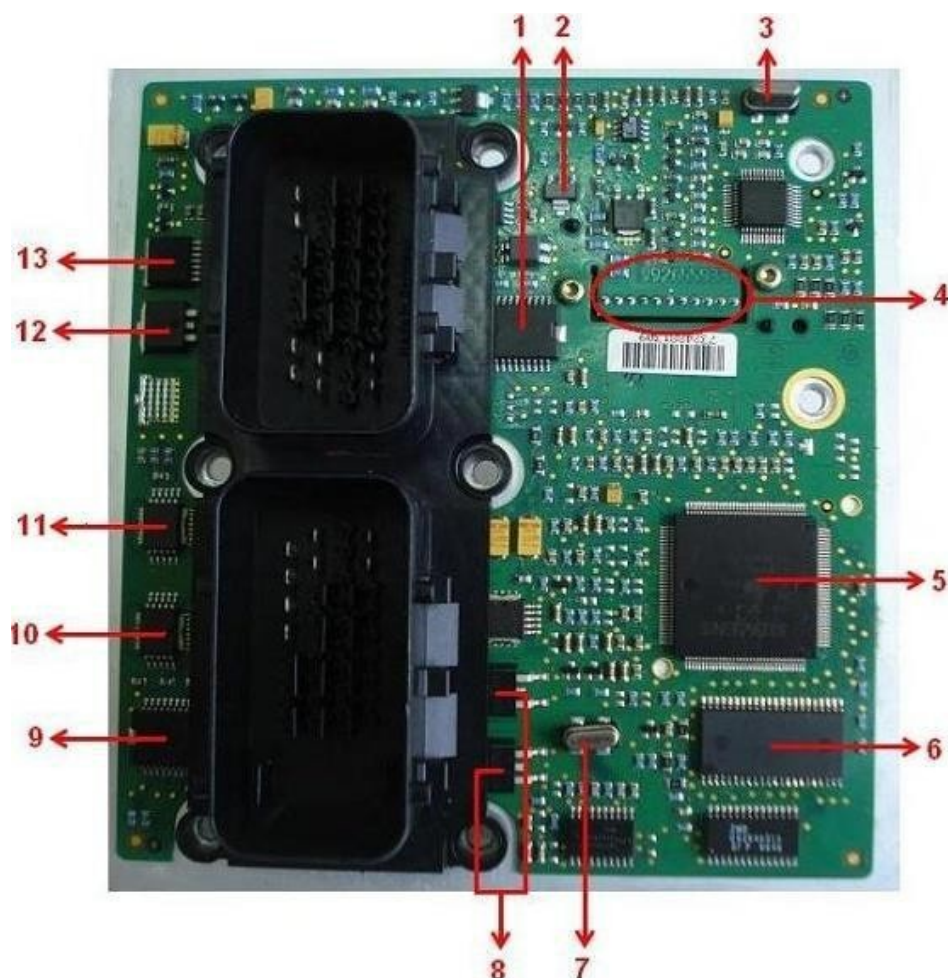


Pin	Component	Function
1 - 20		Grounding
2	Out - Stepper Motor - Coil B	- Pin 42 ECU
3	Out - Stepper Motor - Coil A	- Pin 12 ECU
4		+15
15		+ 5V
17		+ 15
18	Out - Stepper Motor - Coil A	- Pin 41 ECU
19	Out - Stepper Motor - Coil B	- Pin 72 ECU

Common Defects

- Immobilizer lock (must be reset or decoded)
- Breaking of trails near the mouthpiece

Mercedes Class A 160 and 190 - VDO MSM



Main components

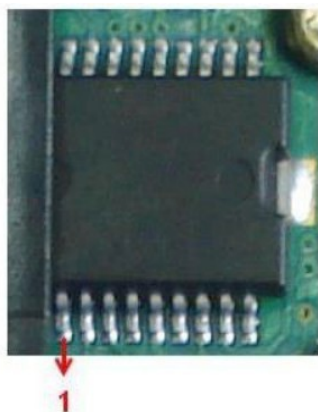
- 1 TLE 5216 G - 20 pins - Multi drive - Pump relay - Canister;*
- 2 Start control drive - line 50;
- 3 Crystal - 8 MHz;
- 4 Air flow meter; (under the terminals)
- 5 Processor
- 6 29F200 BA - Memory Rom;
- 7 Crystal - 4 MHz;
- 8 IRSFS 3010 S - DC motor drives;

- 9 TLE 5216 G - 20 pins - Multi drive of the injectors; *
- 10 VB 027 A - Ignition drive - Cylinder 2 and 3;
- 11 VB 027 A - Ignition drive - Cylinder 1 and 4;
- 12 LZ 24 NS - heating control of the lambda probe;
- 13. 4267 G / F901 - 7 pins;

Common Defects

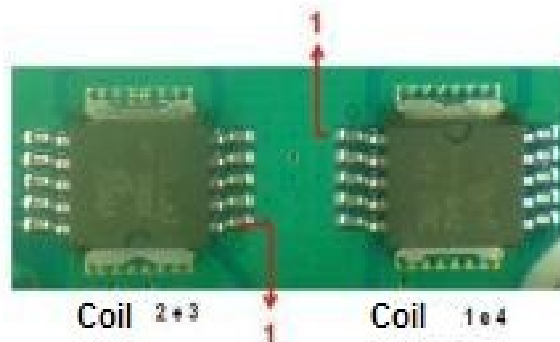
- Start Error (from the immobilizer);
- Air mass sensor.

TLE 5216 G - 20 pins - Multi drive - Pump relay - canister



Pin	Component	Function
3		In - Firing - Injector 3
4		In - Firing - Injector 4
5		Out - Injector 3 - Pin 3 ECU
7		Out - Injector 4 - Pin 4 ECU
14		Out - Injector 1 - Pin 1 ECU
16		Out - Injector 2 - Pin 2 ECU
17		In - Firing - Injector 1
18		In - Firing - Injector 2
1 -10 -11 - 20		ground wire

Ignition drive



ignition coil 1 + 4

Pin Component	Function
9	In - firing - Comes from Pin 49 of the processor
Chassi	Out - pino 8 ECU

Bobina de Ignição 2 + 3

Pin Component	Function
9	In - firing - Comes from Pin 48 of the processor
Chassi	Out - pino 7 ECU

Conclusion

Make sure that all possible tests have been carried out and all correctly, as the module is one of the vital parts of the vehicle and has a high repair cost!